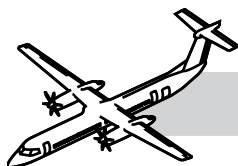


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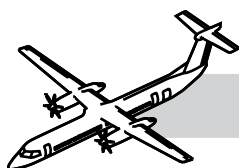
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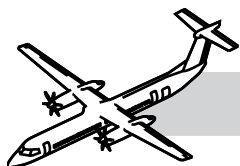
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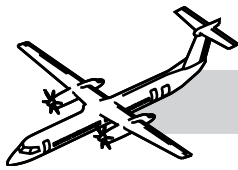
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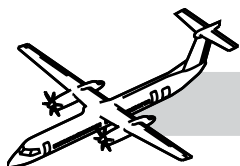
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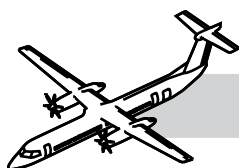
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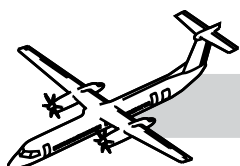


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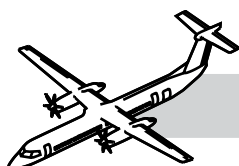


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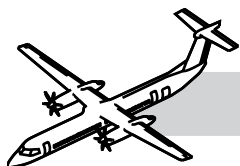


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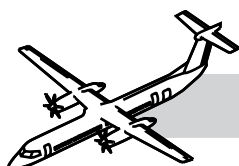
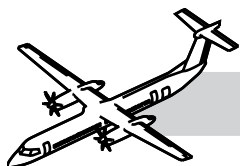
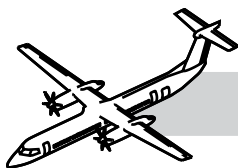


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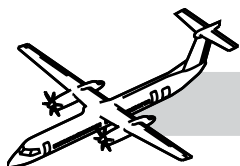
AIR CONDITIONING



21-00-00 AIR CONDITIONING

INTRODUCTION

The air conditioning system supplies the flight compartment and cabin with conditioned air to ensure passenger and crew comfort.



GENERAL

Refer to:

- Figure 21-1. ECS Simplified Block Diagram.
- Figure 21-2. ECS Detail Block Diagram.
- Figure 21-3. Air Conditioning System Control Panels.

The air conditioning system has these subsystems:

- Flow Control
- Temperature Control
- Pressurization Control
- Indications.

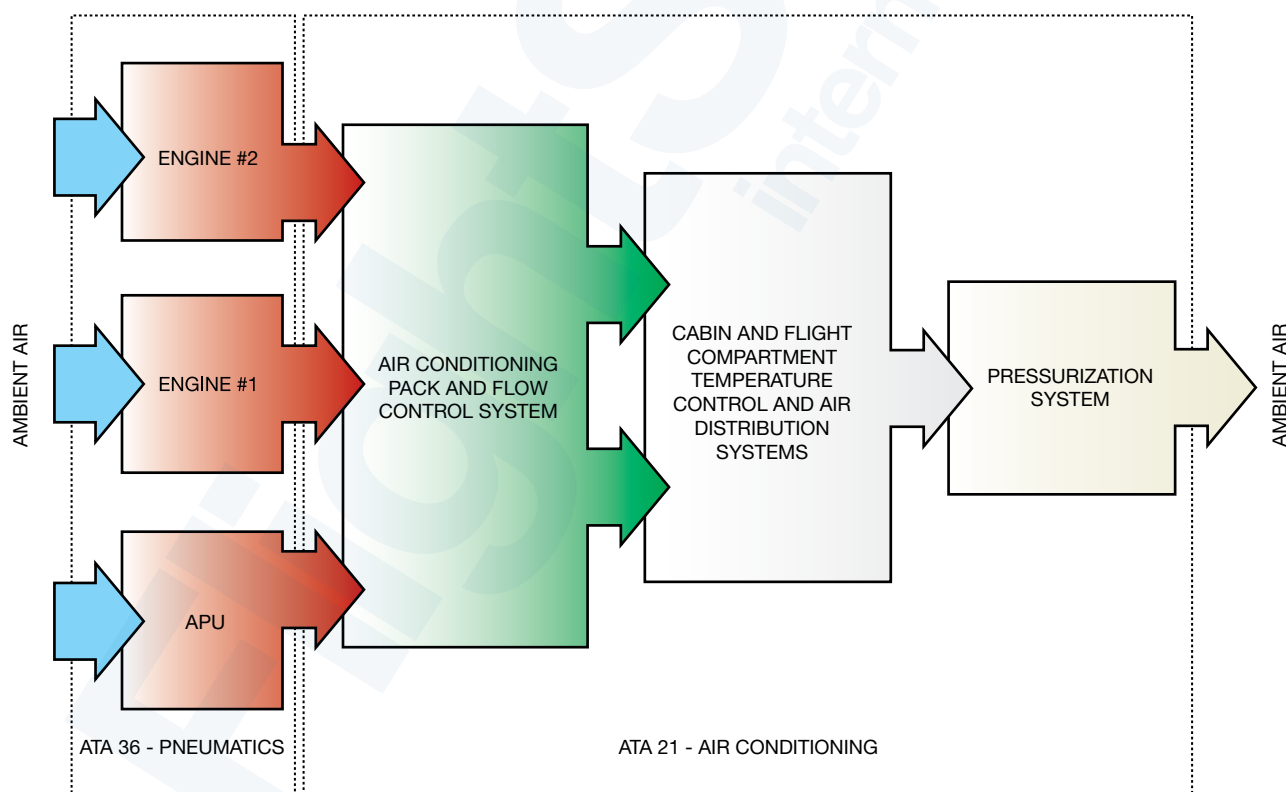


Figure 21-1. ECS Simplified Block Diagram

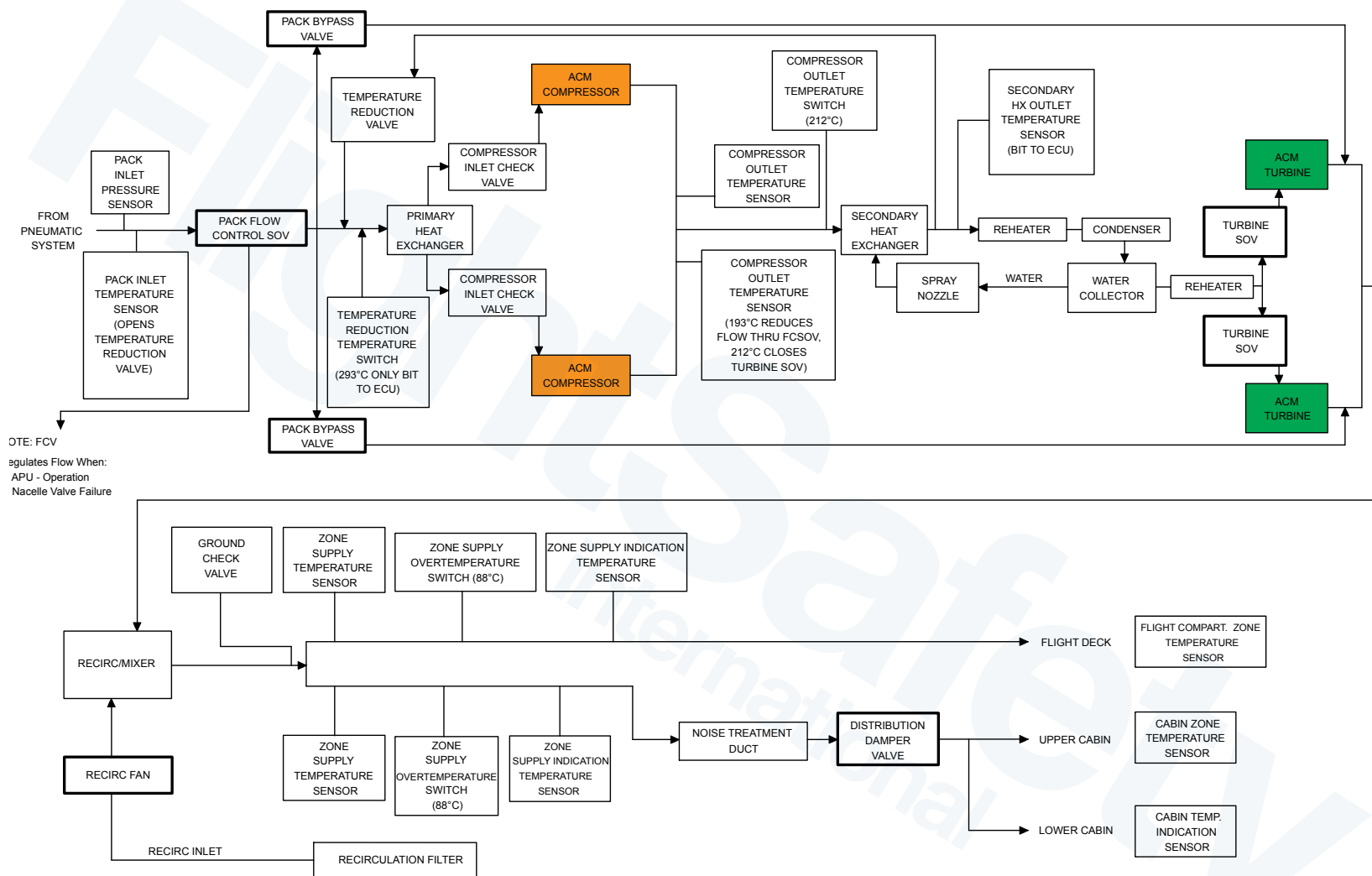
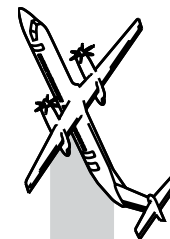
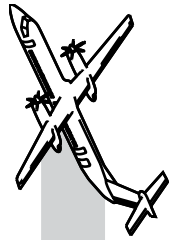


Figure 21-2. ECS Detail Block Diagram





OVERHEAD CONSOLE

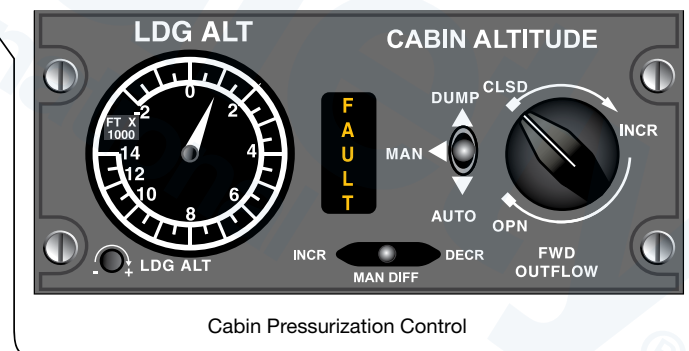
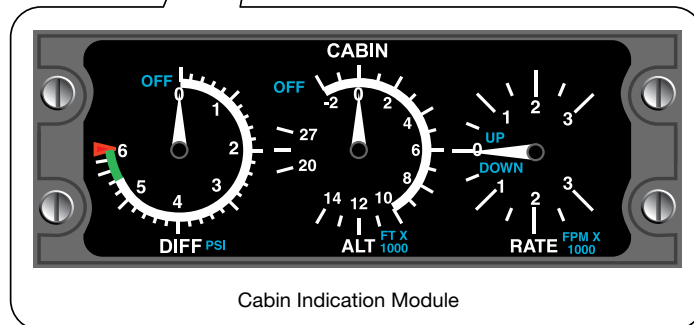
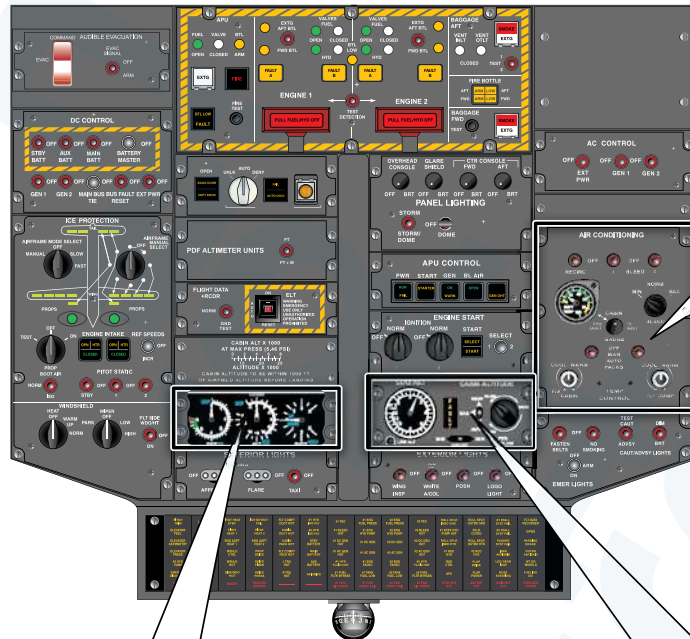
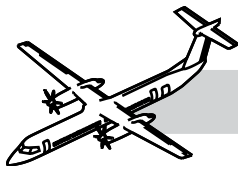


Figure 21-3. Air Conditioning System Control Panels



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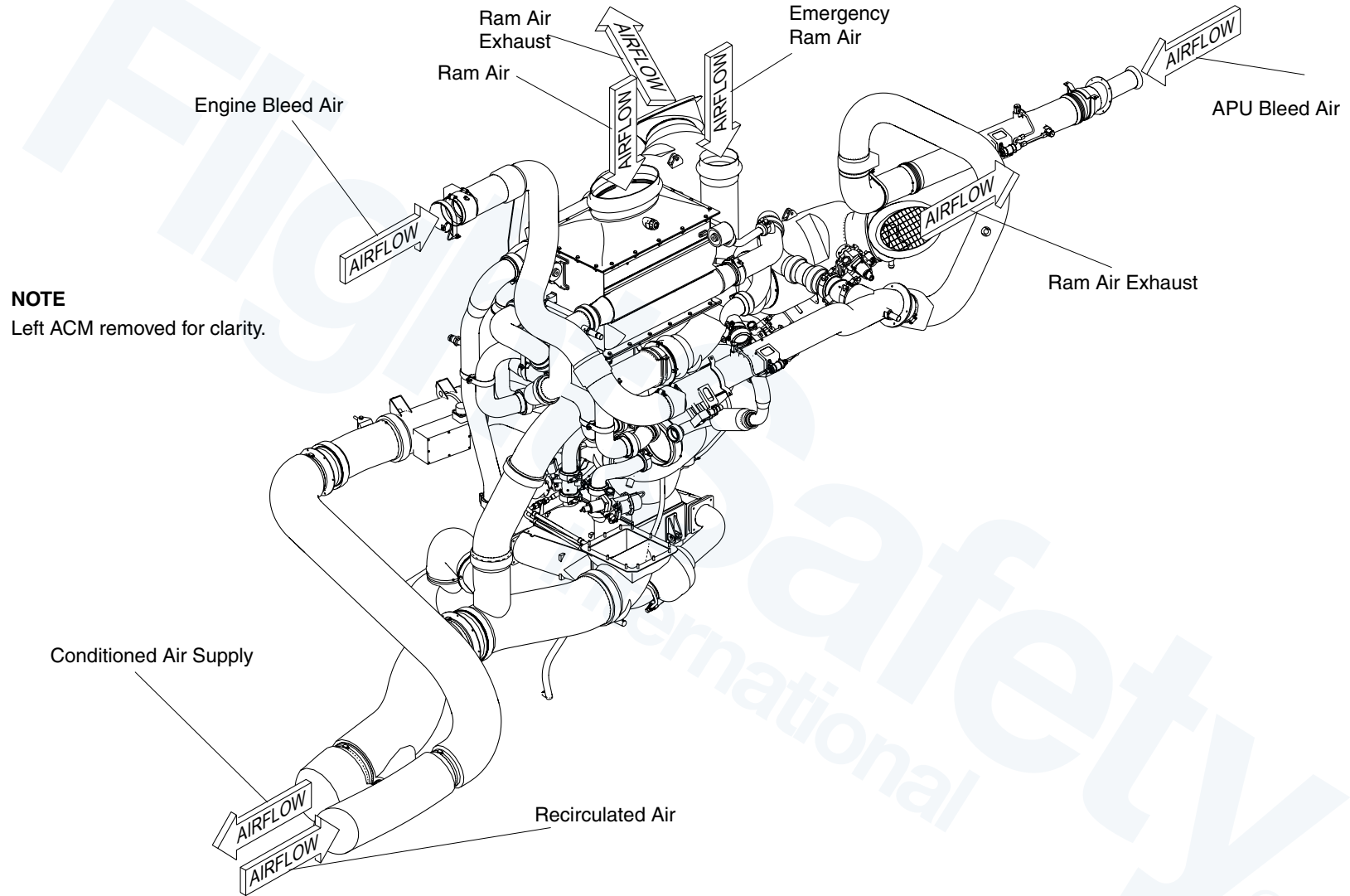
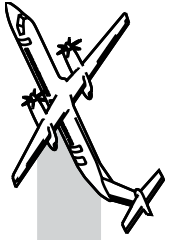
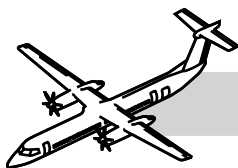


Figure 21-4. Air Conditioning Pack Air Flow



21-51-00 AIR CONDITIONING PACK

GENERAL

Refer to Figure 21-4. Air Conditioning Pack Air Flow.

The air conditioning pack is part of the Environmental Control System (ECS). It uses bleed air from the engines or APU to supply conditioned air to the cabin and flight compartment.

The system responds to flight crew commands and compartment temperature controls to supply a comfortable temperature environment for passengers and crew.

Two Air Cycle Machines (ACM), with one dual heat exchanger located in the aft fuselage, cool the bleed air coming from the two engines. This configuration provides the redundancy of two packs while allowing access to a much larger dual heat exchanger during operation with a single ACM.

The air conditioning pack has the following components:

- Heat-dual exchanger
- Condenser/reheater/mixer
- Spray nozzles
- Water collector
- Air cycle machine
- Fan inlet diffuser housing
- Pack bypass valve
- Compressor inlet check valve
- Turbine shut-off valve
- Fan bypass check valve
- Ram outlet check valve
- Ram inlet check valve
- Temperature reduction shut-off valve

- Air conditioning pack ducts
- Air conditioning pack struts
- Air conditioning pack mounts
- Temperature reduction valve
- Overtemperature switch compressor
- Outlet temperature sensor compressor and
- Secondary heat exchanger sensor outlet temperature.

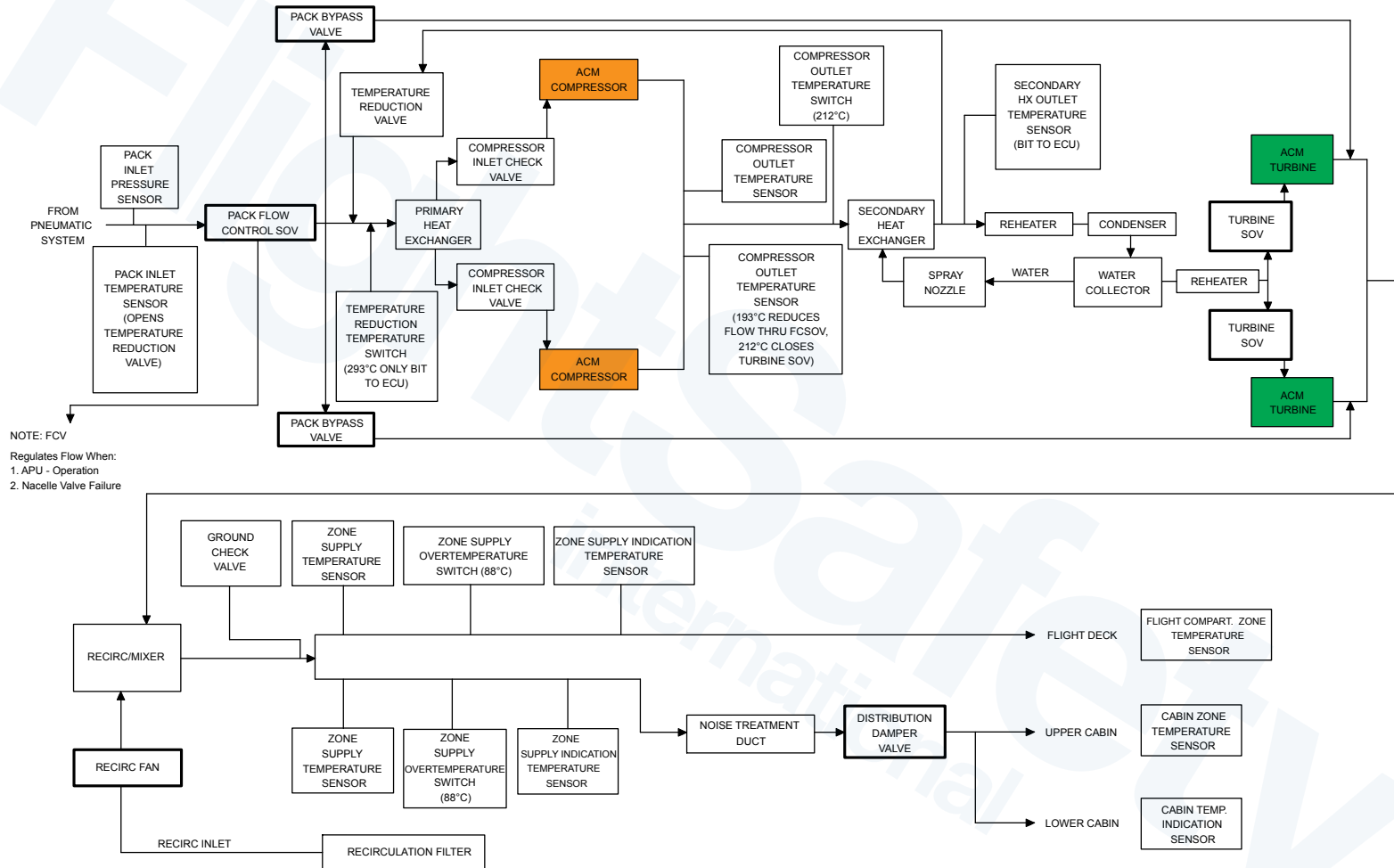
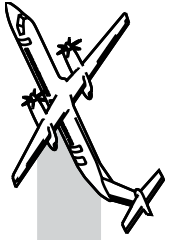
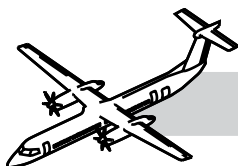


Figure 21-5. ECS Detail Block Diagram



SYSTEM DESCRIPTION

Refer to:

- Figure 21-5. ECS Detail Block Diagram.
- Figure 21-6. Air Conditioning Pack Schematic.

The ECS Electronic Control Unit (ECSECU) monitors and controls the operation of the air conditioning packs.

The ECU has input from the selector switches on the AIR CONDITIONING control panel and from the sensors and switches located throughout the system.

First the air passes through the primary stage of the heat exchange. The ACM then compresses the air and sends the warm compressed air to the secondary circuit of the dual heat exchanger for cooling. The air then goes through the condenser section of the condenser/reheater/mixer, where water in the air is condensed into droplets.

The cooled air then goes through the water collector where the water droplets are collected to flow to a nozzle where it is sprayed against the ram inlet face of the dual heat exchanger to assist with cooling.

The air continues through the re-heater, where it is warmed. This relatively warm, compressed air is then expanded in the turbine section of the ACM, cooling it to cold air.

The temperature at the pack outlet is modulated by the pack bypass valves that introduce warm bleed air into the turbine outlet to mix with the cold ACM air.

The signals that control the positions of these valves come from the ECU.

When necessary, the temperature of the bleed air at the pack inlet is lowered by opening the temperature reduction valve.

This valve allows cooler air downstream of the secondary heat exchanger to flow into the pack inlet.

The temperature switch of the temperature reduction valve sends a signal to the ECU that is compared with the signal from the pack inlet temperature sensor.

The ECU then determines if the temperature reduction valve is operating correctly.

The compressor outlet overtemperature sensor supplies a signal to reduce the pack flow if the temperature in the compressor is too high.

The compressor outlet overtemperature switch turns the ACM off if the compressor temperature is too high. The switch also brings on a PACK HOT caution light after a time delay. The PFCSOV will be closed if the temperature remains high.

The secondary heat exchanger outlet temperature sensor senses the temperature at the secondary heat exchanger outlet. It is used to monitor the efficiency of the heat exchanger and is used for maintenance only.

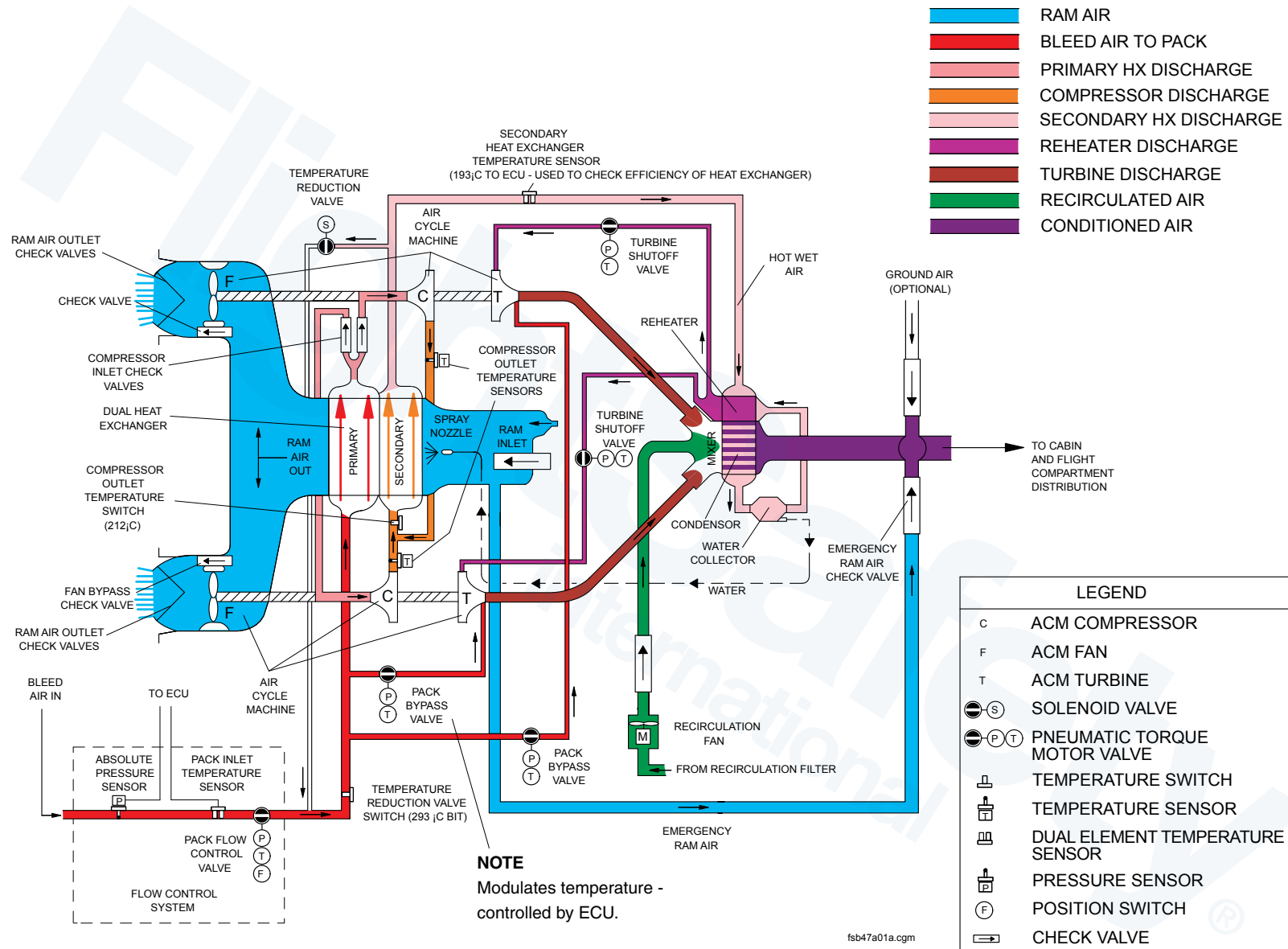
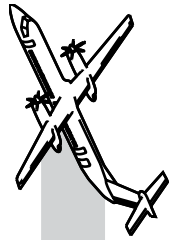
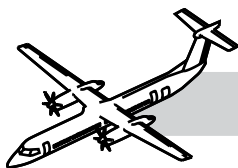


Figure 21-6. Air Conditioning Pack Schematic



COMPONENT DESCRIPTION

Heat-Dual Exchanger

Refer to Figure 21-7. Dual Heat Exchanger.

The dual heat exchanger is an aluminum plate-and-fin, primary crossflow/secondary counterflow heat exchanger. It is mounted at the top of the air conditioning pack between the ram air inlet duct and fan inlet diffuser housing.

It provides primary cooling of the pack inlet air and secondary cooling of the air from the ACM compressors.

The function of the dual heat exchanger is to use ram air to cool the engine bleed air to levels compatible with the ACM and system performance requirements.

Ram Air Inlet

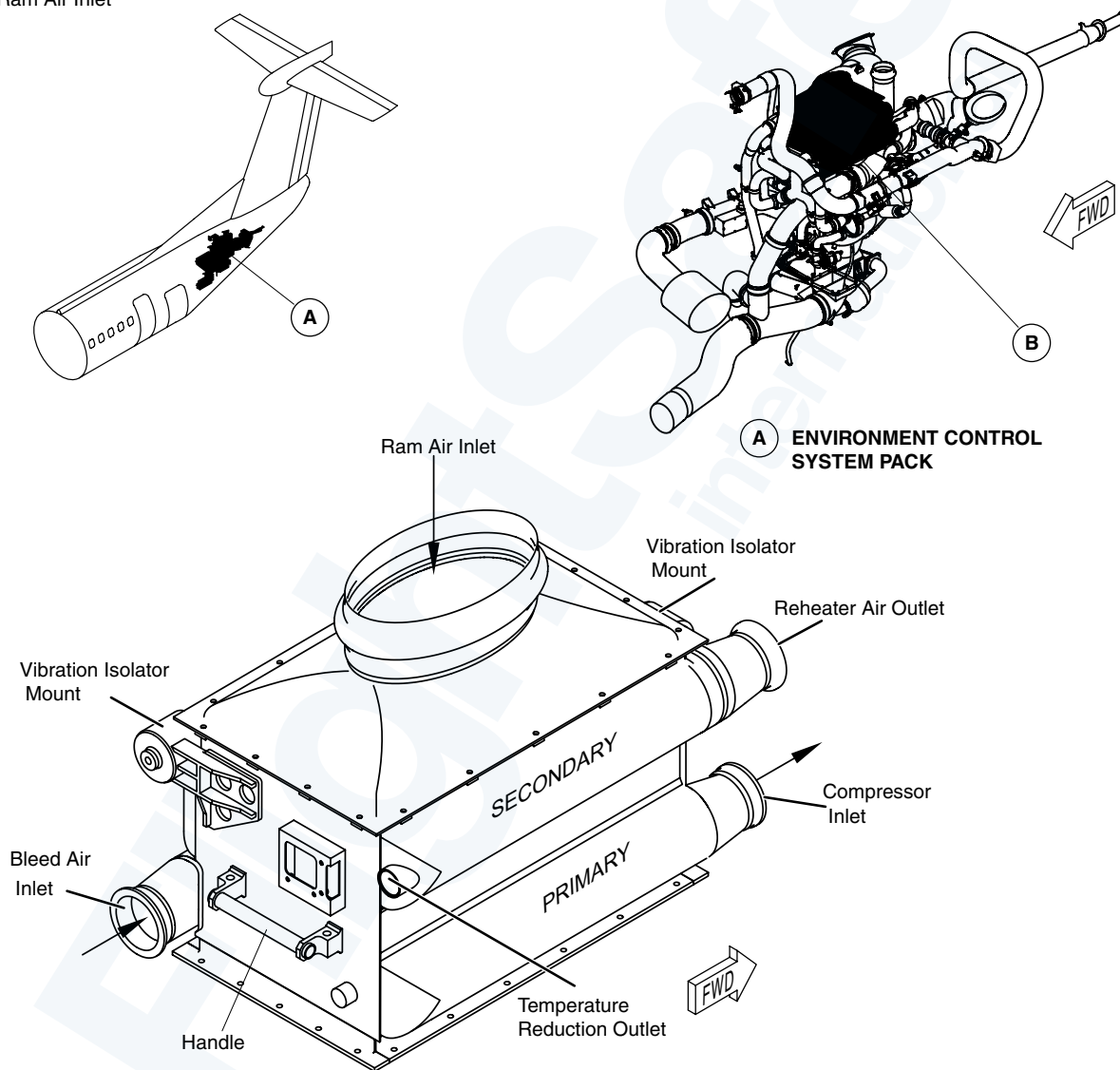
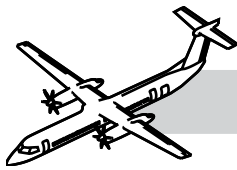


Figure 21-7. Dual Heat Exchanger



Condenser, Reheater and Mixer

Refer to Figure 21-8. Condenser, Reheater and Mixer.

The condenser, reheater and mixer, located at the bottom of the air conditioning pack, is a dual-turbine single recirculation inlet with two condenser outlets.

The condenser, reheater and mixer is a crossflow plate and fin, aluminum heat exchanger. It has three air circuits as follows:

- The condenser condenses the water out of the cabin supply air prior to entering the turbines
- The reheater function is to raise the temperature of the air entering the turbine
- The mixer combines recirculated air from the passenger compartment and air from the two turbine exit ducts.

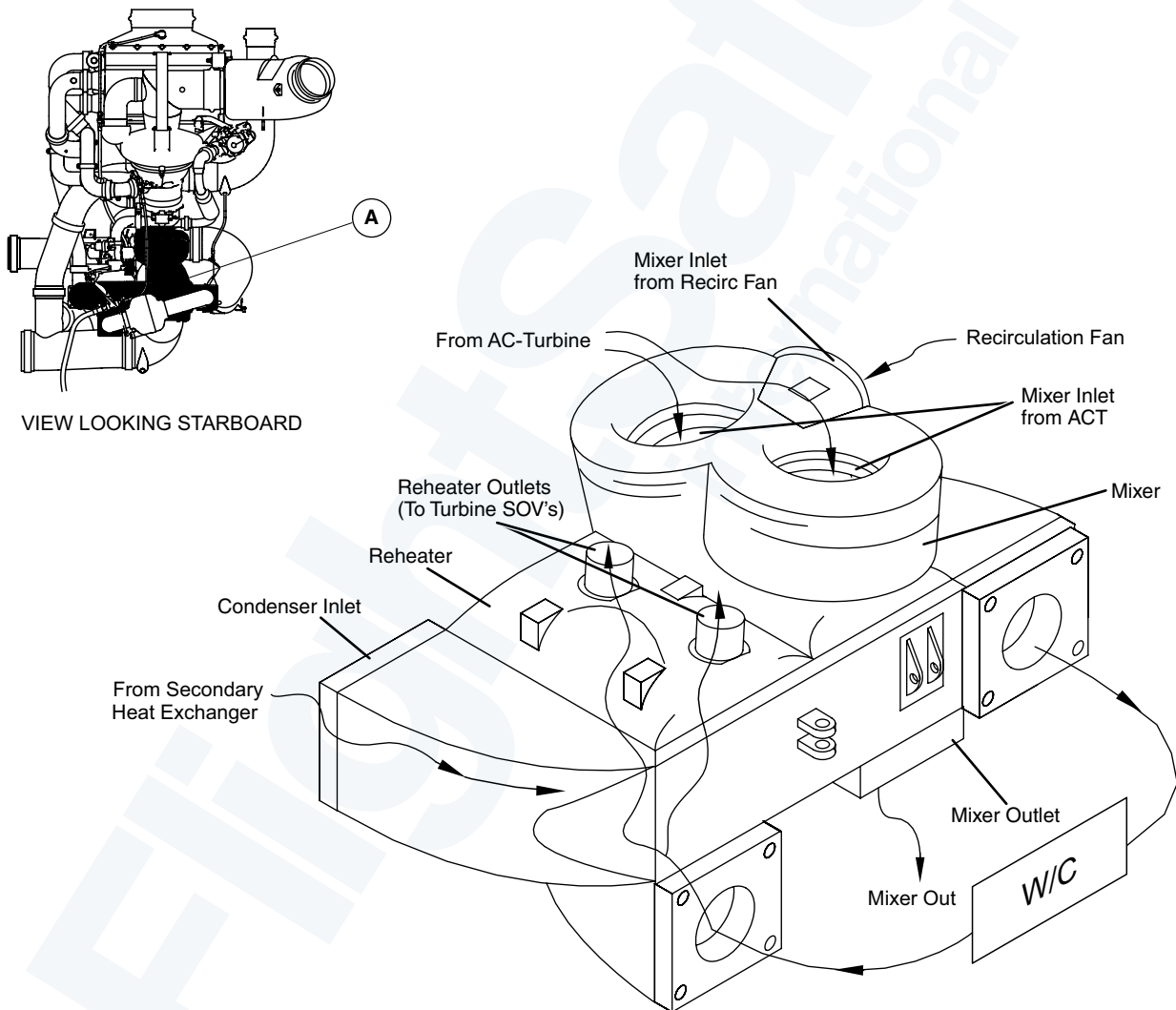
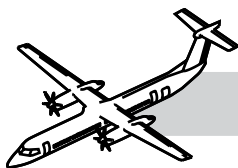


Figure 21-8. Condenser, Reheater and Mixer



Water Collector

Refer to Figure 21-9. Water Collector.

The water collector, installed on the left side of the condenser, removes water from the bleed air.

The water collector has swirl vanes that use centrifugal action to push the condensed water to its inside walls.

The water goes out through a tube to the spray nozzle. A small overflow provides an outlet for excess water.

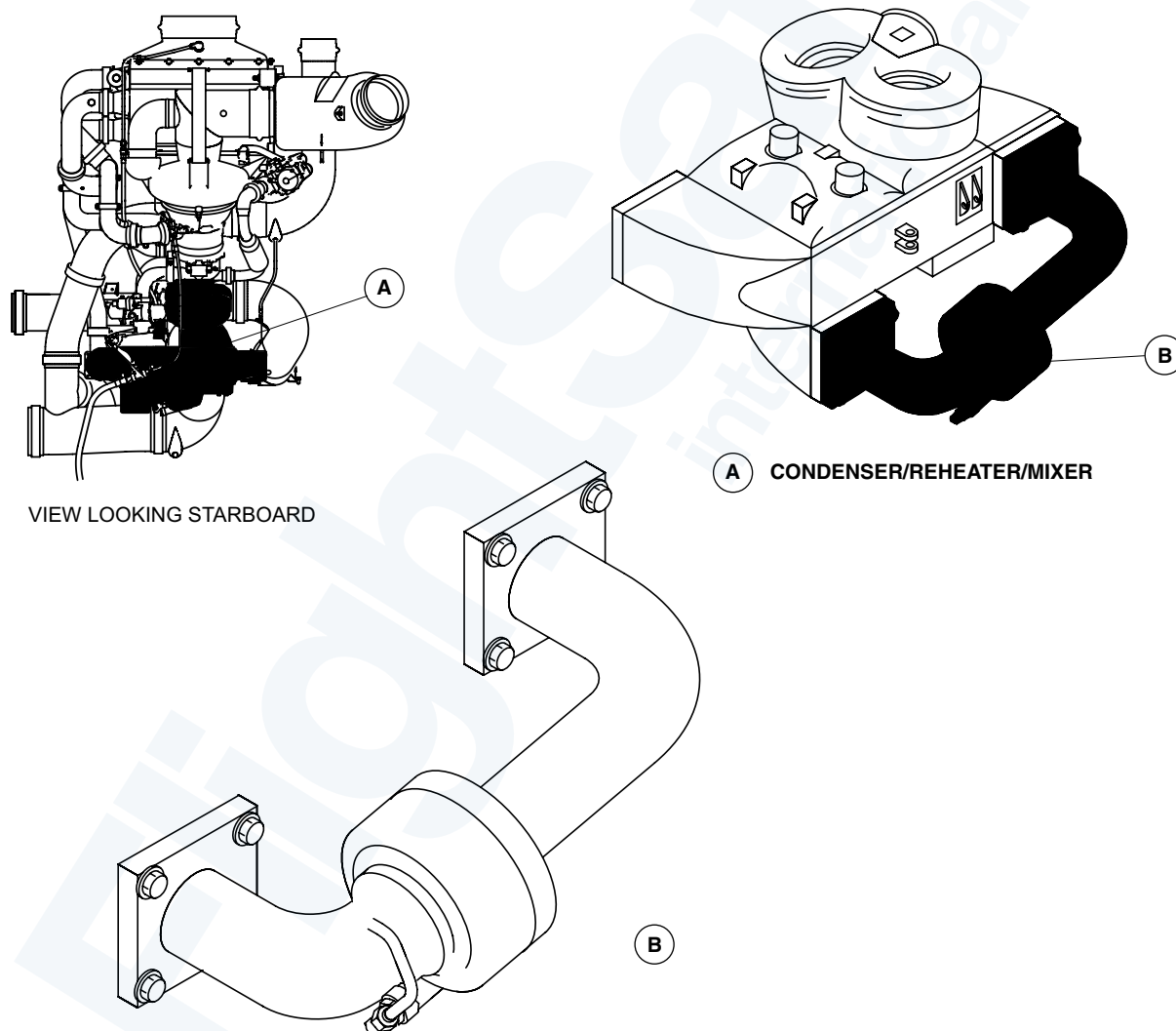


Figure 21-9. Water Collector

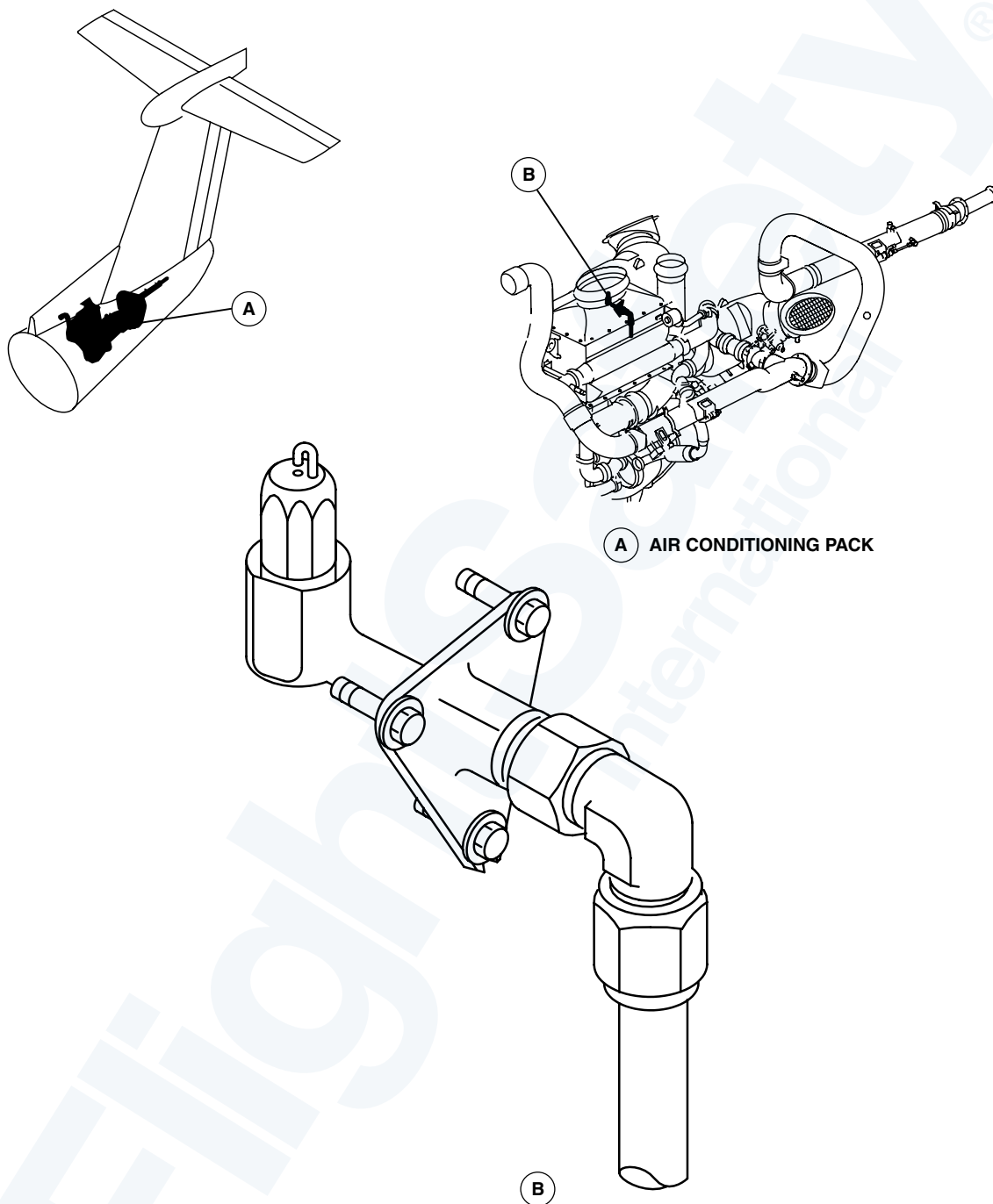
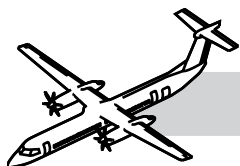
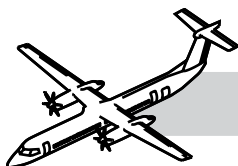


Figure 21-10. Spray Nozzle



Spray Nozzle

NOTES

Refer to Figure 21-10. Spray Nozzle.

The spray nozzle, installed on the top left side of the ram air header duct, sprays the moisture that has been removed onto the secondary heat exchanger, thereby increasing its efficiency.

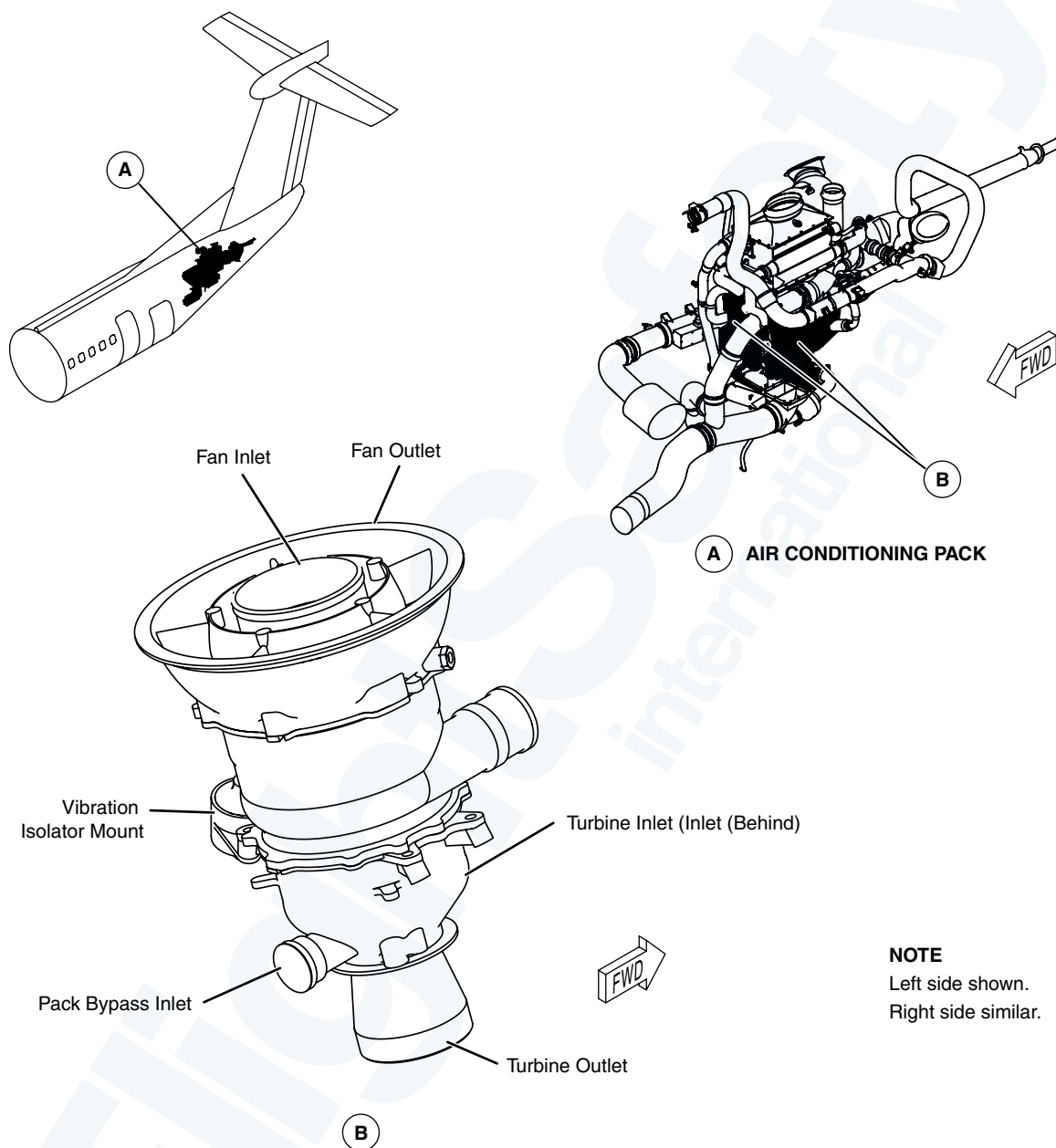
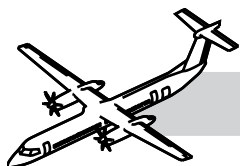
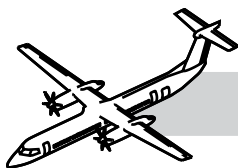


Figure 21-11. Air Cycle Machine Synoptic



Air Cycle Machine

Refer to:

- Figure 21-11. Air Cycle Machine Synoptic.
- Figure 21-12. Air Cycle Machine.

Two ACMs are located below the fan inlet diffuser housing. Each has a vibration isolator mount to support and locate the lower part of the pack.

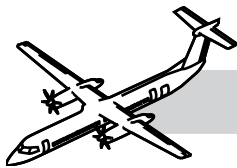
The ACM is a three-wheeled air bearing device that provides cooling of the warm bleed air.

There are three rotating parts in each ACM:

- Compressor
- Turbine
- Fan.



Figure 21-12. Air Cycle Machine



Fan Inlet Diffuser Housing

Refer to Figure 21-13. Fan Inlet Diffuser Housing.

The fan inlet diffuser housing, located in the air conditioning pack between the dual heat exchanger outlet and the ACM fans, ducts the ram flow from the dual heat exchanger cold side outlet to the fan inlets.

The housing has a common duct used for cooling airflow from the dual heat exchanger to the ACM fans and to the ram air exhaust outlets.

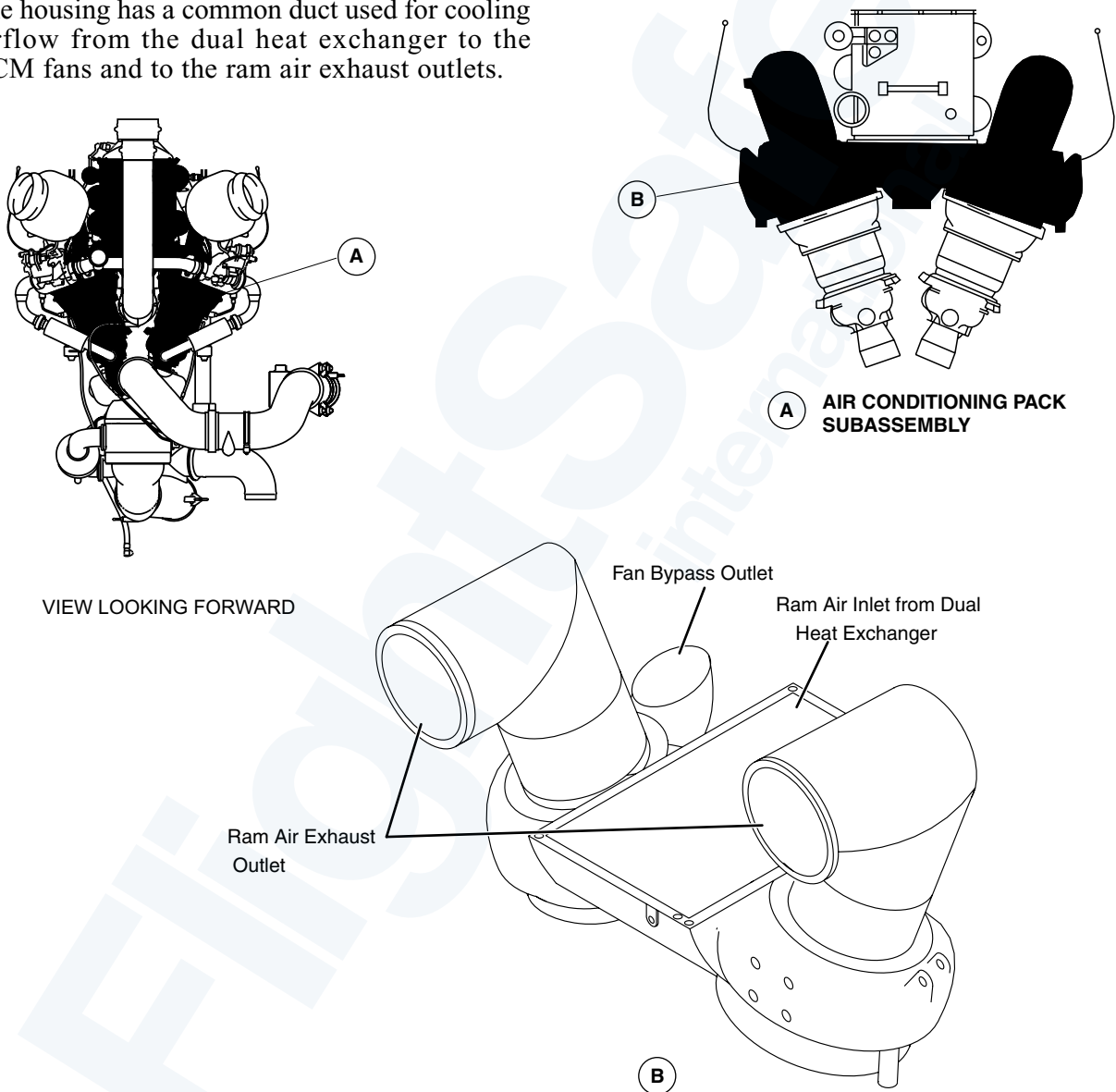
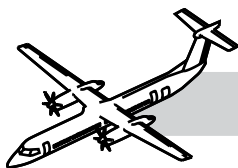


Figure 21-13. Fan Inlet Diffuser Housing



Pack Bypass Valve

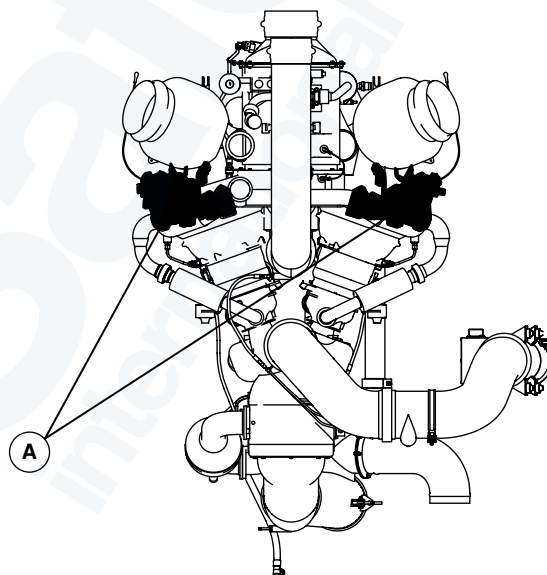
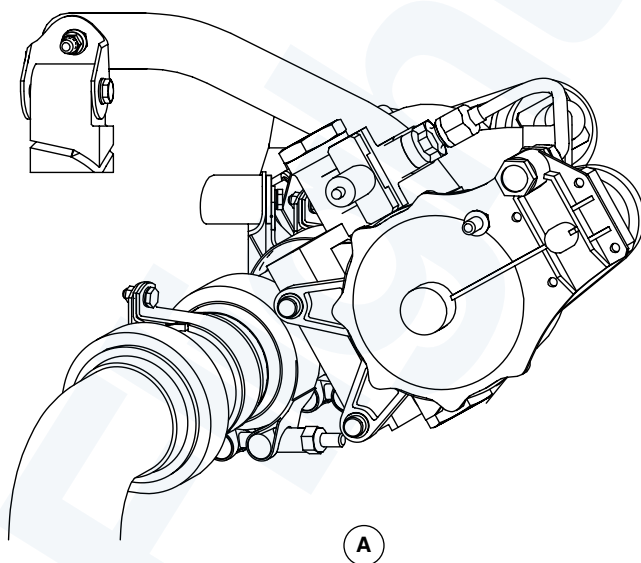
Refer to Figure 21-14. Pack Bypass Valve.

The pack (ACM) bypass valves located between the bleed air inlet and the ACM turbine outlets, divert the warm air around the ACM and send it to the cabin and flight compartments.

Each pack bypass valve is an ECU electrically-controlled pneumatic valve that consists of:

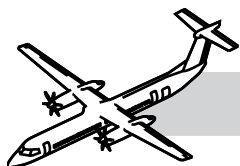
- Torque motor
- Pneumatic actuator
- Butterfly valve.

Each valve is fitted with a metal wire mesh filter in the servo air line. The valves also have a locking pin. When in the locked position, the locking pins bleed servo air. This action vents and mechanically locks the valves in the closed position.



VIEW LOOKING FORWARD

Figure 21-14. Pack Bypass Valve



Turbine Shut-off Valve

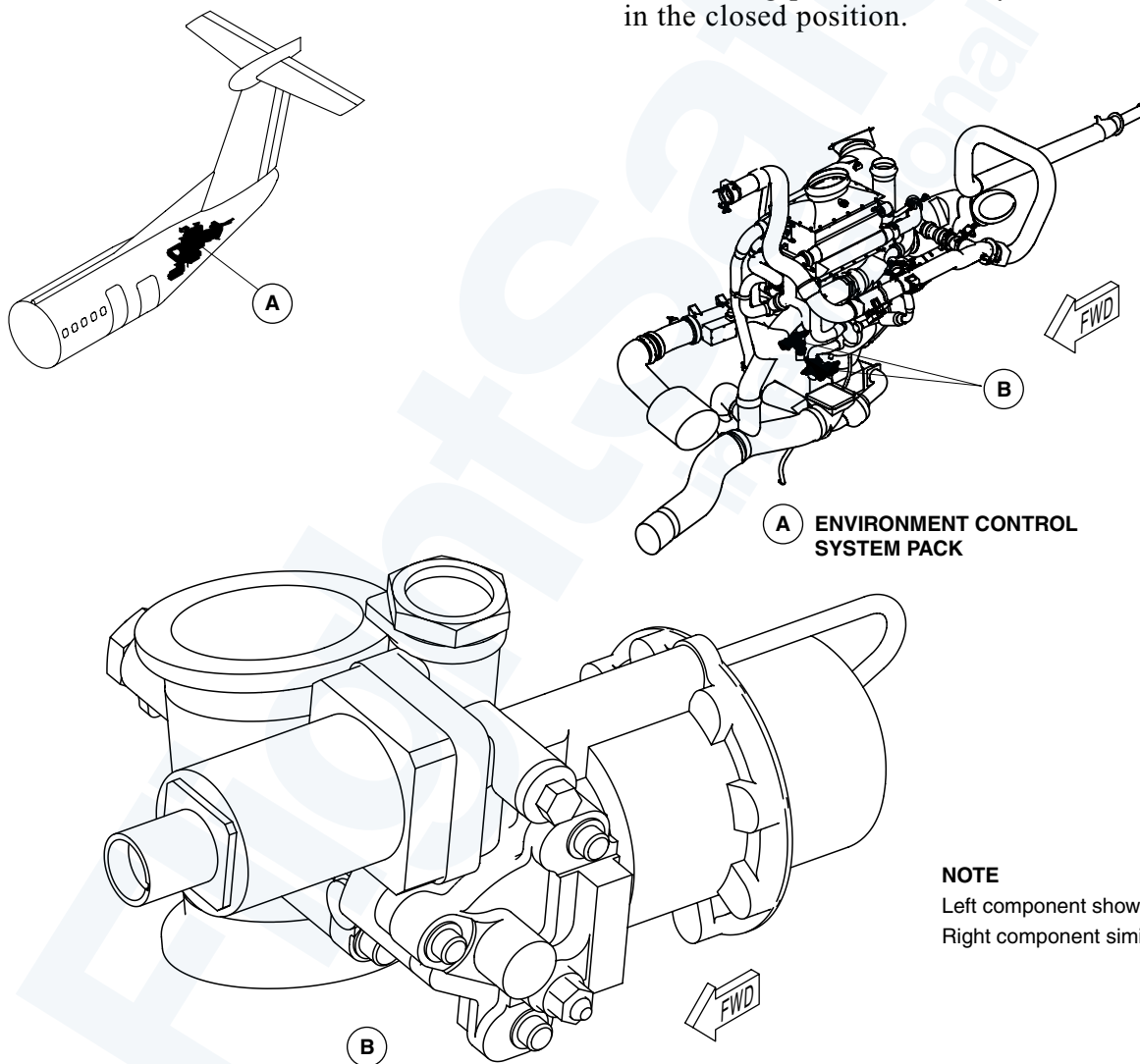
Refer to Figure 21-15. Turbine Shut-Off Valve.

The turbine shut-off valves, located between the condenser/reheater/mixer and the ACM turbine inlets, control the flow of air to each ACM turbine.

The turbine shut-off valves are ECU electrically-controlled and pneumatically operated. Each valve consists of:

- A torque motor
- Pneumatic actuator
- Butterfly valve.
- The valve has a metal wire mesh filter in the servo air line and a locking pin that, when in the locked position, bleeds servo air to ambient air.

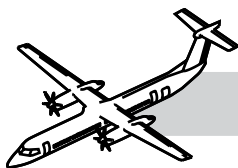
The locking pin mechanically locks the valve in the closed position.



NOTE

Left component shown.
Right component similar.

Figure 21-15. Turbine Shut-Off Valve



Temperature Reduction Shut-off Valve

Refer to Figure 21-16. Temperature Reduction Shut-off Valve.

The temperature reduction shut-off valve, installed on the aft side of the heat exchanger, allows cooled secondary heat exchanger discharge air to be recirculated to the pack inlet.

The temperature reduction shut-off valve is a pneumatically actuated, solenoid-controlled gate type device.

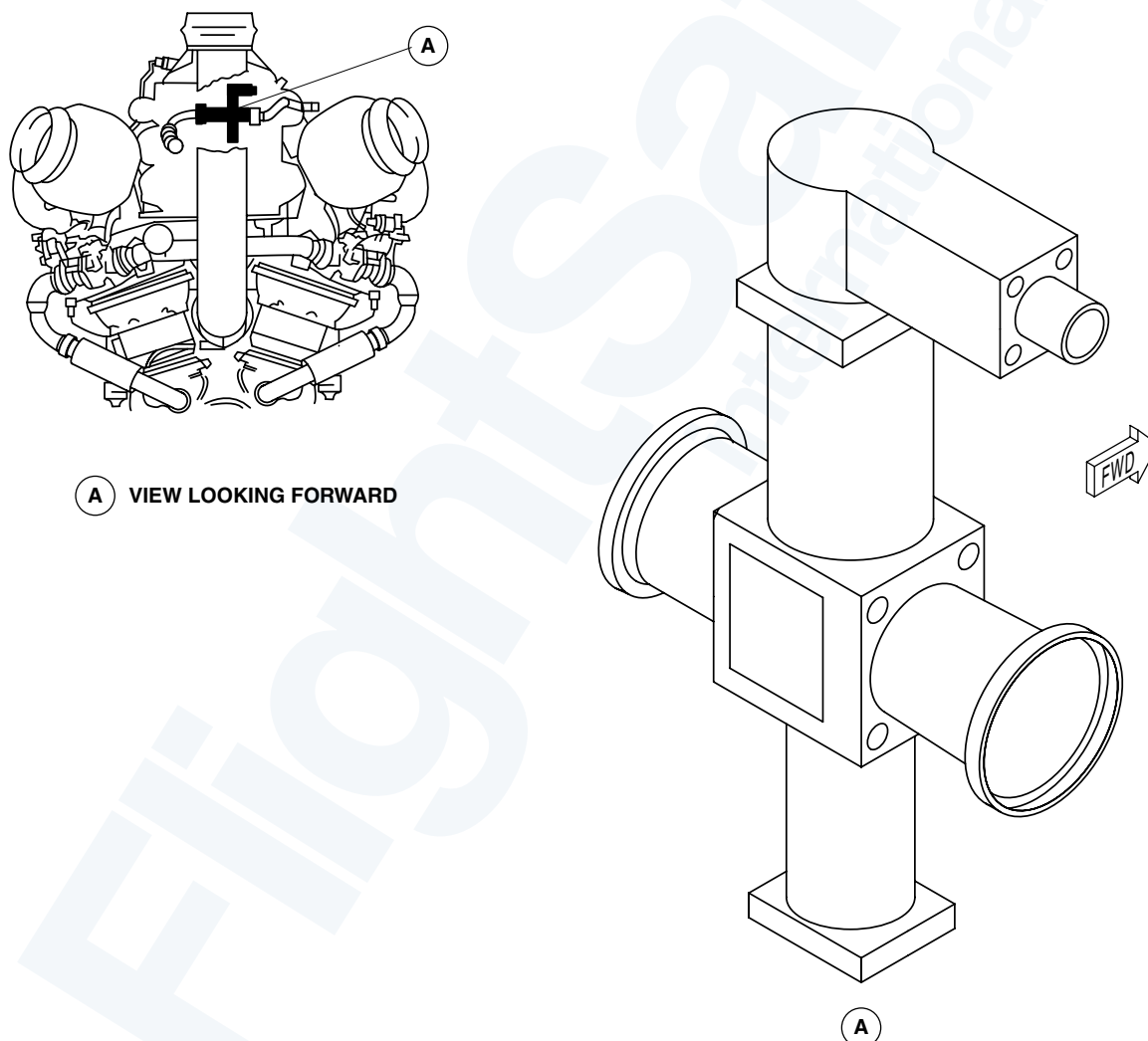
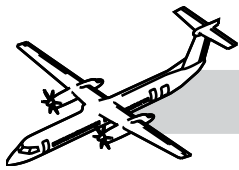


Figure 21-16. Temperature Reduction Shut-off Valve



Temperature Reduction Valve Switch

Refer to Figure 21-17. Temperature Reduction Valve Switch.

The temperature reduction valve switch, installed on the bleed air inlet ducting on the aft side of the air conditioning pack, determines if the temperature reduction valve has failed to open.

The temperature reduction valve switch is a thermal switch that opens when the pack inlet temperature exceeds 560°F (293°C).

This action signals the ECU to generate a fault code to the CDS for a temperature reduction valve failure.

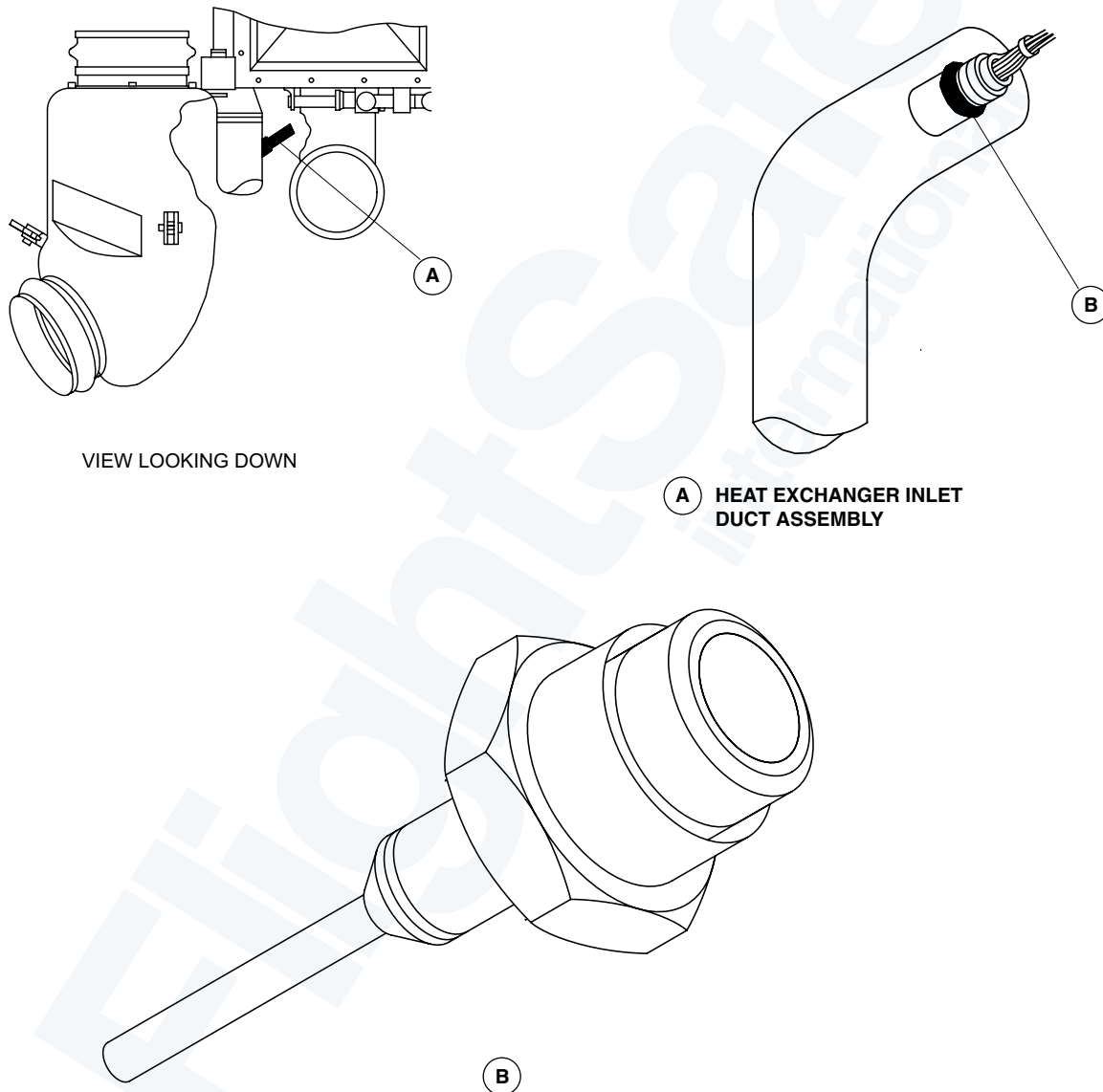
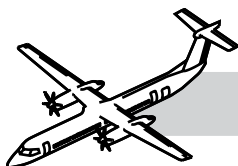


Figure 21-17. Temperature Reduction Valve Switch



Compressor Outlet Temperature Switch

Refer to Figure 21-18. Compressor Outlet Temperature Switch.

The compressor outlet temperature switch, mounted in the secondary heat exchanger inlet duct, monitors air flow temperature entering the secondary heat exchanger.

It is a thermal switch that trips when the compressor outlet temperature exceeds 415°F (212°C).

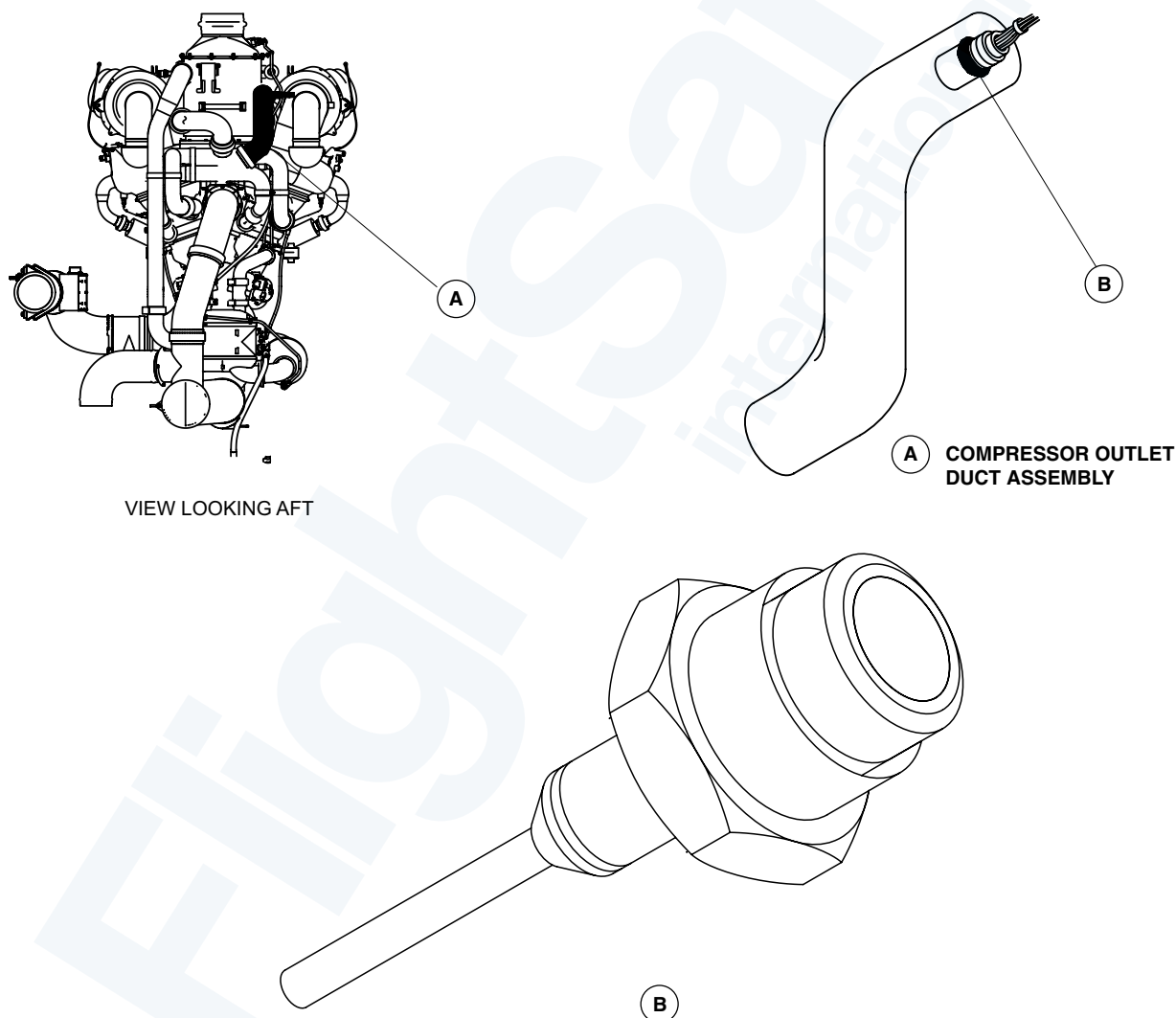
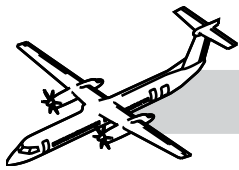


Figure 21-18. Compressor Outlet Temperature Switch



Compressor Outlet Temperature Sensors

Refer to Figure 21-19. Compressor Outlet Temperature Sensors.

The compressor outlet temperature sensors, located in the ACM compressor outlet ducting, monitor the temperature of the air leaving the ACM compressors.

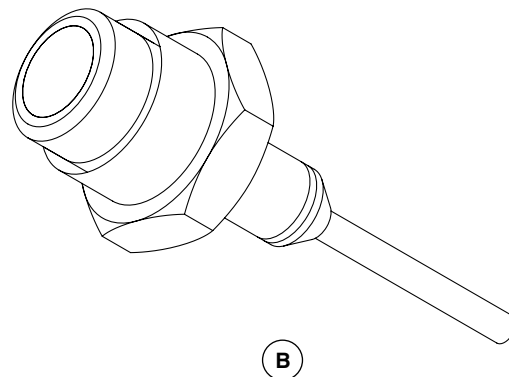
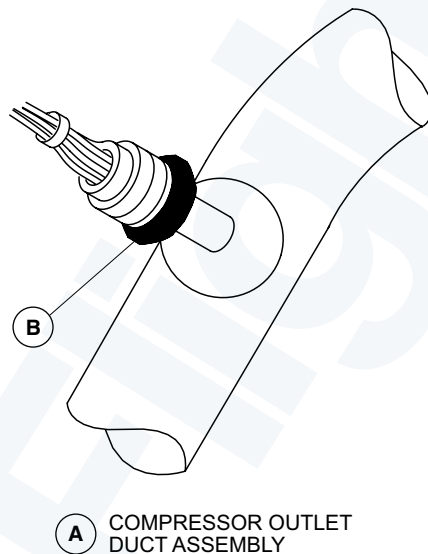
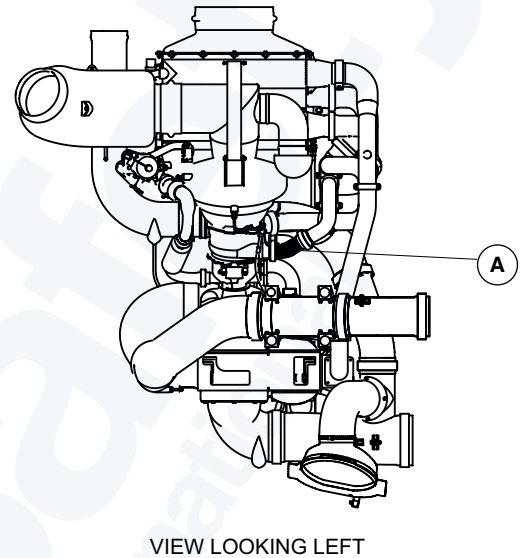
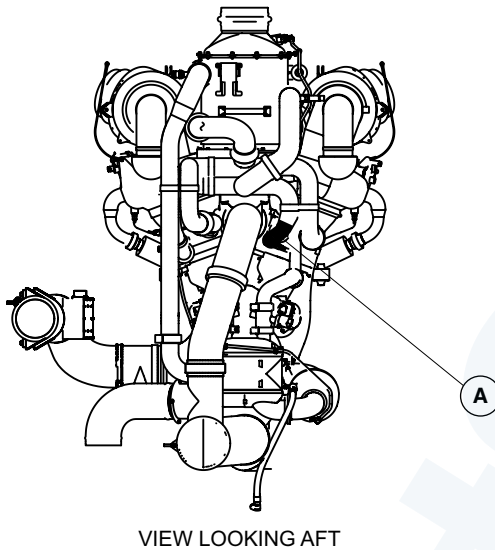
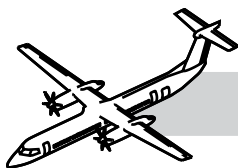


Figure 21-19. Compressor Outlet Temperature Sensors



Secondary Heat Exchanger Outlet Temperature Sensor

Refer to Figure 21-20. Secondary Heat Exchanger Outlet Temperature Sensor.

The secondary heat exchanger outlet temperature sensor, installed in the secondary heat exchanger outlet duct, senses the temperature of the air leaving the heat exchanger.

The ECU compares the signal from this sensor with the signal from the pack inlet temperature sensor and other inputs. This action determines whether the dual heat exchanger is operating efficiently.

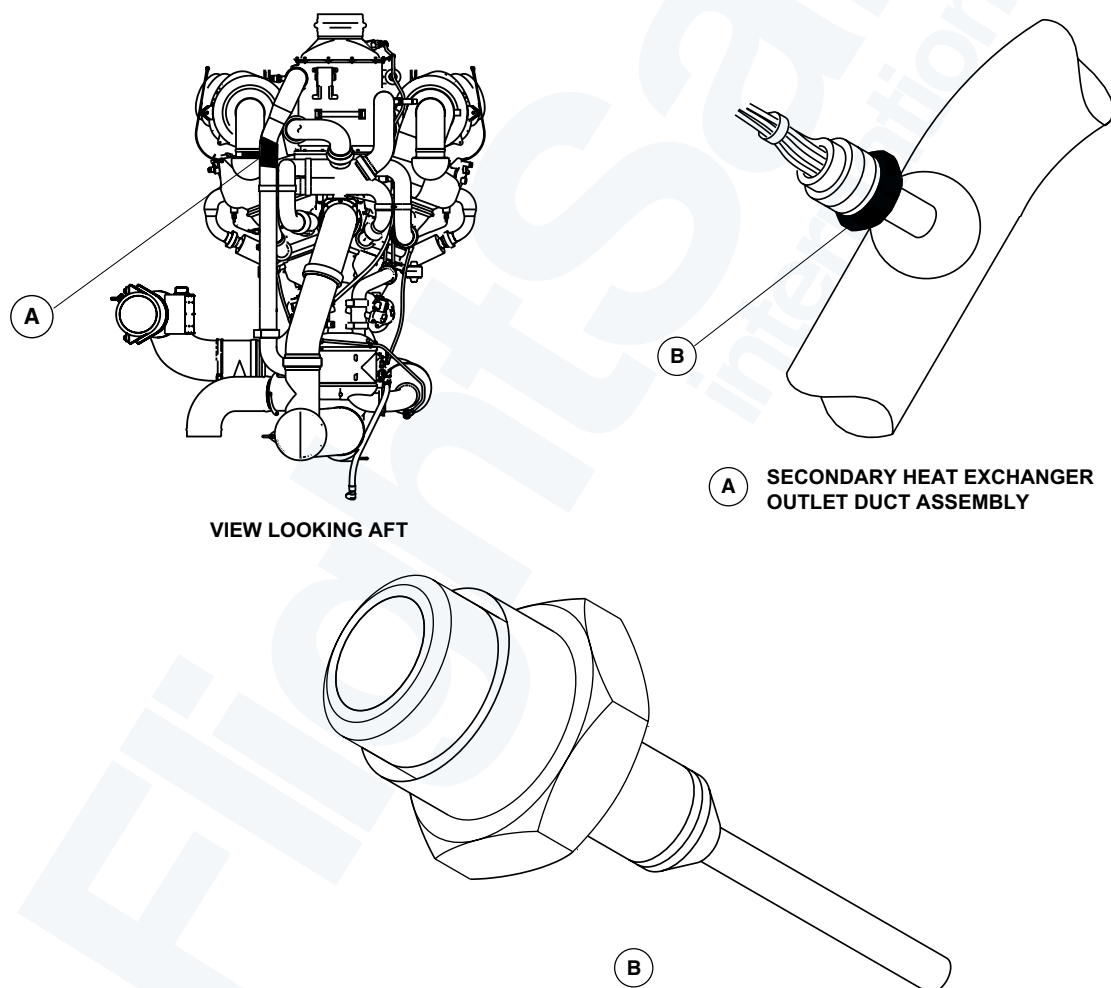
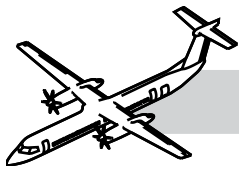


Figure 21-20. Secondary Heat Exchanger Outlet Temperature Sensor



Air Conditioning Pack Struts and Air Conditioning Pack Mounts

Refer to Figure 21-21. Air Conditioning Support System.

The mounting straps attach to either side of the fan inlet diffuser housing and the airframe.

They support the weight of the air conditioning pack.

The support system has the following components:

- Vibration Isolators (4)
- Muffler Support Struts (4)
- Support Struts (12) and
- Pack Support Straps (2).

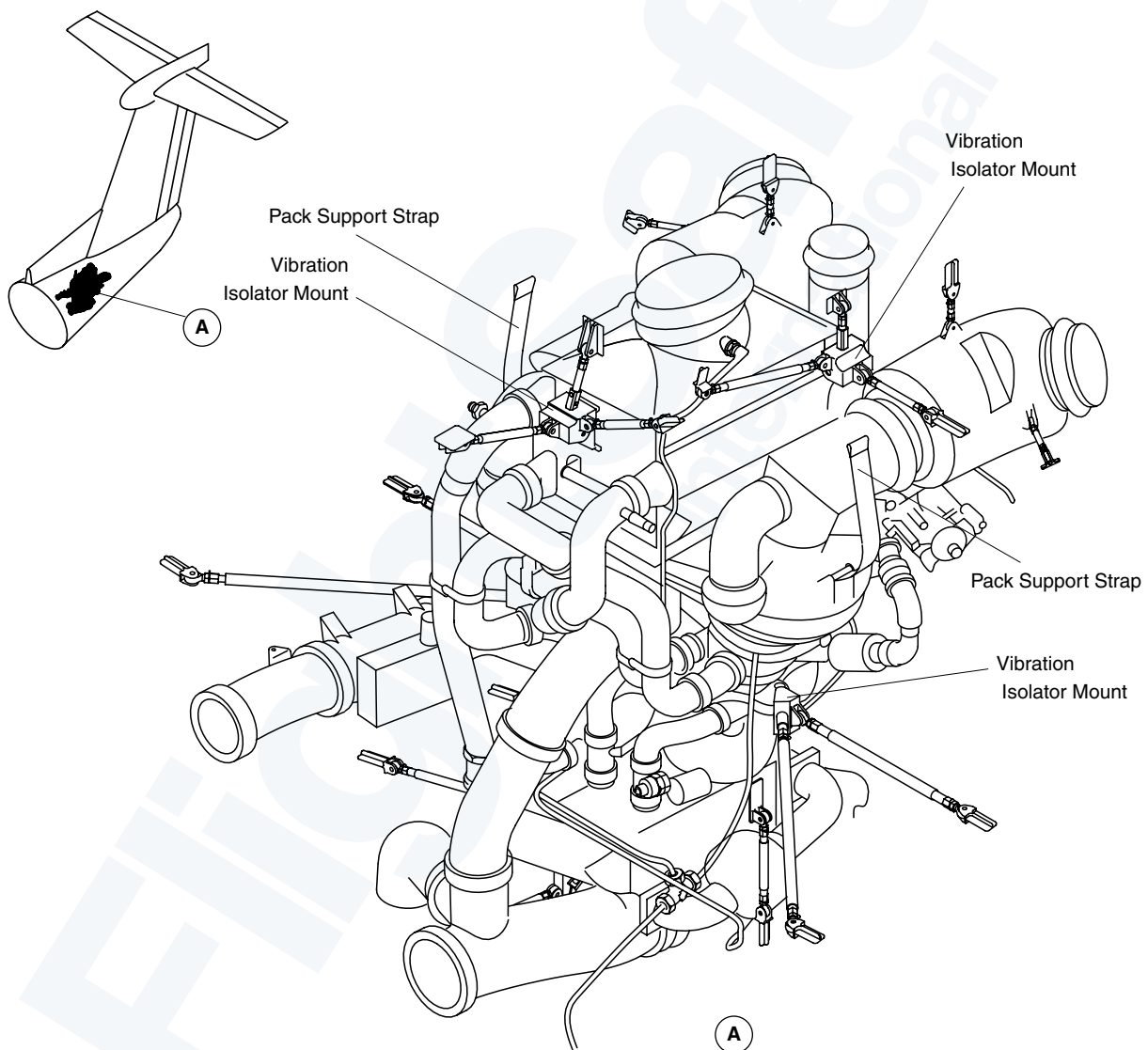
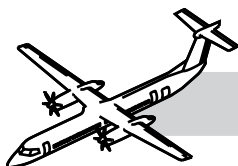


Figure 21-21. Air Conditioning Support System



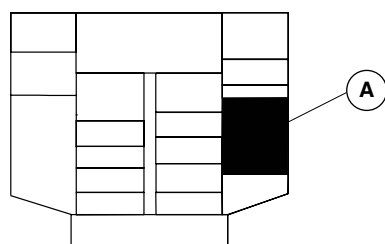
CONTROLS AND INDICATIONS

Packs Control Switches

Refer to Figure 21-22. Air Conditioning Control Panel - Pack Control Switches.

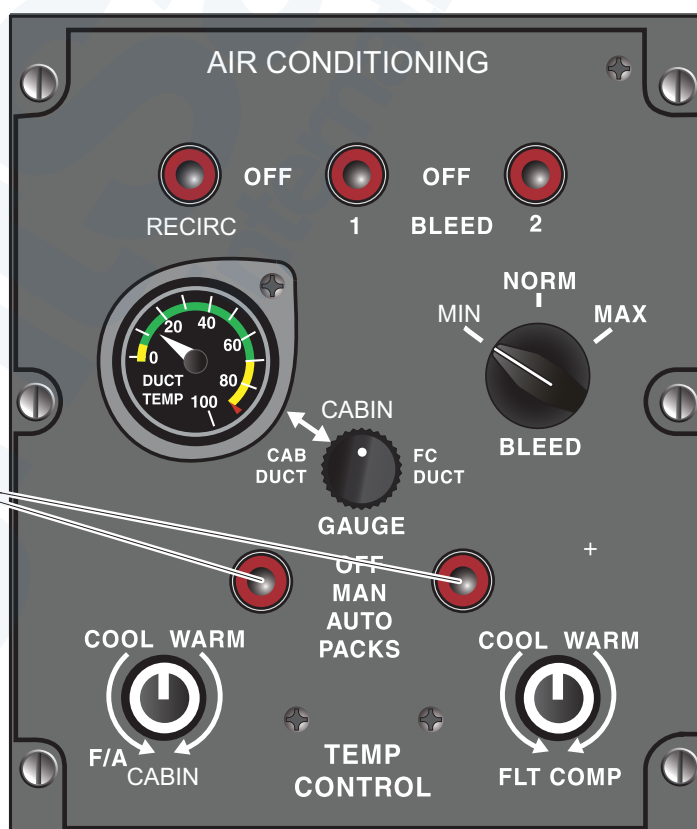
The three-position packs control switches are located on the AIR CONDITIONING control panel.

- OFF - Closes related Turbine Shut-off Valve (TSOV).
- MAN - ECU temperature control based on related flight compartment or cabin rotary switch temperature selection.
- AUTO - ECU temperature control based on actual cabin temperature modulates the pack bypass valve.



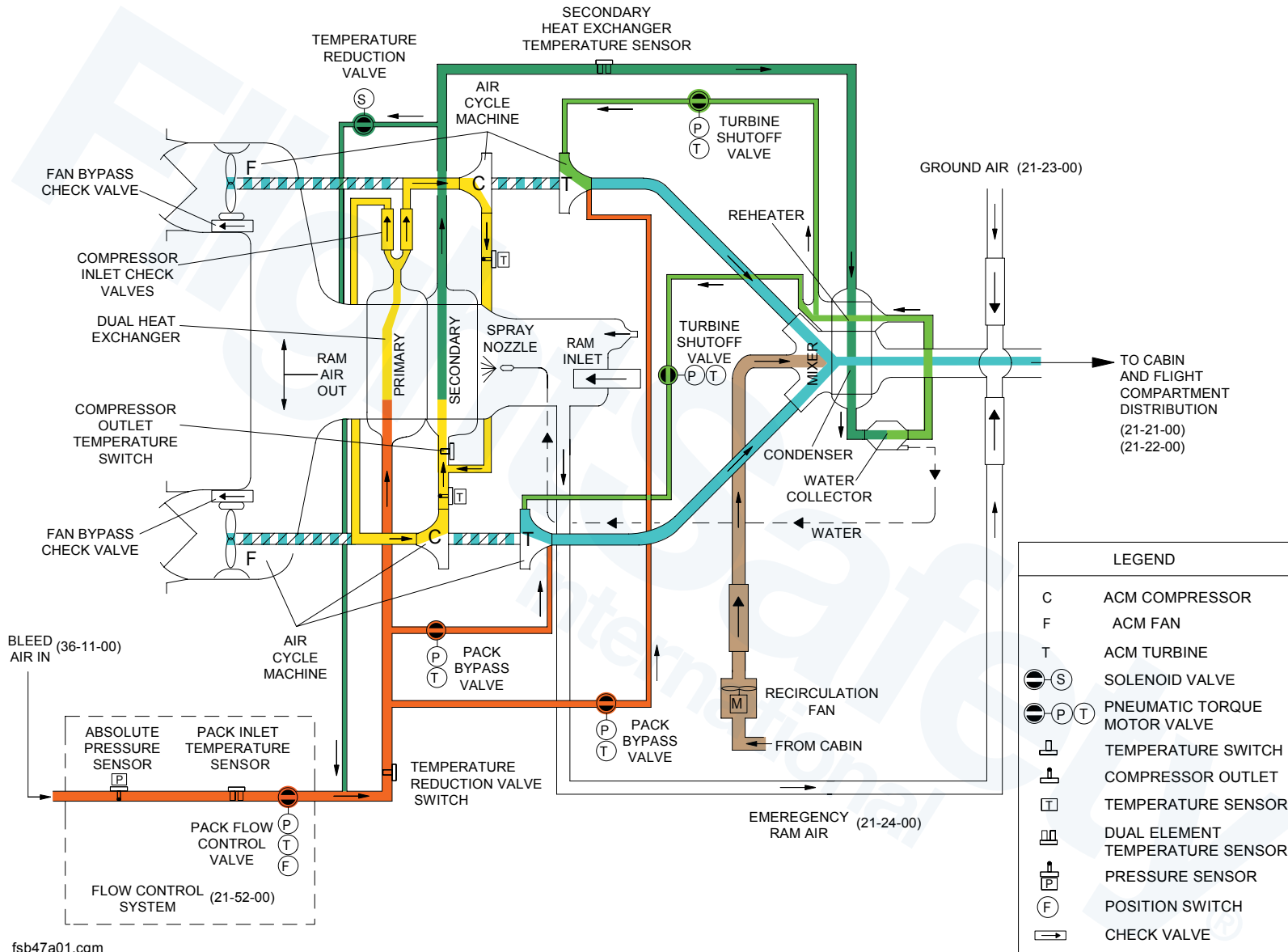
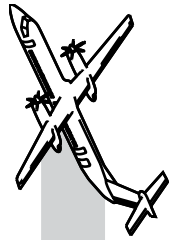
OVERHEAD CONSOLE

PACKS CONTROL SWITCHES



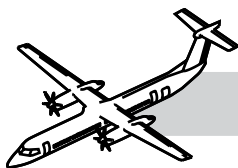
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Figure 21-22. Air Conditioning Control Panel - Pack Control Switches



fsb47a01.cgm

Figure 21-23. Air Conditioning Pack



OPERATION

Normal System Operation

Refer to Figure 21-23. Air Conditioning Pack.

The Packs

Selection of the BLEED 1 and/or 2 switches on the AIR CONDITIONING panel provides power to the ECS ECU. The ECS ECU automatically controls the packs airflow through the modulation of the PFCSOV and the selected NFCSOV.

In normal operation the PFCSOV modulates at approximately 10% > NFCSOVs' output.

Selection of the PACKS switches to AUTO, causes:

- ECU monitoring of temperatures, pressures, and flow rates and
- Adjusting of the pack valves to achieve the selected conditioned air temperatures and flow rates.

The modulated bleed air flows through the primary heat exchanger.

The heat exchanger uses ram air to remove heat from the bleed air. The primary cooled bleed air is then divided into two air flows.

The air flows through one check valve for each of the two compressors of the ACM. These compressors increase the temperature and pressure of the bleed air.

The two high pressure air flows are combined into one flow to pass through the secondary heat exchanger.

The secondary heat exchanger uses cool ram air to remove heat from the ACM compressed bleed air.

The secondary heat exchanger temperature sensor measures bleed air as it leaves the secondary heat exchanger. This data is used by the ECU to determine the effective cooling

effects of the secondary heat exchanger.

Each compressor outlet temperature sensor sends data to the ECS ECU in order to maintain the outlet temperature below 380°F (193°C). If the temperature reaches 380°F, the ECU will modulate the PFCSOV towards the closed position. If the temperature reaches 210°C (415°F), the ECU closes the TSOV and the bypass valve.

After a time delay of 10 seconds, if the temperature is still high, the PFCSOV will be commanded closed.

The cooler air from the ACM turbine enters the condenser causing the water moisture in the hot wet air to accumulate and be removed by the water collector.

The removed water is sent to the secondary heat exchanger spray nozzle.

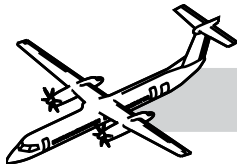
After the air passes through the condenser, it is ducted back through the reheater section.

The reheater warms the air and vaporizes any remaining water before it is used to power the ACM turbines.

The turbine SOV controls the bleed air flow into the ACM turbine. The energy to power the turbines and the air expansion absorb heat from the air.

After passing through the turbine, the cold discharge air is ducted into the mixer to combine with the recirculation fan air. The recirculation fan takes air from the cabin. After the mixer, the air is ducted into the cabin and flight compartments.

The ECU controls two pack bypass valves to regulate the temperature of the conditioned air sent to the cabin and flight compartment. These valves can send warm, high pressure bleed air from the pack inlet to the outputs of each ACM turbine. Such action raises the temperature of the air supplied to the cabin and flight compartments.

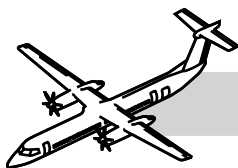


Closing the pack bypass valve decreases the air flow, decreasing the air temperature flowing to the cabin and flight compartments.

The turbines also drive a fan which sends the ram air flow through the dual heat exchanger. This is required when there is no ram air flow through the dual heat exchanger (During ground runs).

When the aircraft is in-flight, the ACM fans can limit the flow of ram air. During flight the ram air increases, causing the ram air pressure to increase.

The increase in pressure opens two bypass valves in the ram air outlet ducts, allowing maximum flow of ram air.



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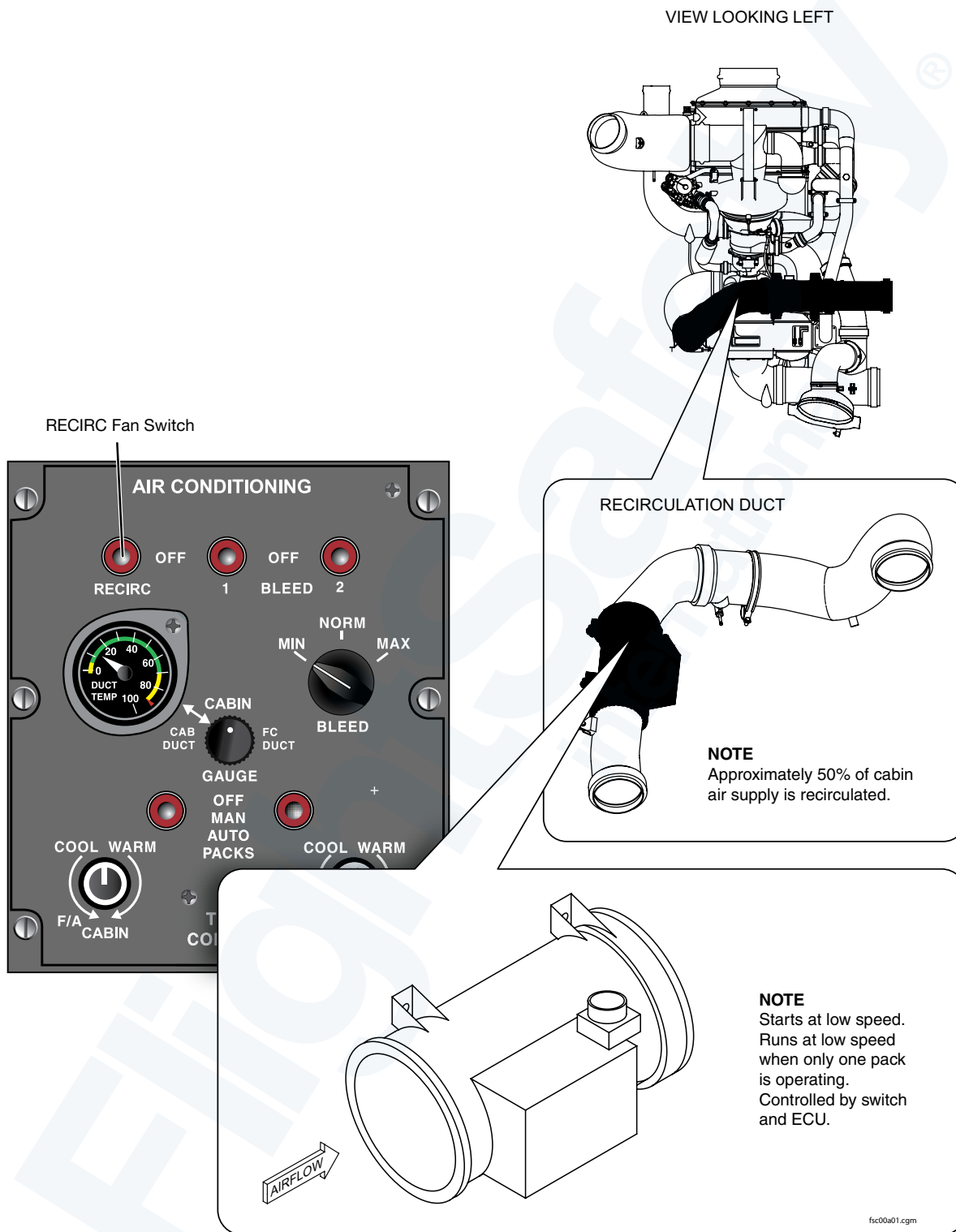
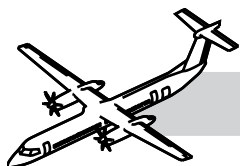
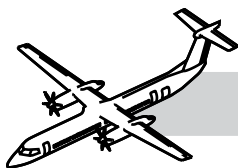


Figure 21-24. Recirculation Fan Synoptic



Recirculation Fan

Refer to:

- Figure 21-24. Recirculation Fan Synoptic. Synoptic.
- Figure 21-25. Recirculation Fan..

The recirculation fan, installed in the recirculation duct located on the right side of the air conditioning pack in the aft equipment bay, draws air from the passenger cabin through a filter located between the rear wall of the baggage compartment and the rear pressure bulkhead.

The fan returns this air to the mixing box in the air conditioning pack.

The recirculation ducting contains a check valve that prevents reverse flow. The recirculation fan on/off switch on the AIR CONDITIONING control panel controls the on/off operation of the recirculation fan. Fan speed is controlled by the ECS ECU.



Figure 21-25. Recirculation Fan

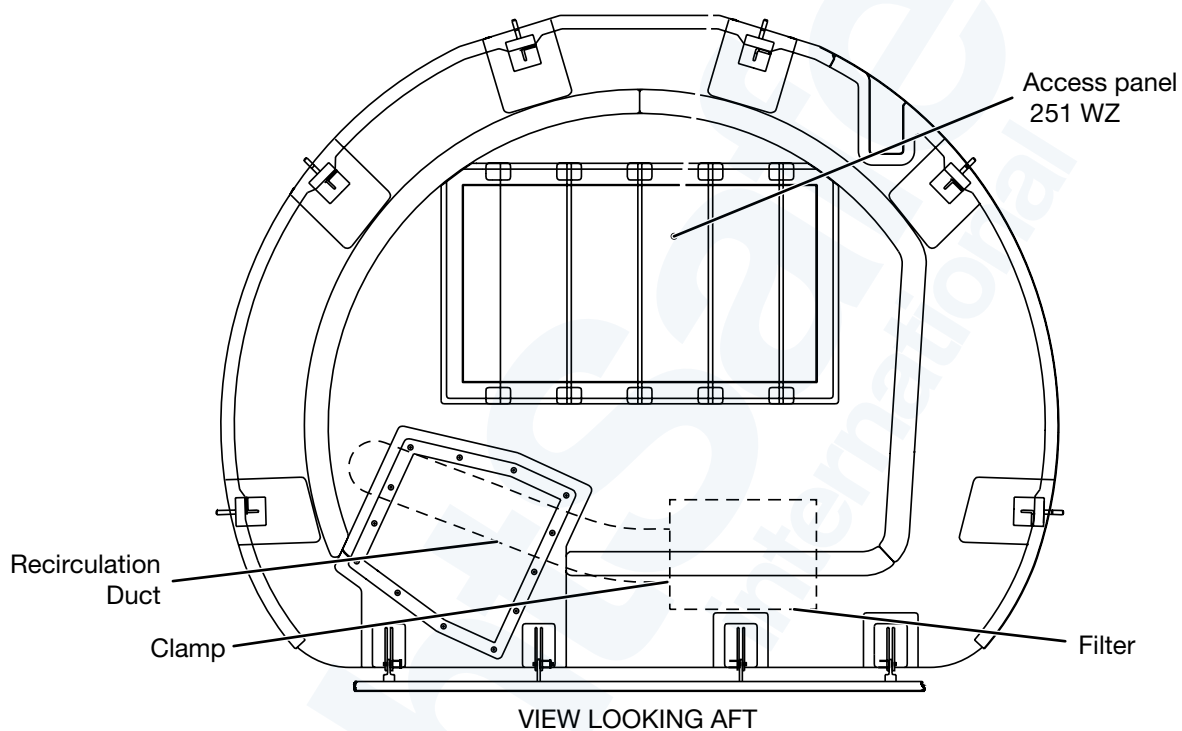
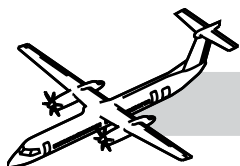
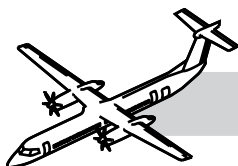


Figure 21-26. Recirculation Fan Filter Removal



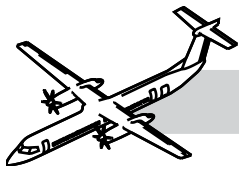
Recirculation Fan Filter Removal

NOTES

Refer to Figure 21-26. Recirculation Fan Filter Removal.

NOTE

The recirculation filter can only be replaced. Used filters should be discarded.



Recirculation Filter

Refer to:

- Figure 21-27. Recirculation Filter.
- Figure 21-28. Recirculation Filter Synoptic.

The recirculation HEPA filter is installed on the aft face of the baggage canted bulkhead and forward of the aft pressure bulkhead in the aft baggage compartment.

The recirculation duct coming from the recirculation fan inlet is connected to the recirculation filter. The fan inlet is located in the aft equipment bay.



Figure 21-27. Recirculation Filter

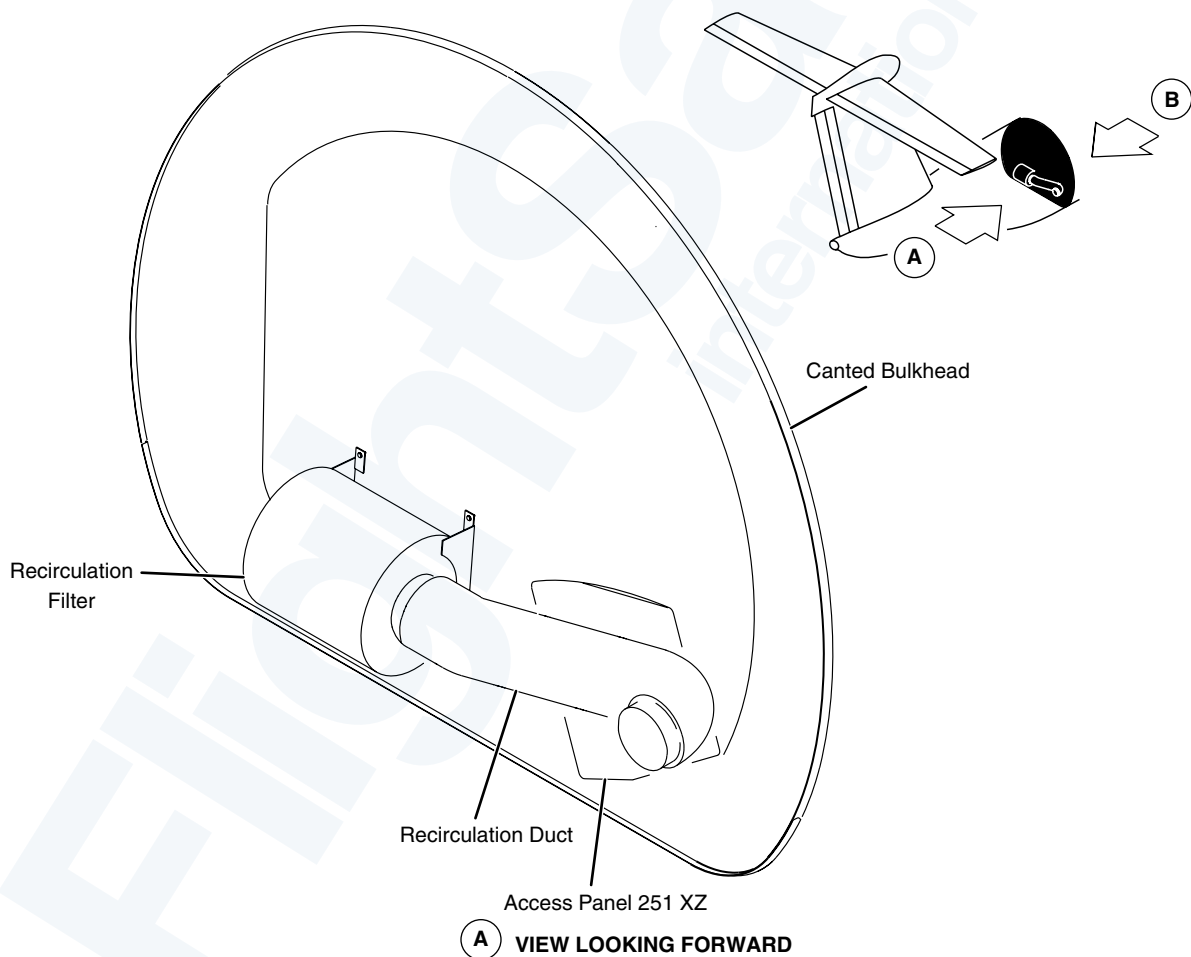
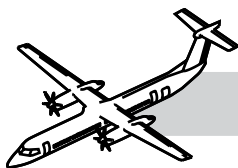


Figure 21-28. Recirculation Filter Synoptic



21-52-00 FLOW CONTROL SYSTEM

INTRODUCTION

The ECU regulates the airflow based upon temperature and pressure signals.

SYSTEM DESCRIPTION

Refer to Figure 21-29. Flow Control System.

Pack inlet pressure and temperature sensors, located upstream of the pack flow control and shut-off valve, are used in regulating airflow to the air conditioning pack.

When both engines are operating and supplying bleed air, the ECU keeps the pack flow control and shut-off valve open approximately 10% more than Nacelle Flow Control Shut-off Valve (NFCSOV).

If the left or right side of the bleed air system is turned off or the APU is supplying the bleed air, the ECU regulates bleed airflow with the pack flow control and shut-off valve.

The pack inlet pressure and temperature sensor signals are supplied to the ECU. Both digital channels of the ECU use these signals to share control of the pack flow control and shut-off valve.

During a flight, one digital channel gets full control of the pack flow control and shut-off valve (the other channel gets full control during the next flight).

If the digital channel in control loses power or fails, the other digital channel takes control of the pack flow control and shut-off valve.

The analog backup channels do not control the pack flow control and shut-off valve.

The digital channel modulates the pack flow control and shut-off valve to provide either minimum, normal, or maximum pack flow.

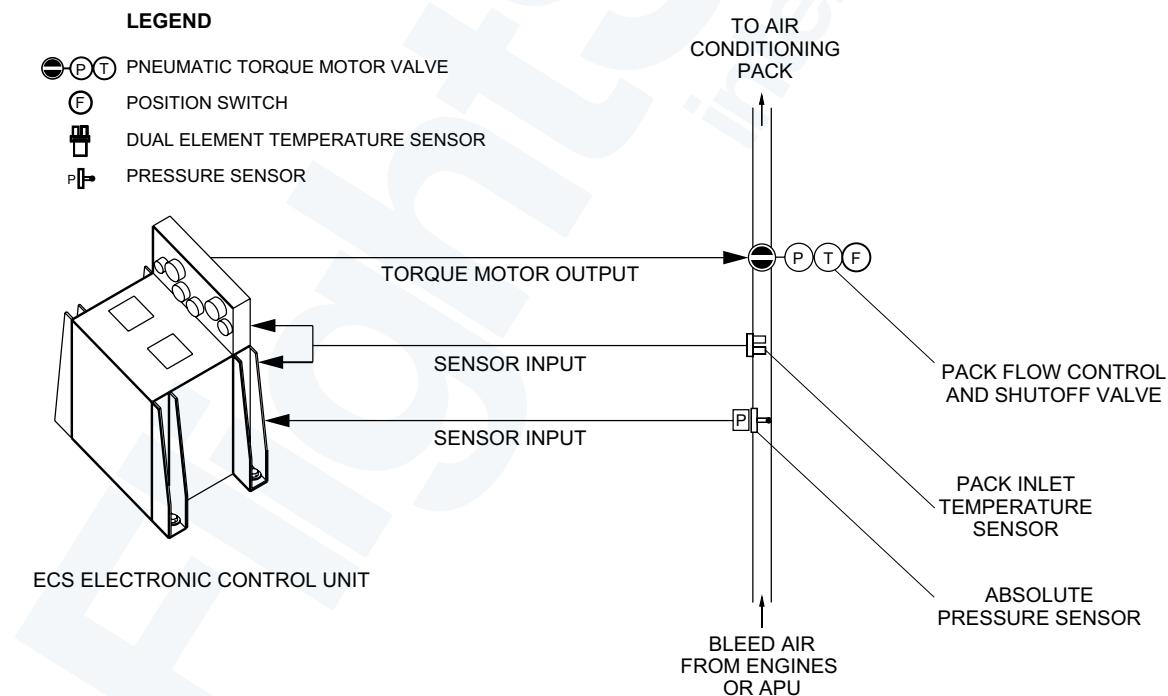


Figure 21-29. Flow Control System

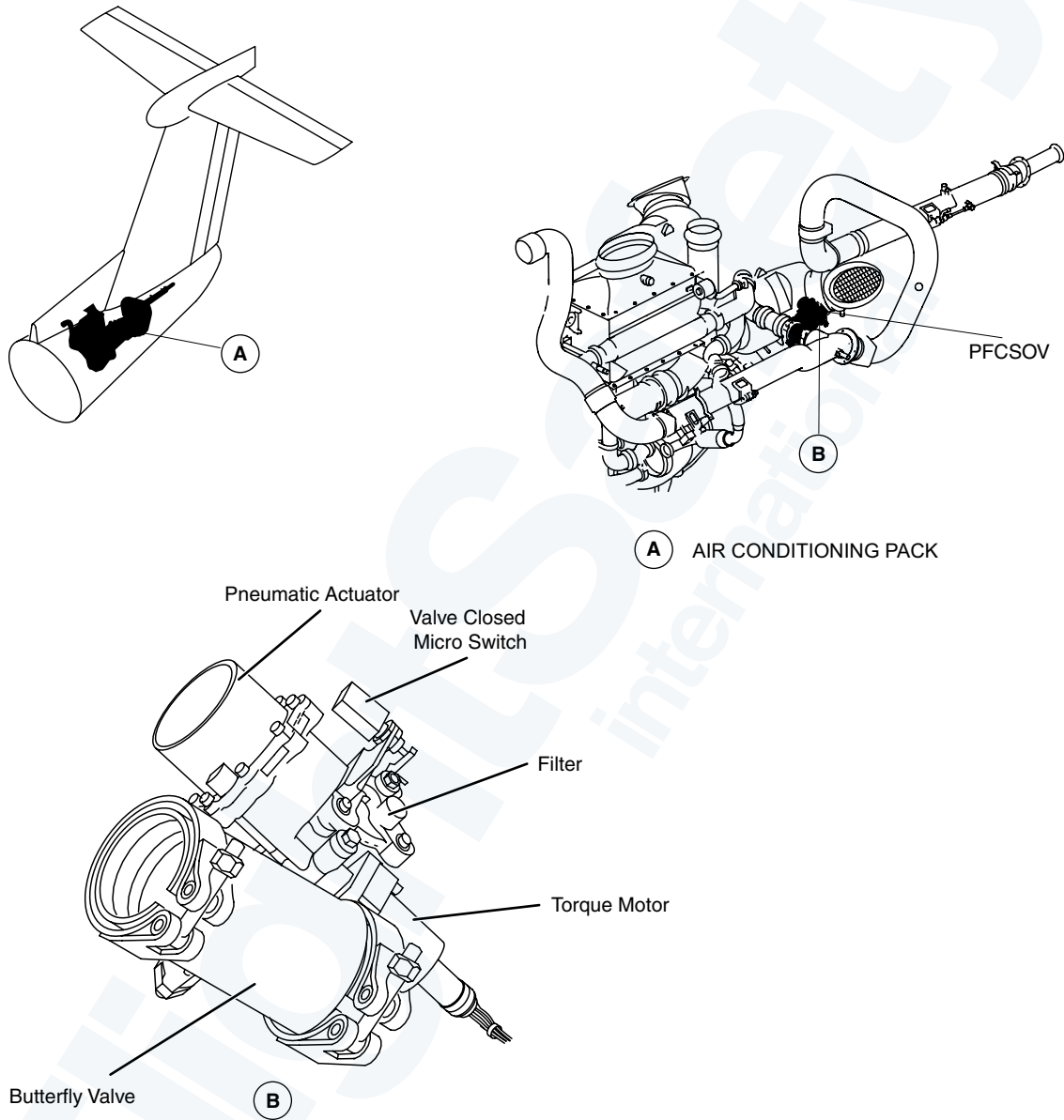
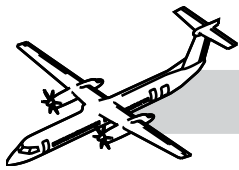
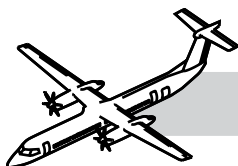


Figure 21-30. Pack Flow Control and Shut-off Valve Synoptic



COMPONENT DESCRIPTION

Pack Flow Control and Shut-off Valve

Refer to:

- Figure 21-30. Pack Flow Control and Shut-off Valve Synoptic.
- Figure 21-31. Pack Flow Control and Shut-off Valve.

The pack flow control and shut-off valve is installed in line with the bleed air supply ducting on the left aft side of the air conditioning pack.

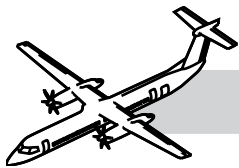
This digital controlled shared component is a pneumatically operated, torque motor actuated butterfly valve. It has a position switch that signals the ECU.

Operational Check of the PFCSOV

A faulty PFCSOV detected by the left or right digital channel will cause one or more PACK FCV faults to appear on the CDS.



Figure 21-31. Pack Flow Control and Shut-off Valve



Absolute Pressure Sensor

Refer to Figure 21-32. Absolute Pressure Sensor.

The absolute pressure sensor, installed in the bleed air supply ducting on the left side of the air conditioning pack, supplies a linear pressure signal to the ECU.

It is a dual channel absolute pressure sensor. It senses the pressure of the bleed air as it flows into the air conditioning pack.

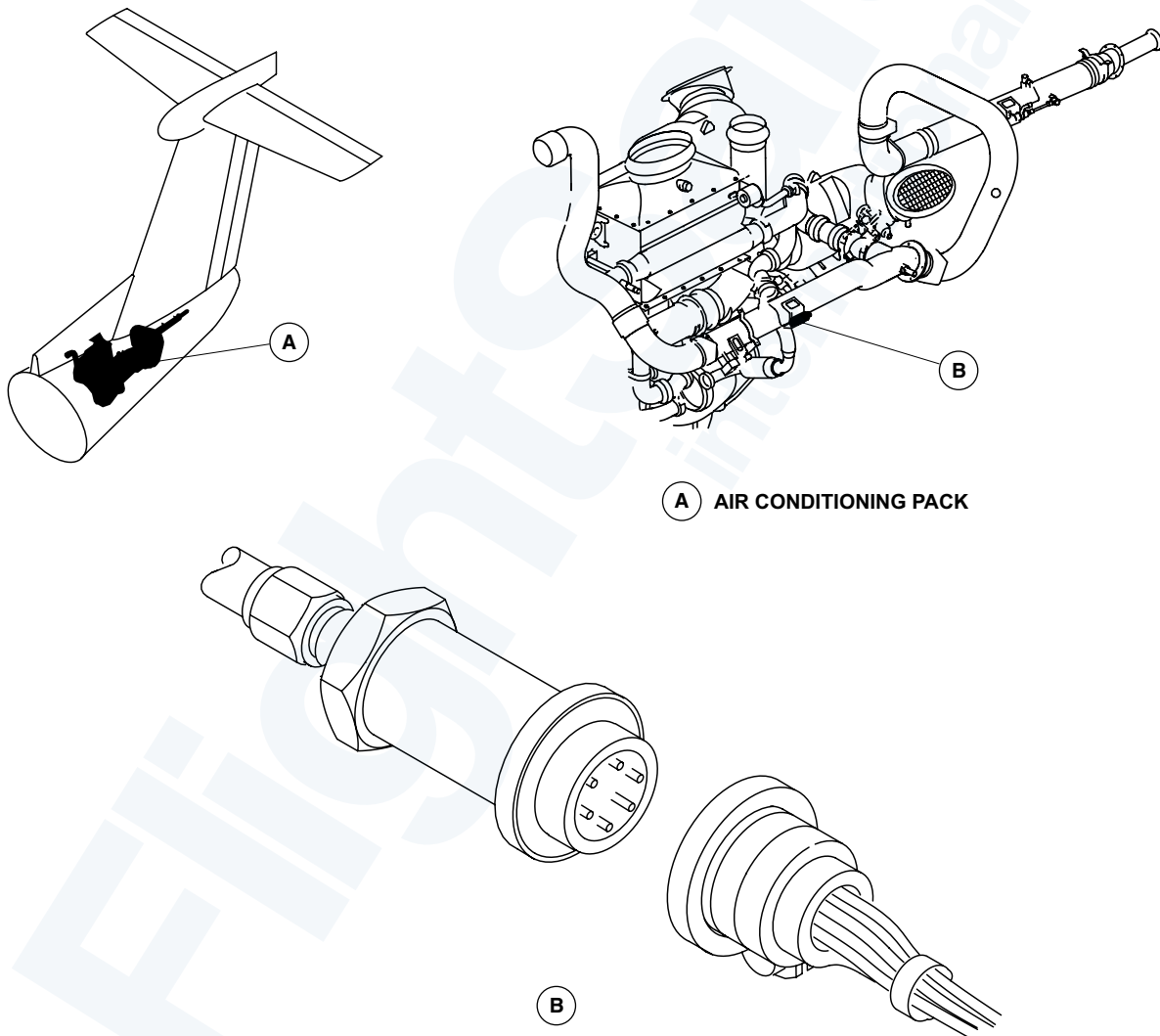
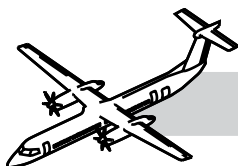


Figure 21-32. Absolute Pressure Sensor



Pack Inlet Temperature Sensor

Refer to Figure 21-33. Pack Inlet Temperature Sensor.

The pack inlet temperature sensor, installed in the bleed supply ducting on the left side of the air conditioning pack, supplies two independent temperature signals to the ECU.

This sensor is a dual Resistive Temperature Device (RTD). It senses the temperature of the bleed air as it flows into the air conditioning pack.

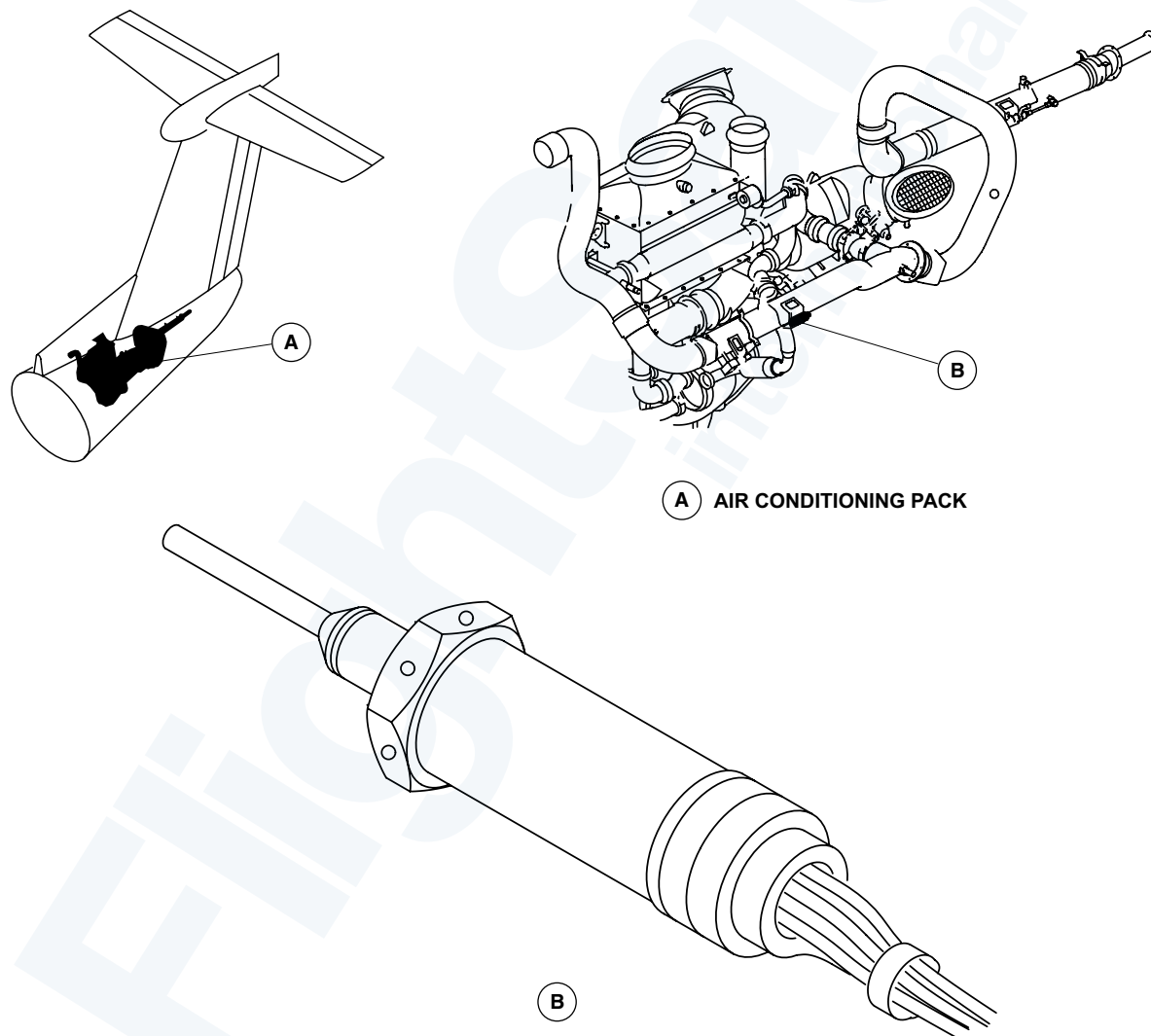
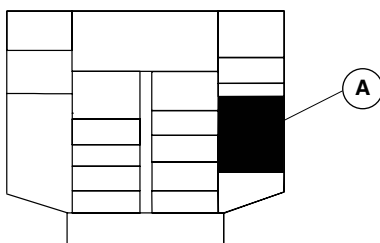
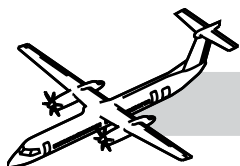


Figure 21-33. Pack Inlet Temperature Sensor



OVERHEAD CONSOLE

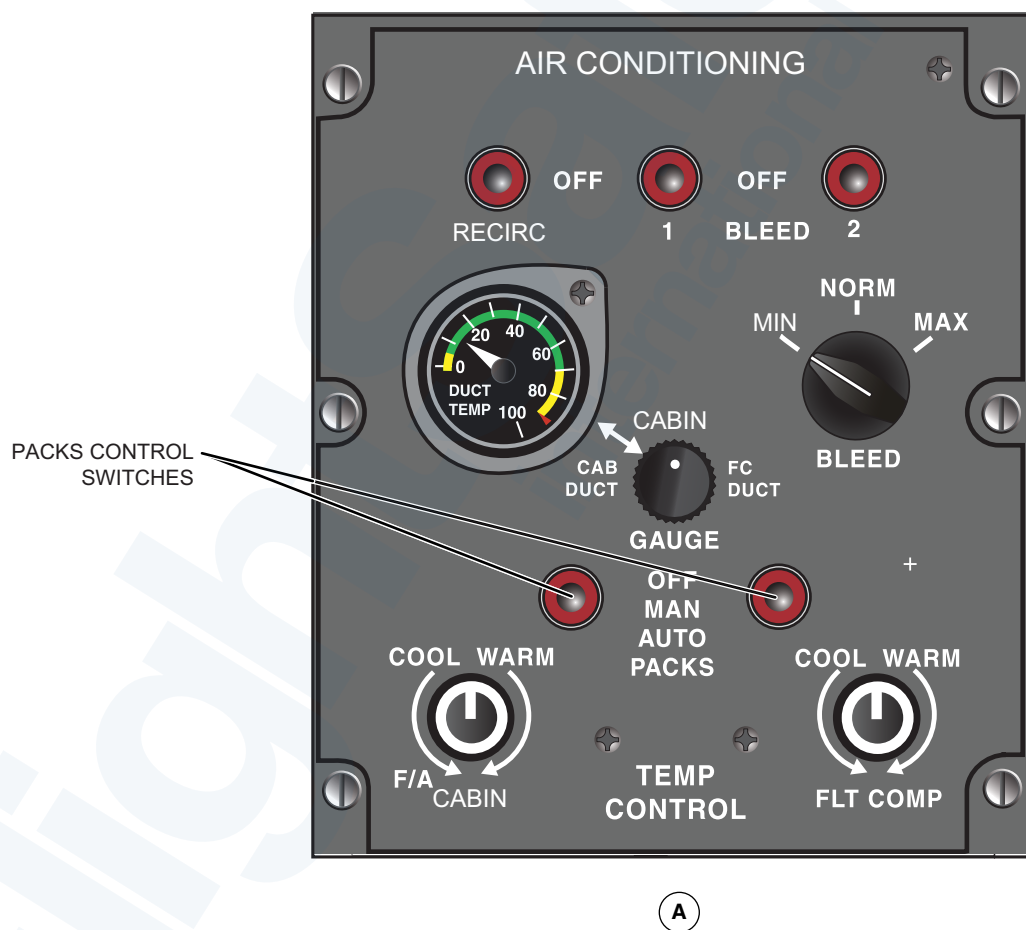
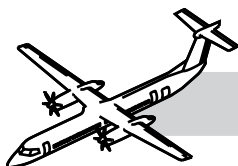


Figure 21-34. Pack Control Switches



CONTROLS AND INDICATIONS

NOTES

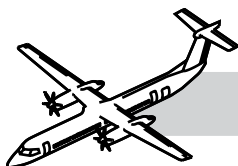
Packs Control Switches

Refer to Figure 21-34. Pack Control Switches.

The packs control switches are located on the AIR CONDITIONING control panel in the flight compartment.

Refer to Figure 21-22 in 21-51-00 for details.





OPERATION

Normal System Operation

Refer to Figure 21-35. ECS Flow Control System.

The aircraft flow control system is automatically controlled.

Selection of the BLEED 1 and/or 2 switches on the AIR CONDITIONING control panel provides a signal to the ECS ECU. The ECU will open the selected engine NFCSOV.

The ECS ECU automatically controls airflow to the ECS Packs through modulation of the Pack Flow Control and Shut-Off Valve (PFCSOV) or the selected NFCSOV.

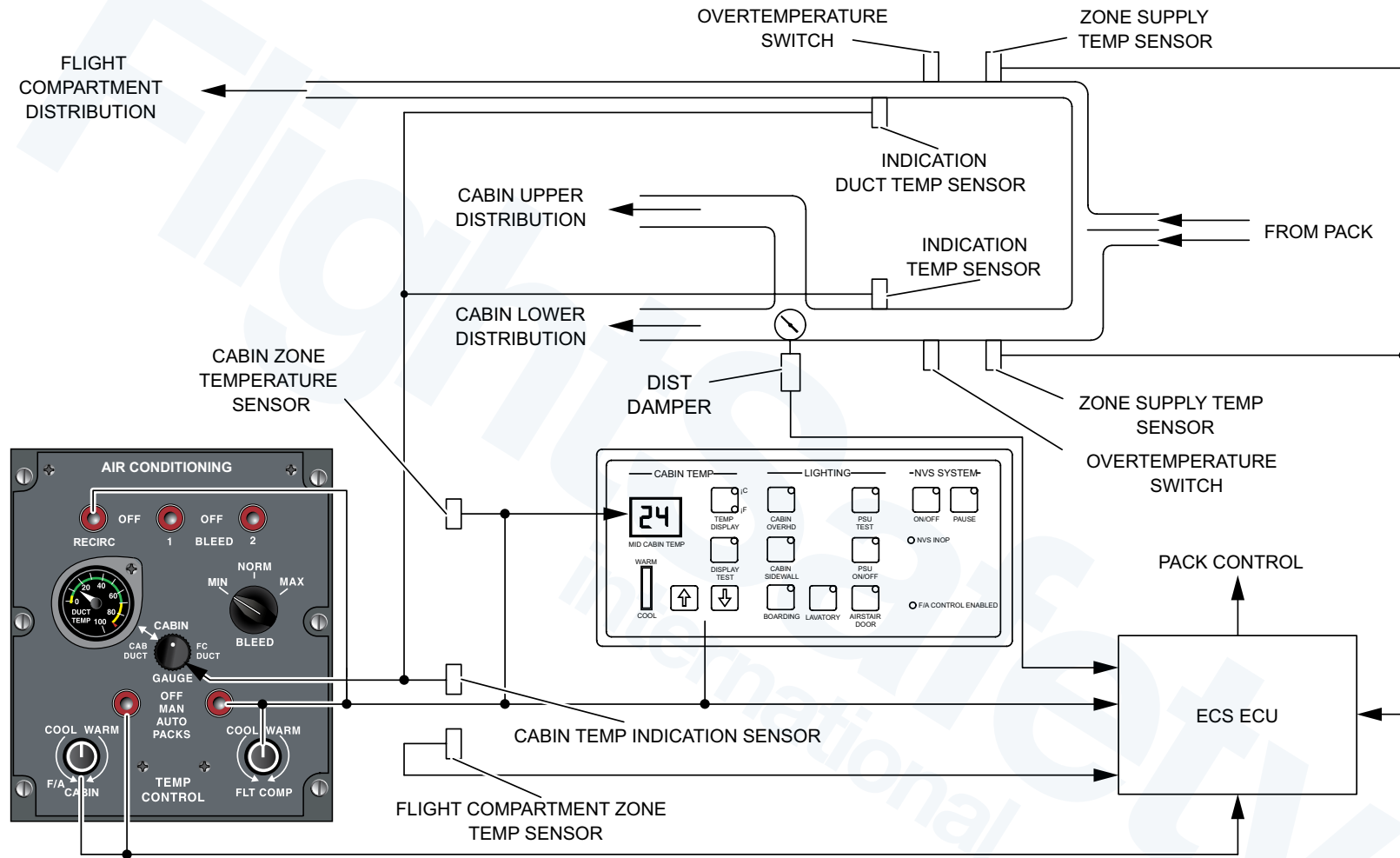
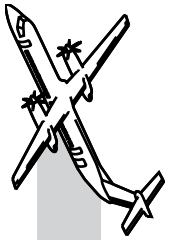
During a single failure of a digital channel, the other digital channel will take over control of the PFCSOV and the analog control will operate for that side. If both digital channels fail, the analog channel is unable to control the PFCSOV and it will close.

Operation with one channel on analog control will cause the related NFCSOV to open and flow to be controlled through the PFCSOV. This is also the case with both pack switches selected to MAN.

An ECU failure or power loss causes the PFCSOV to close and stop the ECS operation.

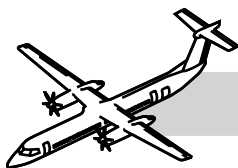
The ECU uses four input sensors to calculate the airflow into the packs:

- The left wing differential pressure sensor signal comes from connector P8 pins 3 and 4 to the ECU connector 2100-P3 pins 50 and 54.
- The right wing differential pressure sensor signal comes from connector P25 pins 3 and 4 to the ECU connector 2100-P4 pins 50 and 54.
- The APU differential pressure sensor signal comes from connector 2121-P9 pins 3 and 4 to the ECU connector 2100-P4 pins 49 and 53.
- The Pack inlet pressure sensor signal comes from connector P26 pin 3 to the ECU connector P3 pin 1.



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Figure 21-36. Temperature Control and Indication Block Diagram



21-61-00 TEMPERATURE CONTROL AND INDICATION

GENERAL

Refer to Figure 21-36. Temperature Control and Indication Block Diagram.

The temperature control and indication system:

- Provides temperature control of the ECS
- Provides temperature indication to the crew and
- Detects and records malfunctions of ECS.

Switches on the AIR CONDITIONING control panel are used to select the desired cabin and flight compartment temperatures.

The ECU uses sensors and switches to monitor:

- Temperature
- Pressure
- Flow rates and
- Valve positions.

The ECU then operates the appropriate valves to maintain the selected temperature setting.

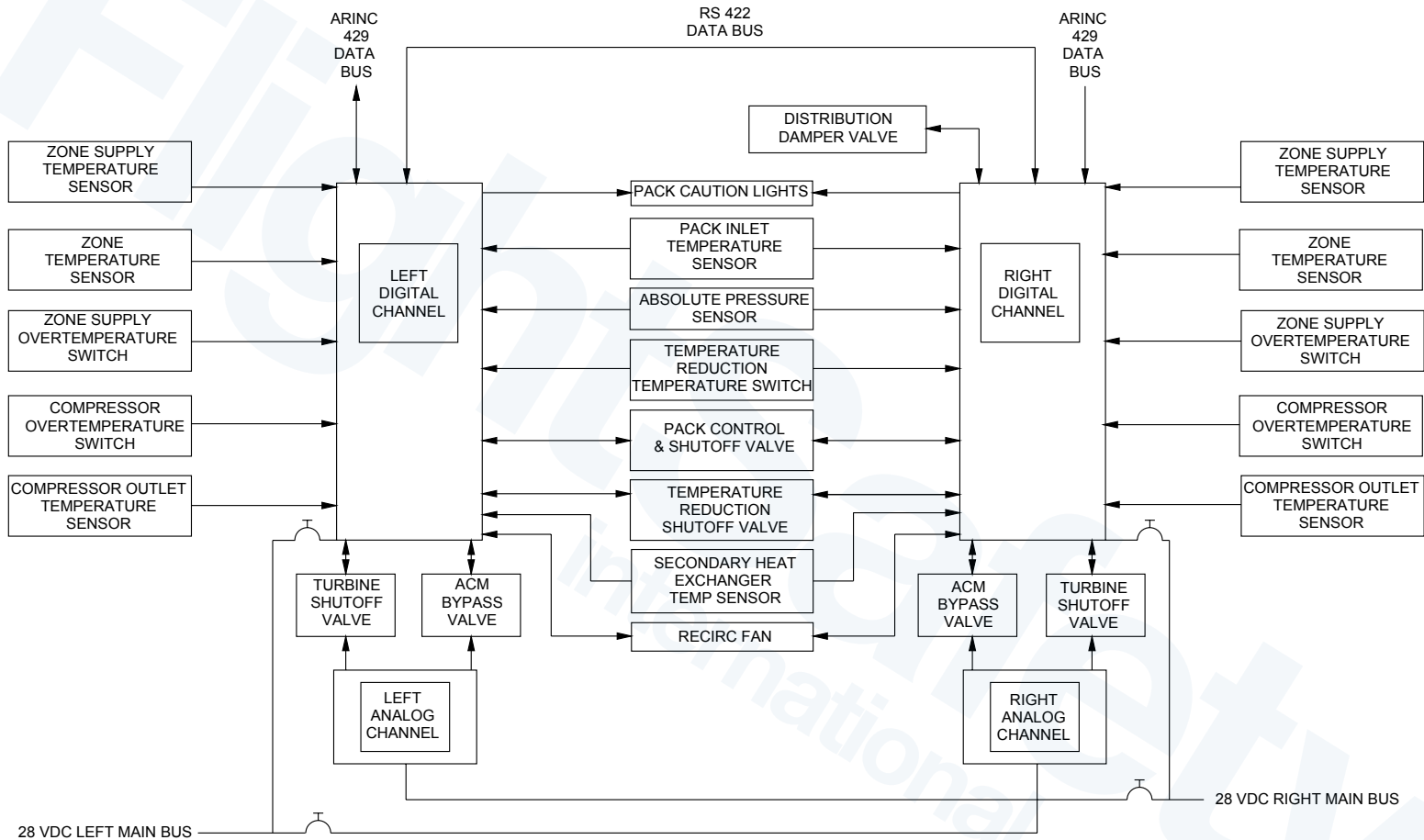
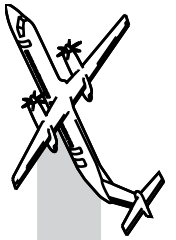
The ECU uses the pilot and crew commands to control the air conditioning system in the following conditions:

- Manual or automatic operation
- One or two ACM operation
- Bleed source selection.

The ECU and the air conditioning system have redundant control systems for continued ECS operation with mechanical and electrical component malfunctions.

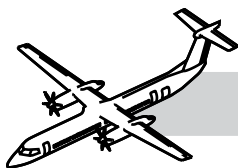
Components of the temperature control and indication system are listed below:

- Electronic control unit - environmental control system zone supply indication sensors
- Temperature sensors
- Zone supply temperature sensors
- Zone supply overtemperature switches
- AIR CONDITIONING control panel
- Temperature selector - Cabin
- Temperature selector - Flight compartment
- Control switches - Packs
- Ducts temperature gauge
- Ducts temperature switch.



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Figure 21-37. Electronic Control Unit Block Diagram - Air Conditioning



SYSTEM DESCRIPTION

Refer to Figure 21-37. Electronic Control Unit Block Diagram - Air Conditioning.

The ECU is the heart of the temperature control system. It automatically controls the temperature of the conditioned air for the aircraft.

The ECU is the interface between the AIR CONDITIONING panel and the air conditioning system's mechanical and electrical components.

It is divided into a left digital channel with a left backup analog channel and a right digital channel with a right backup analog channel.

The left digital channel and the right analog channel receive power from the left 28 VDC bus. The right digital channel and the left analog channel receive power from the right 28 VDC bus.

The air conditioning system operates in MANUAL mode or in AUTO mode.

The left and right digital channels share control of the pack flow control and shut-off valve, the temperature reduction valve, and the recirculation fan.

One channel is selected to be the digital channel in control. The digital channel in control is responsible for all functions applicable to the shared components.

The control channel changes to the other channel at each power up. Both digital channels share control of the PFC SOV. If the control channel fails, the other channel takes control of the PFC SOV.

The analog backup channels do not control the PFC SOV.

The shared components are used to control the PACK as a single unit. The non-shared components are used to control the individual ACM of the PACK. Loss of the digital channel in control causes the other digital channel to take control of the shared components.

Auto Mode

The AUTO mode uses the left/right digital channels for ECS control. The temperature control and indication system uses the following components to sense the temperature of the conditioned air entering the cabin and flight compartments:

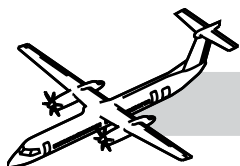
- Zone supply temperature sensors
- Zone temperature sensors
- Zone supply indication sensors
- Zone supply overtemperature switches.

Two zone supply temperature sensors measure the temperature of the air in the supply ducts. The ECU uses these signals to control the temperature of the air leaving the air conditioning pack.

Two zone temperature sensors measure compartment temperatures and send the data to the ECU.

A third sensor supplies cabin temperature to the gauge on the AIR CONDITIONING control panel.

Two other sensors supply flight and cabin duct supply temperatures to the gauge.



Manual Mode

Refer to Figure 21-38. Maintenance Panel ECS Auto Fail.

The ECU reverts to manual mode by selection to MAN. Automatic reversion to manual occurs without indication to the flight compartment when the ECU detects digital control has been lost.

An ECU amber fault light on the Maintenance panel illuminates whenever a major fault is detected by BIT. This indication is only displayed with WOW.

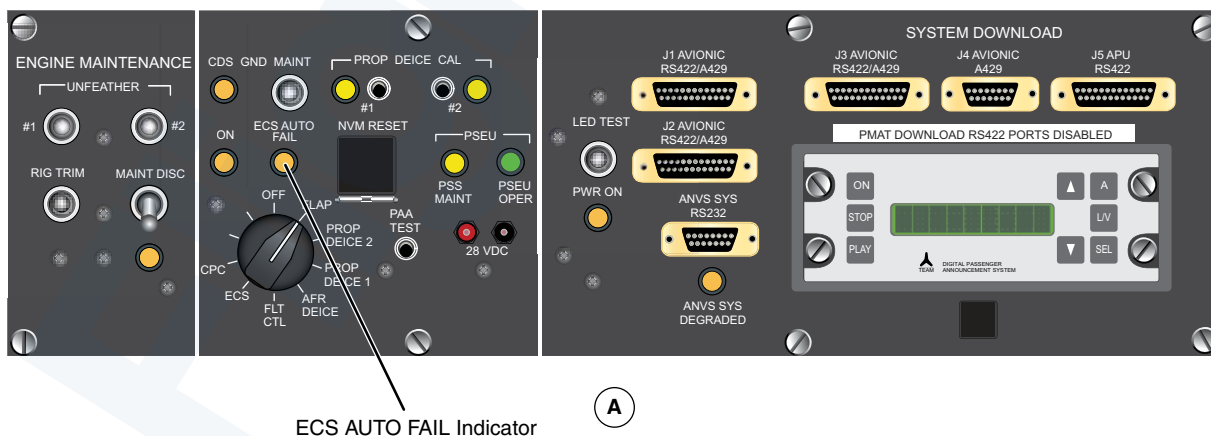
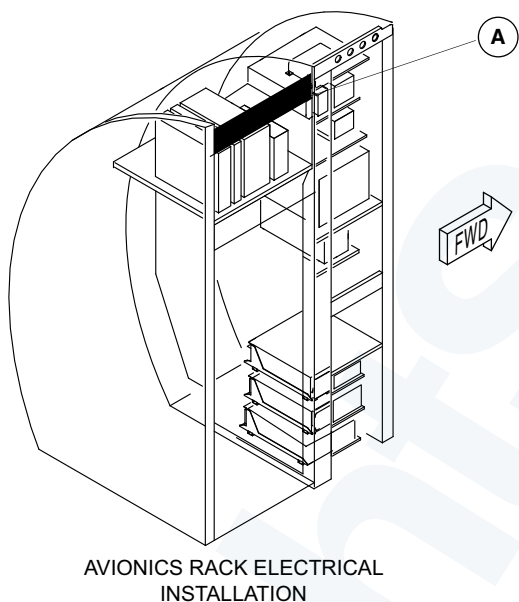
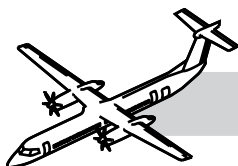


Figure 21-38. Maintenance Panel ECS Auto Fail



COMPONENT DESCRIPTION

Environmental Control System Electronic Control Unit (ECS ECU)

Refer to Figure 21-39. Electronic Control Unit (ECS ECU).

The ECU is installed in the avionics bay under the floor boards.

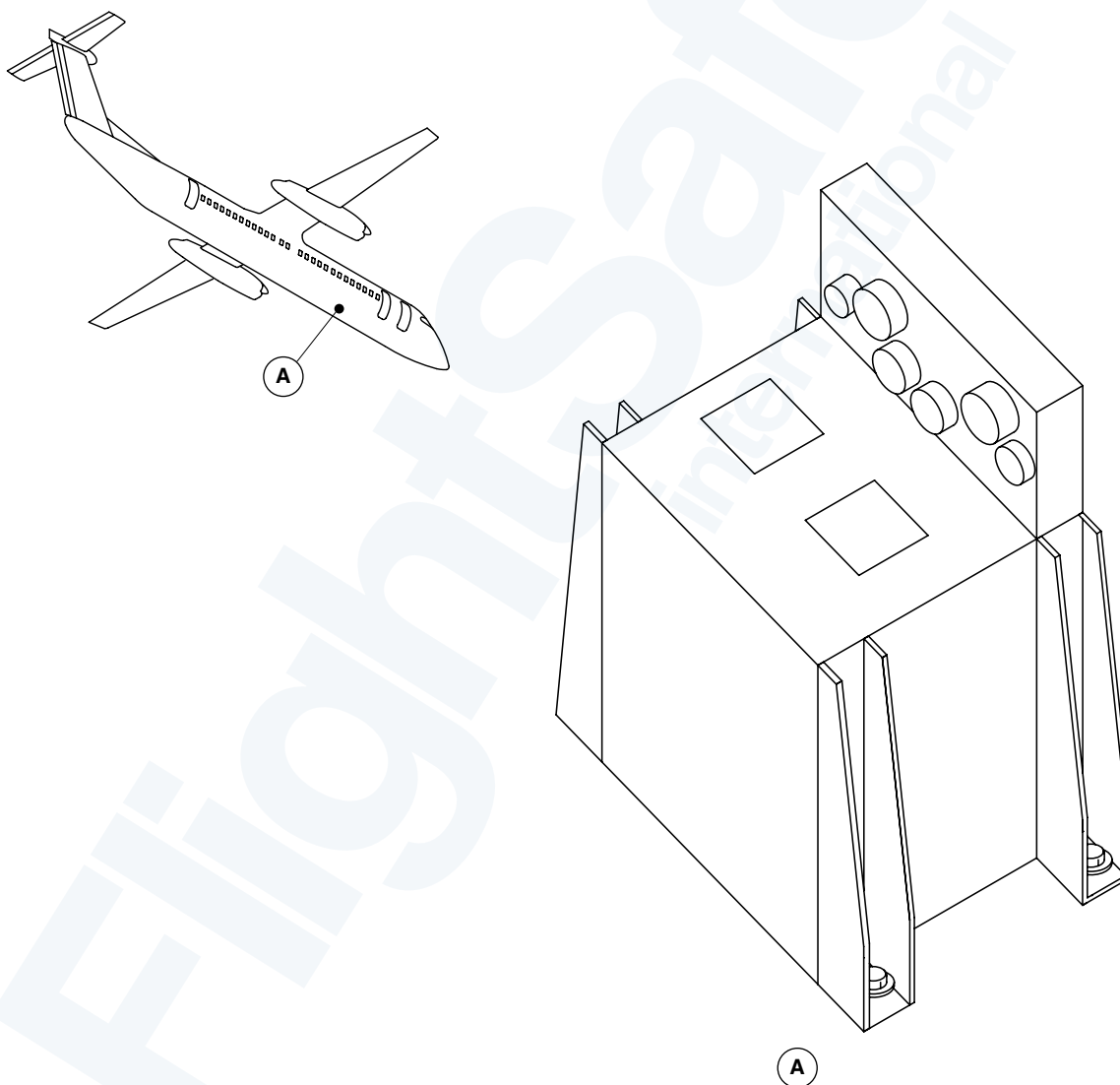


Figure 21-39. Electronic Control Unit (ECS ECU)

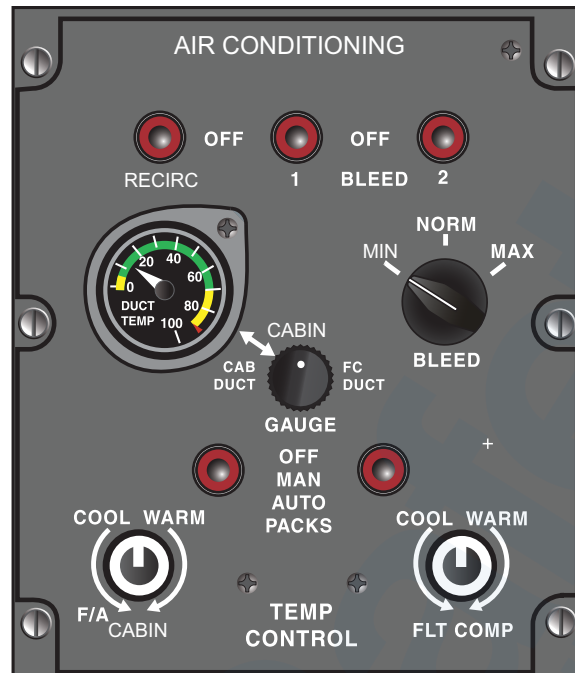
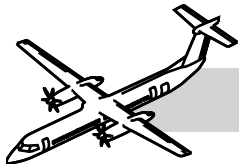


Figure 21-40. Air Conditioning Control Panel

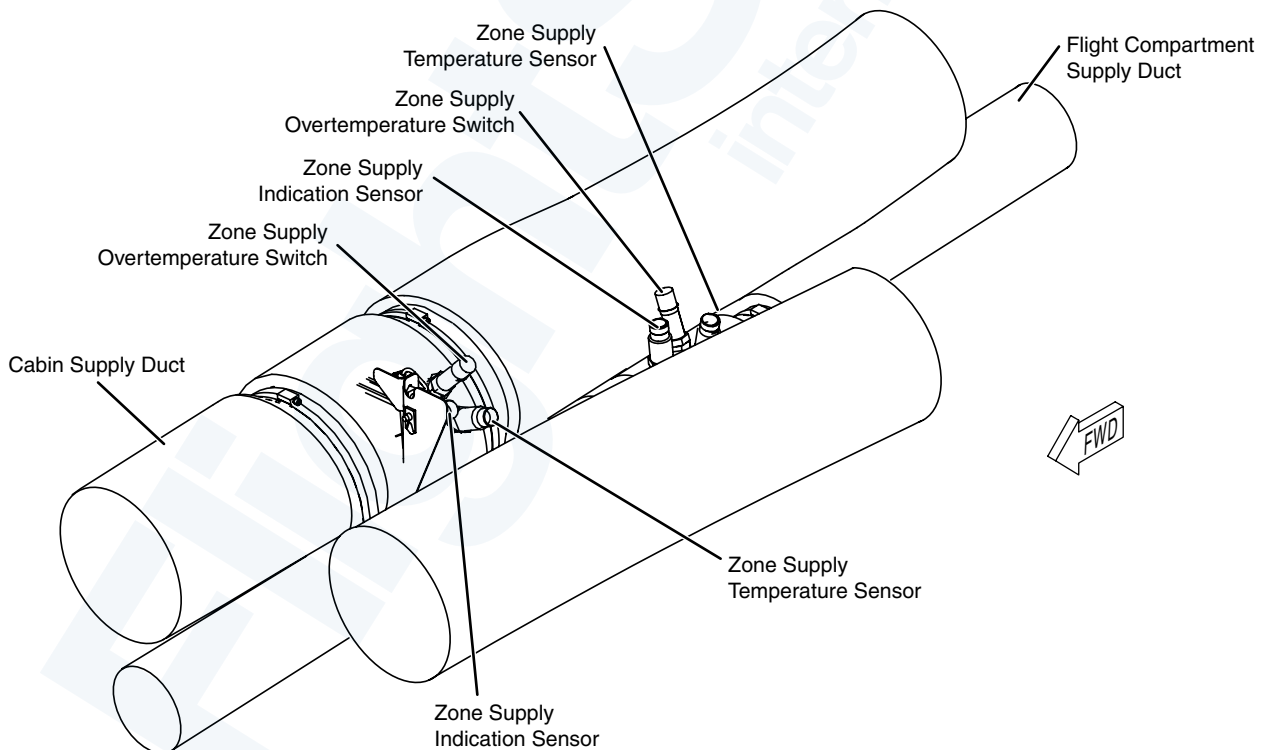
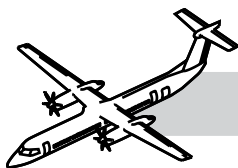


Figure 21-41. Zone Supply Temperature Sensors and Switches



Zone Supply Indication Sensors

Refer to Figure 21-40. Air Conditioning Control Panel.

There are two sensors located at Stn X765 in the flight compartment and cabin supply ducts.

The zone supply indication sensors are RTDs. An RTD gives a resistance which is proportional to the temperature it senses.

These sensors supply temperature signals directly to the DUCT TEMP gauge on the AIR CONDITIONING control panel.

Zone Supply Temperature Sensors

Refer to Figure 21-41. Zone Supply Temperature Sensors and Switches.

The zone supply temperature sensors are installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor. These sensors supply signals to the ECU for ECS control.

The zone supply temperature sensors are dual thermistor sensors that supply a signal to the digital channel of the ECU.

Zone Supply Overtemperature Switches

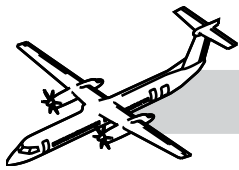
Refer to Figure 21-41. Zone Supply Temperature Sensors and Switches.

The zone supply overtemperature switches, installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor, supply discrete signals to the ECU when the temperature in either supply duct exceeds 190°F (87.8°C).

This discrete signal causes the ECU to illuminate the related CABIN DUCT HOT or FLT COMP DUCT HOT caution light on the CAWP.

Temperature Sensors

Two zone temperature sensors for the flight and cabin compartments are used for zone temperature control.



Cabin Zone Temperature Sensors

Refer to:

- Figure 21-42. Cabin Zone Sensors.
- Figure 21-43. Cabin Zone Temperature Sensor and Indication Sensor Synoptic.

The cabin zone temperature sensors, located on the cabin temperature ejector by the fourth dado panel (Stn X 352) on the left side of the aircraft, measure the cabin zone temperature and provide signals to the ECU for control and indication.

The zone temperature sensors are dual thermistor sensors. Cabin supply air flows through a restriction in the dado duct.

This accelerates the flow of air and causes a low pressure air which draws cabin air over the zone temperature sensor. This ensures an accurate measurement of the cabin ambient air.



Figure 21-42. Cabin Zone Sensors

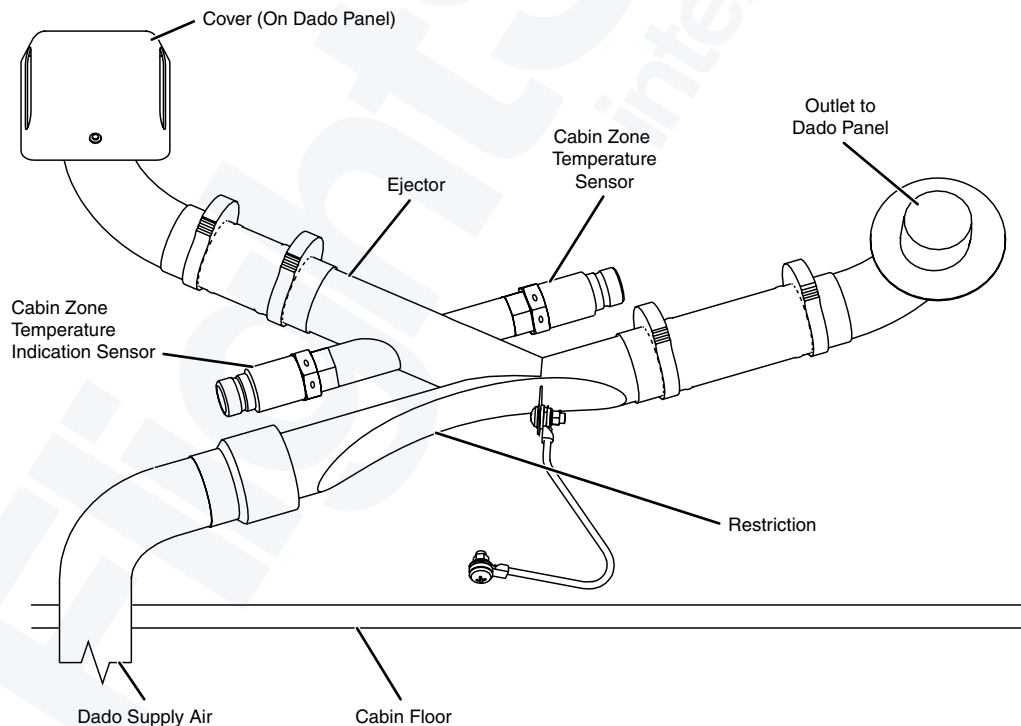
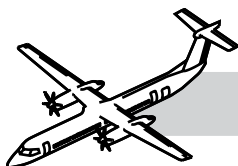


Figure 21-43. Cabin Zone Temperature Sensor and Indication Sensor Synoptic



Flight Compartment Zone Temperature Sensor

Refer to Figure 21-44. Flight Compartment Zone Temperature Sensor.

The flight compartment zone temperature sensor, installed in the right-hand forward underfloor avionics cooling duct below the co-pilot rudder-pedal cover assembly, measures the flight compartment zone temperature and provides a feedback signal to the ECU.

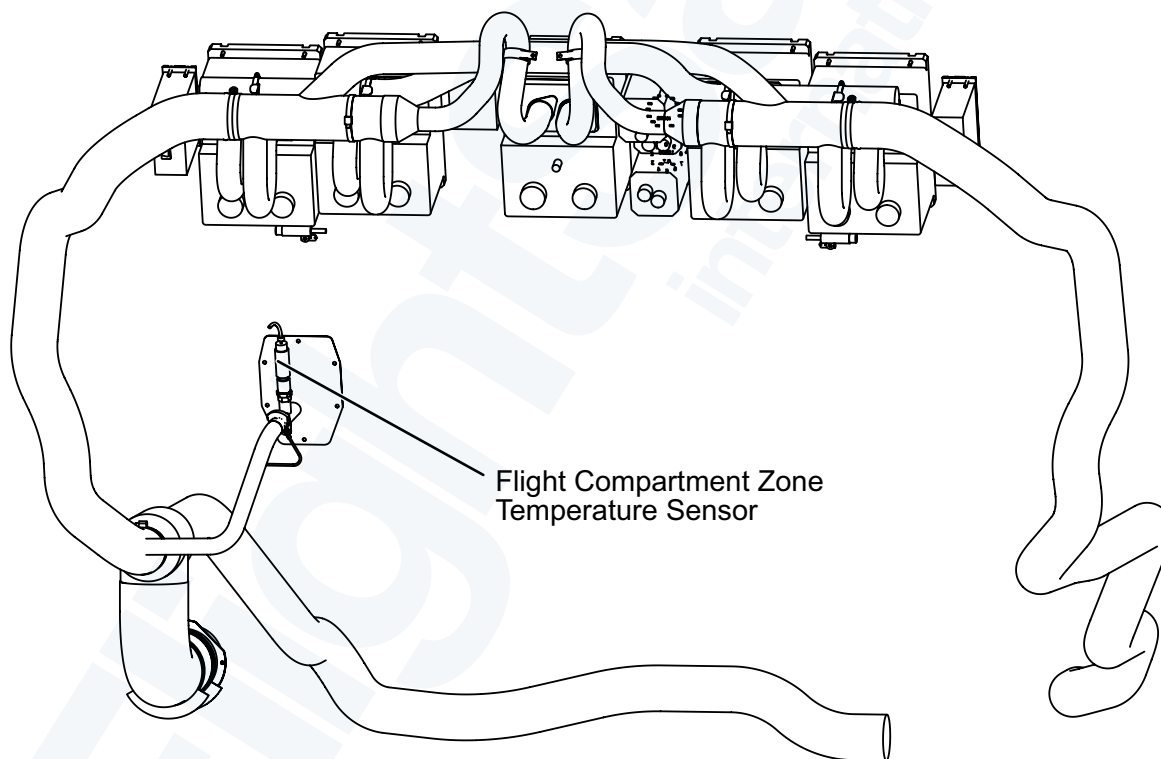


Figure 21-44. Flight Compartment Zone Temperature Sensor

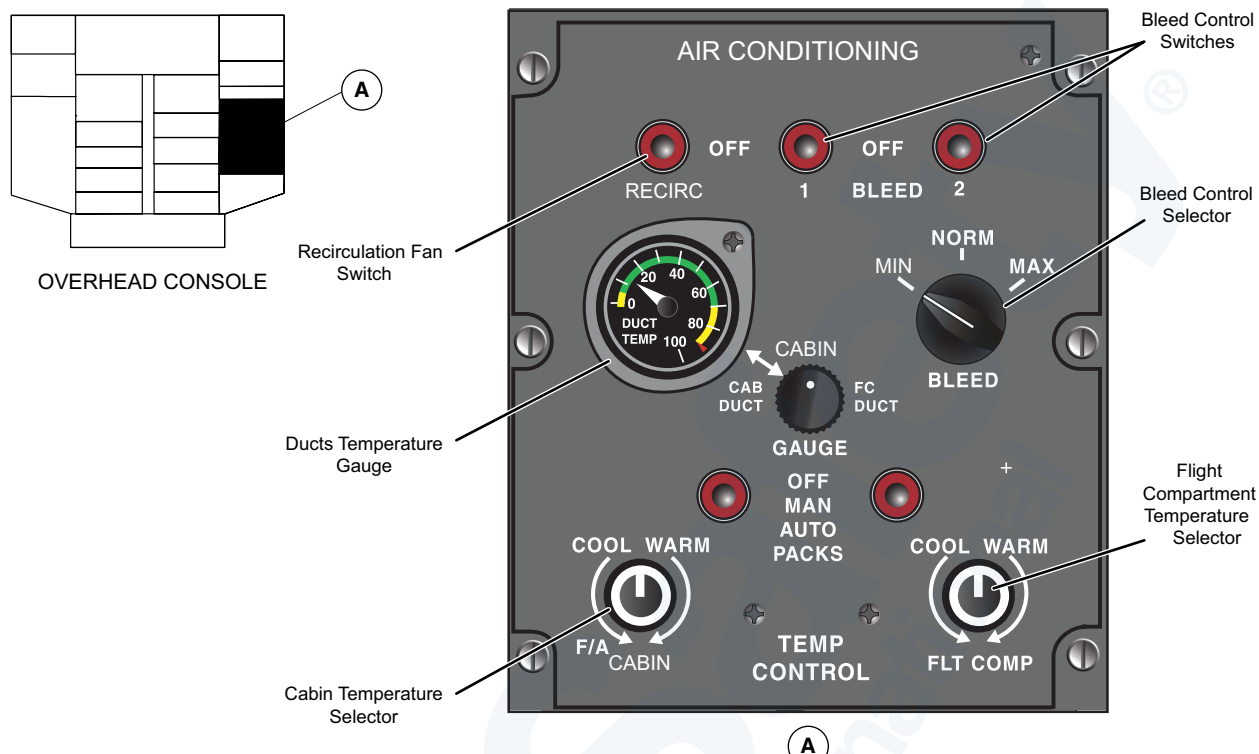
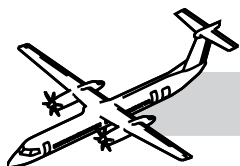


Figure 21-45. Air Conditioning Control Panel - Detail

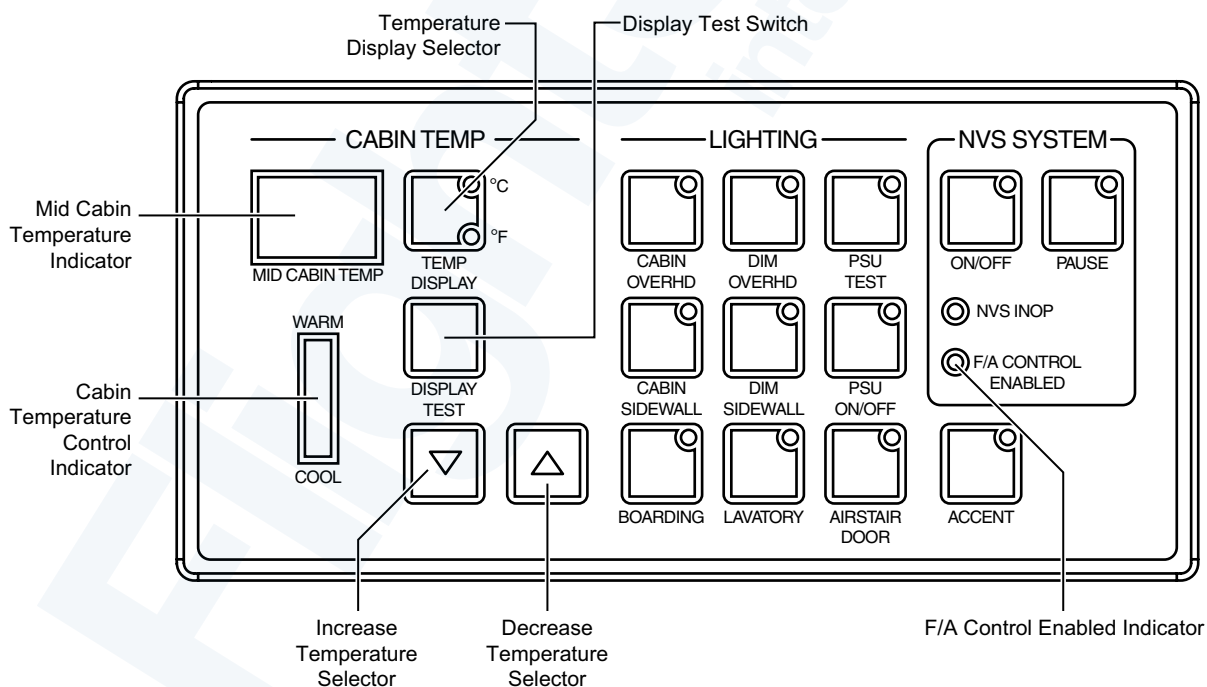
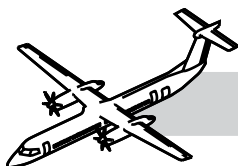


Figure 21-46. Flight Attendant's Panel - Cabin Temperature Control



CONTROLS AND INDICATIONS

Temperature Control and Indication - Air Conditioning Control Panel

Refer to:

- Figure 21-45. Air Conditioning Control Panel - Detail.
- Figure 21-46. Flight Attendant's Panel - Cabin Temperature Control.

RECIRC FAN Switch - The RECIRC fan switch controls the operation of the fan.

Duct Temp Gauge -The DUCT TEMP gauge has a scale from 0 through 100 to show the temperature selection in degrees Celsius.

Cabin Temperature Selector - The CABIN compartment temperature control rotary knob is rotated to change the requested temperature between COOL and WARM.

BLEED CONTROL Switches - Starts bleed airflow from the No. 1 and/or No. 2 engine to the air conditioning packs.

The CABIN temperature control rotary knob is turned fully counterclockwise (CCW) to the F/A position to allow the flight attendant panel control of the cabin temperature.

Ducts Temperature Gauge Switch - The air conditioning system has a GAUGE rotary knob to make the DUCT TEMP gauge show the cabin duct, cabin, flight compartment duct or passenger cabin area temperature.

Packs Control Switches -The PACKS control switches are three position toggle switches with the following positions:

- **OFF** - Closes related Turbine Shut off Valve (TSOV)
- **MAN** - Temperature control based on analog positioning of bypass values. NO FEEDBACK

- **AUTO** - ECU temperature control based on actual cabin temperature modulates the pack bypass valves.
- **FLT COMP Temperature Selector** -The FLT COMP temperature control rotary knob is rotated to change the flight compartment temperature request between COOL and WARM.

The Flight Attendant's Panel has the following controls and indications:

- **MID CABIN TEMP INDICATOR** - Displays temperature from the cabin zone temperature sensor.
- **TEMP DISPLAY SELECTOR** - Toggles the MID CABIN TEMP indicator between degrees Celsius and degrees Fahrenheit.
- **F/A CONTROL ENABLED INDICATOR** - A amber LED that comes on when the CABIN temperature selector is rotated to the F/A position in the Air Conditioning Control Panel.
- **DECREASE TEMPERATURE SELECTOR** - A down arrow selector that decreases the temperature request.
- **INCREASE TEMPERATURE SELECTOR** - An up arrow selector that increases the temperature request.
- **CABIN TEMPERATURE CONTROL INDICATOR** - An LED bar graph that displays the temperature request in red bar segments from COOL to WARM.

Interfaces

Refer to Figure 21-47. ECS Interfaces.

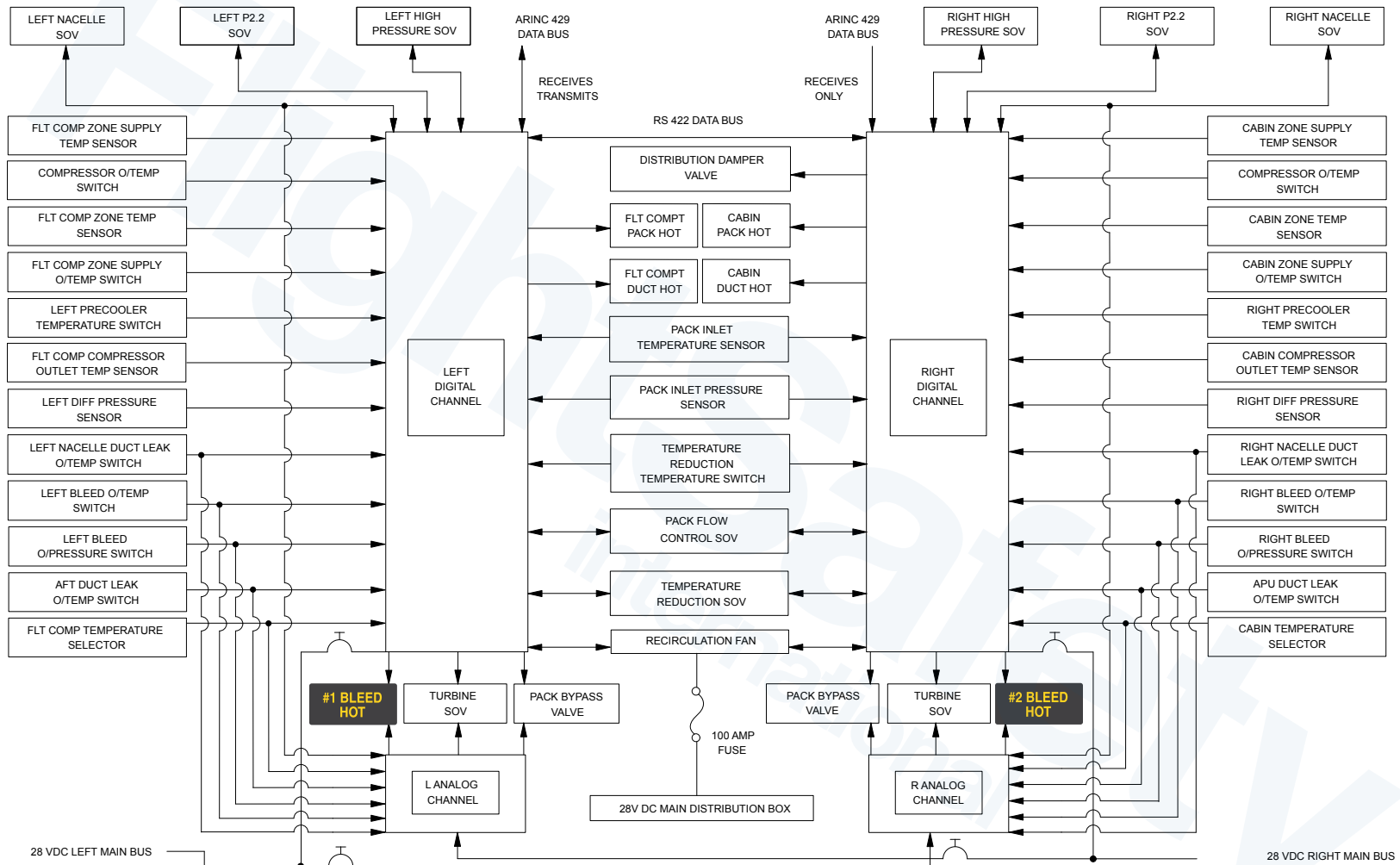
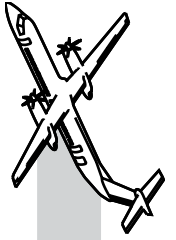
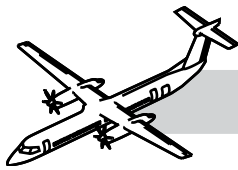


Figure 21-47. ECS Interfaces



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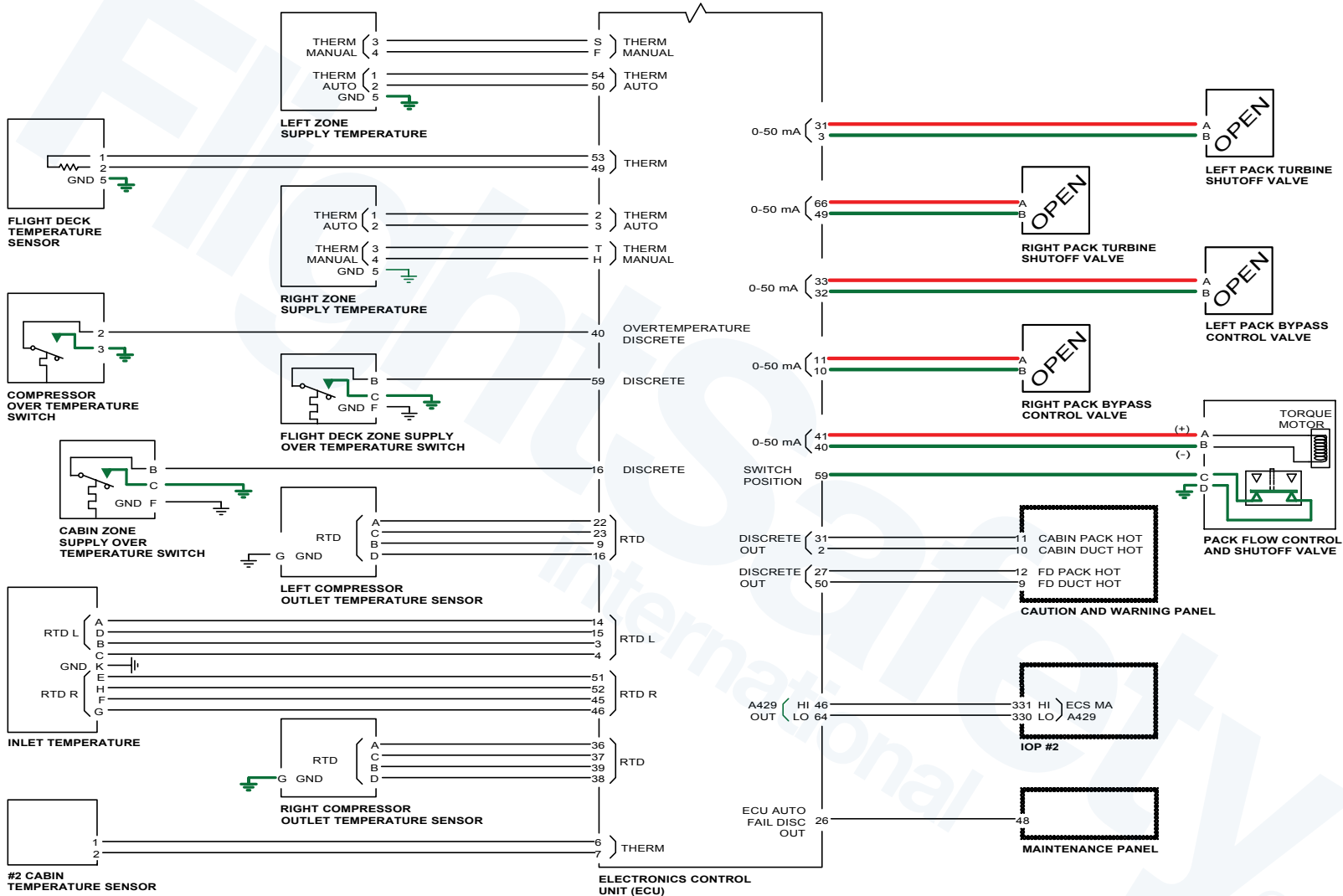
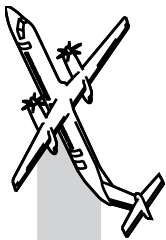
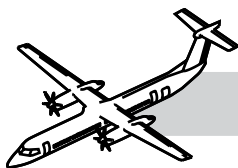


Figure 21-48. Temperature Control Electrical Schematic



OPERATION

Normal System Operation

Refer to:

- Figure 21-48. Temperature Control Electrical Schematic.
- Figure 21-48. Temperature Control and Indication Electrical Schematic.

The ECS ECU provides solenoid power and pressure sensor excitation. The ECU controls the units as follows:

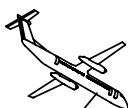
- Recirculation fan speed
 - PFCSOV modulation
 - Pack temperature reduction SOV
 - Pack turbine SOV
 - Pack Bypass SOV and
 - Distribution damper valve
1. The BLEED switches are set to the 1 and 2 position (PACK switches are not in the OFF position):
 - This discrete will cause the ECU to open the Nacelle Flow Control Shutoff Valve (NFCSOV), the PFCSOV and the two Turbine SOVs. ACMs will start to operate.
 2. The PACK switches are set to the AUTO position:
 - Pack switches send a ground discrete to connector J5 pin 37 and connector J2 pin 41 on the ECU
 - Send an open circuit discrete to connector J5 pin 10 and connector J2 pin 16 on the ECU
 - The signals will cause the ECU to start controlling the two ACMs to maintain the selected compartment temperatures and pack air flow rate.

During normal operation, the ECU logic will automatically:

- Monitor the two supply duct and two zone temperature sensors and perform temperature control calculation
- Monitor two pack over-temperature sensors and reduce the pack flow if the temperatures are too high
- Monitor the pack over-temperature switch to stop the pack flow if the temperature is too high
- Open the two turbine SOVs
- Modulate the two pack bypass valves to warm or cool the pack outlet air
- Open temperature reduction SOV to cool the pack inlet air when required
- Modulate the PFCSOV to control air flow into the packs. The PFCSOV open switch sends a ground signal from the PFCSOV connector P23 pin C to the ECU connector P5 pin 59
- Perform continuous system BIT

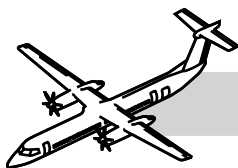
The two PACK switches are set to the MAN position to give manual control of the packs:

- These actions will send open circuit discretes to the ECU (Figure 21-48. Temperature Control and Indication Electrical Schematic.):
 - Connector J5 pins 10 and 37
 - Connector J2 pins 16 and 41
 - The ECU logic uses these discretes to send an analog (0 to 50 ma) signal to open or close the pack bypass valve
 - The ECU signals are sent from the: (refer to Figure 21-48)
 - Connector P4 pins 10 and 11 to the right bypass valve and
 - Connector P3 pins 33 and 32 to the left bypass valve
 - The ECU digital channel transfers the control of the control of the NFCSOV to the analog channel



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FOR TRAINING PURPOSES ONLY



- The ECU logic uses these discretes to send an analog (0 to 50 ma) signal to open or close the pack bypass valve
- The ECU signals are sent from the: (Refer to Figure 21-48)
 - Connector P4 pins 10 and 11 to the right bypass valve and
 - Connector P3 pins 33 and 32 to the left bypass valve
- The ECU digital channel transfers the control of the NFCSOV to the analog channel.

Refer to Figure 21-49. Temperature Control and Indication Electrical Schematic.

Rotating the CABIN control knob to the full WARM position raises the cabin temperature.

- An open circuit discrete is sent to connector J5 pin 78 on the ECU and connector J3 pin 37 on the Flight Attendant (F/A) panel
 - This signal provides cabin temperature primary control and indication to the A/C panel
 - The A/C panel will use the No.1 temperature sensor data from connector P101 pins 4 and 5
 - The ECU sends a 50 mA signal to the torque motor to close the turbine SOV
- If AUTO is selected, a low resistance discrete is sent to connector J4 from pin 27 to pin 20 on the ECU
- If MAN is selected, a low resistance discrete is sent to connector J6 from pin U to pin K on the ECU.

In the full COOL (F/A) position, the A/C panel output is as follows:

- A ground discrete is sent to connector J5 pin 78 on the ECU and connector J3 pin 37 on the flight attendant panel

- This gives cabin temperature primary control and indication to the F/A panel
- The ENABLED light on the F/A panel will illuminate
- The F/A panel will use the No.2 cabin temperature sensor data on connector J3 pins 5 and 24
- If AUTO is selected, a 10 kΩ discrete is sent to connector J4 from pin 27 to pin 20 on ECU
- If MAN is selected, a 10 kΩ discrete is sent to connector J6 from pin U to pin K on ECU.

The Crew rotates the FLT COMP control knob to the full WARM position.

- If AUTO is selected, a low resistance discrete is sent to connector J3 from pin 27 to pin 35 on ECU
- If MAN is selected, a low resistance discrete is sent to connector J1 from pin R to pin D on ECU
 - The ECU uses this signal to send a 50 mA signal to the torque motor to close the turbine SOV.

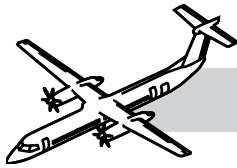
In the full COOL position, the output is:

- If AUTO is selected, a 10 kΩ discrete is sent to connector J3 from pin 27 to pin 35 on ECU
- If MAN is selected, a 10 kΩ discrete is sent to connector J1 from pin R to pin D on ECU.

The GAUGE knob shows three temperature selections on the A/C panel DUCT TEMP gauge. The temperature indicator received 28 VDC from pin C on connector J102.

The indicator measures the current flow through the selected sensor. The current flow is through the A/C panel pins that follow:

- The cabin duct sensor (connector J101 pins 6 and 11)



- The flight compartment duct sensor (connector J101 pins 9 and 13)
- The cabin compartment zone sensor (connector J101 pins 4 and 5)
- The indication ground pins on connector J101 are pins 4, 8, 11 and 13.

The RECIRC switch on the A/C panel will start the recirculation fan. The A/C panel sends out the discretes signal that follow:

- A ground discrete to the recirculation fan (connector J22 pin D)
 - Fan starts at low speed
 - The fan stays at slow speed during single pack operation
 - The fan operates at high speed during all other operations
- A ground discrete to the ECU (connector J2 pin 60)
 - The ECU uses this ground to operate the recirculation fan.

- Turn CABIN TEMP CONTROL knob to F/A position
- Verify the F/A control panel is working by pushing the DISPLAY TEST switch and check that temperature indication and WARM/COOL indicator lights come on
- Push the WARM/COOL indicator to full warm and ensure warm air is flowing out to the cabin
- See the opposite happens when you push the WARM/COOL indicator to full cool.

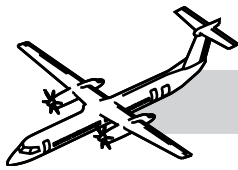
Crew Report

Pilot reported that CABIN DUCT HOT caution light came on during final approach. Fault code 7117 CAB OVR TMP 3 was also found recorded in the CDS. What troubleshooting steps will you take?

Temperature Control System Operational Check

To verify that the ECS temperature control system is functioning correctly, please refer to AMM Task 21-61-00-710-803. Here is a summary of the check:

- Use the APU for bleed air supply
- Do not pressurize the aircraft or turn on the airframe de-ice system
- Select both packs to AUTO and both temperature knob to mid range
- Stable for 5 minutes before recording both FC DUCT and CABIN DUCT temperatures
- Rotate both temperature knobs to full cool and later to full warm
- Verify both duct temperatures do drop below and rise above than what were recorded at mid range for full cool and full warm respectively



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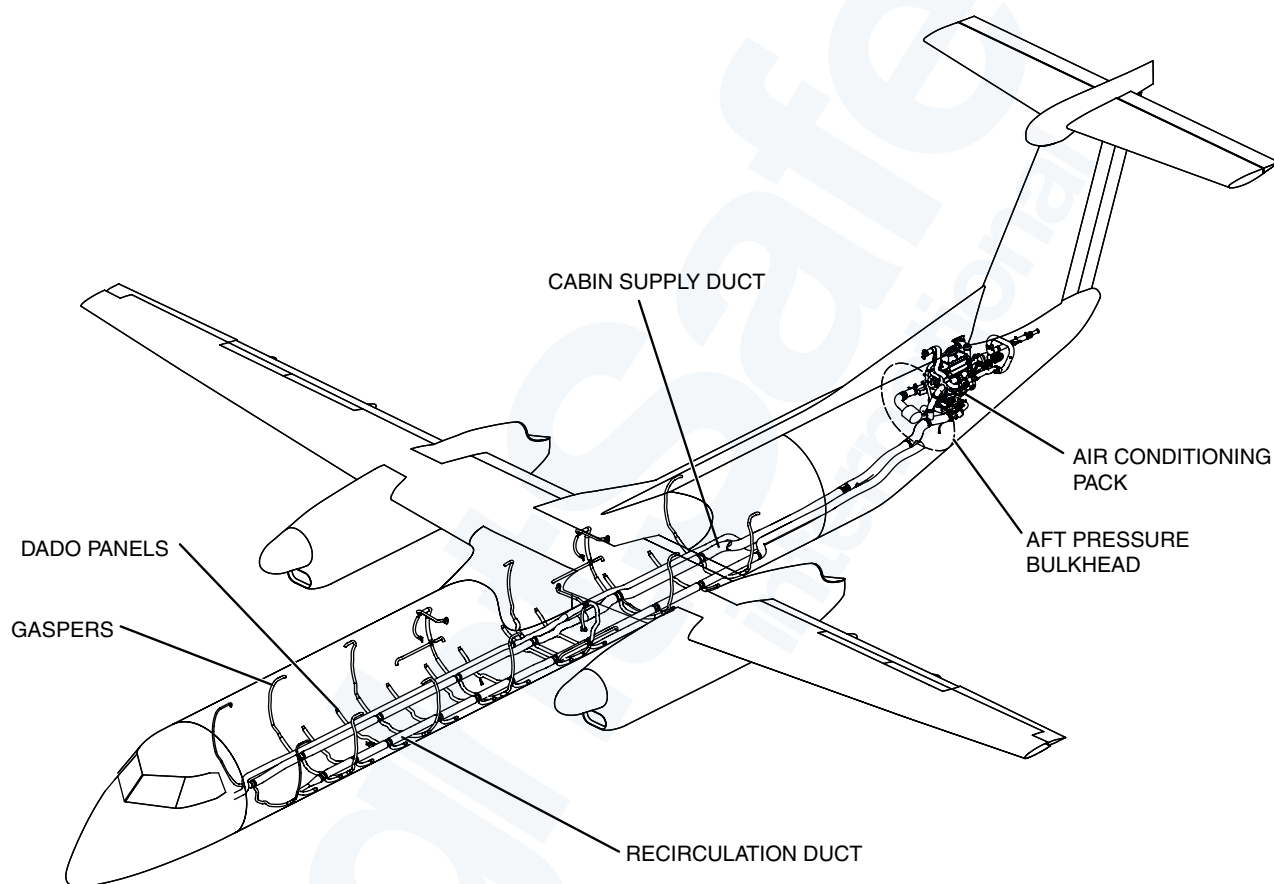
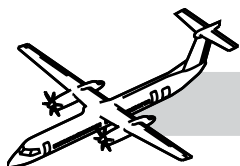
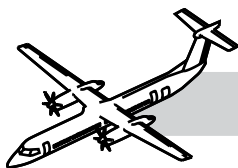


Figure 21-50. General Air Distribution



21-20-00 AIR DISTRIBUTION

INTRODUCTION

The ECS packs supply the flight compartment and cabin air distribution systems with conditioned air.

GENERAL

The distribution system has the following subsystems:

- Air Distribution, Flight Compartment
- Air Distribution, Cabin and Baggage Compartment
- Ground Connection
- Ram Air
- Avionics Cooling.

SYSTEM DESCRIPTION

The air distribution system takes conditioned air from the ECS packs and distributes it throughout the aircraft. Rigid, semi-rigid and flexible ducts are used in the distribution system. The ducts are routed under the cabin floor and have separate distribution systems for the cabin and for the flight compartment.

The conditioned air is distributed to the cabin through lower dado panel vents, overhead vents and passenger gaspers.

A recirculation fan helps to boost the supply of conditioned air.

A gasper system distributes air to individual outlets in the cabin, flight compartment and the lavatory.

Conditioned air is distributed to the flight compartment through lower vents near the rudder pedals, through side console gaspers, and selectable vents for side window demisting. Conditioned air is supplied to the aft baggage compartment through an inlet valve.

An avionics cooling system supplies cooling for the avionics equipment bay and the EFIS/ESID liquid crystal displays.

An optional ground air conditioning connection is available to supply conditioned air from a ground source.

21-21-00 FLIGHT COMPARTMENT AIR DISTRIBUTION

INTRODUCTION

The flight compartment conditioned air distribution system supplies the flight crew with conditioned air.

GENERAL

Air supply to the flight compartment is ducted from the air conditioning pack, through the rear pressure bulkhead, to the left side of a splitter duct. From the splitter duct, the flight compartment air is ducted along the right side of the aircraft below the cabin floor.

At the flight compartment bulkhead, a wye duct splits the airflow into two individual but identical distribution systems, one for the pilot side and the other for the co-pilot side.

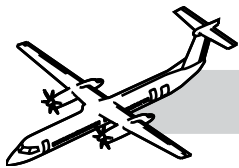
The ECS ECU monitors a supply duct temperature sensor; an overtemperature switch and a flight compartment temperature sensor as part of its control loop.

SYSTEM DESCRIPTION

Refer to Figure 21-50. General Air Distribution.

A duct, branching from the main flight compartment duct, supplies the aft baggage compartment inlet.

Aft of the flight compartment bulkhead, a duct branches off supplying the forward lavatory gasper and the forward flight attendant gasper.



COMPONENT DESCRIPTION

Flight Compartment Supply Ducts

Refer to Figure 21-51. Flight Compartment Supply Ducts.

Flight Compartment Distribution Ducts

The flight compartment distribution ducts consist of the following:

- Flex torso gasper duct
- Small gasper
- Trident duct
- Forward supply duct
- Under floor take-off duct
- Stub duct
- Foot warming supply duct
- Aft main supply duct LH
- Aft main supply duct RH
- Grille side flex duct.

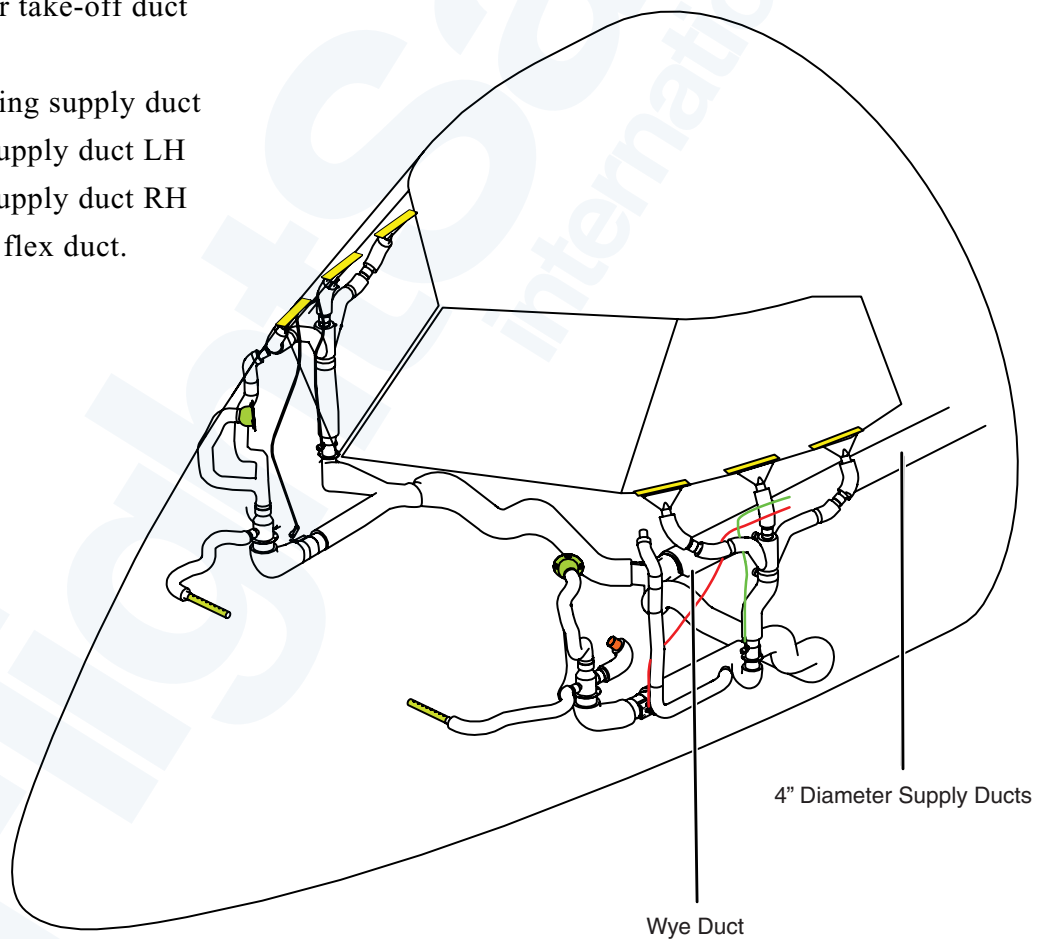
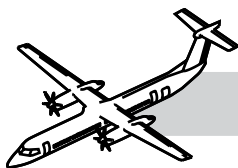


Figure 21-51. Flight Compartment Supply Ducts



OPERATION

Refer to Figure 21-52. Flight Compartment Ducting.

Selecting the SIDE WINDOW DE-MIST lever (3) on the F/C side wall to the OPEN position will open the de-mist flow control valve which will allow conditioned air to blow on the side window.

Selecting the LOW LEVEL VENTS lever (3) to the OPEN position will open the low level flow control valve which will allow conditioned air to blow out of:

- Foot warming piccolo tubes (near the rudder pedals),
- Fixed grilles (near knee height) and
- Large torso gaspers.

There are independent small gaspers near window height on the two sides of the flight compartment that are controlled by the knurled vent knobs on the gaspers.

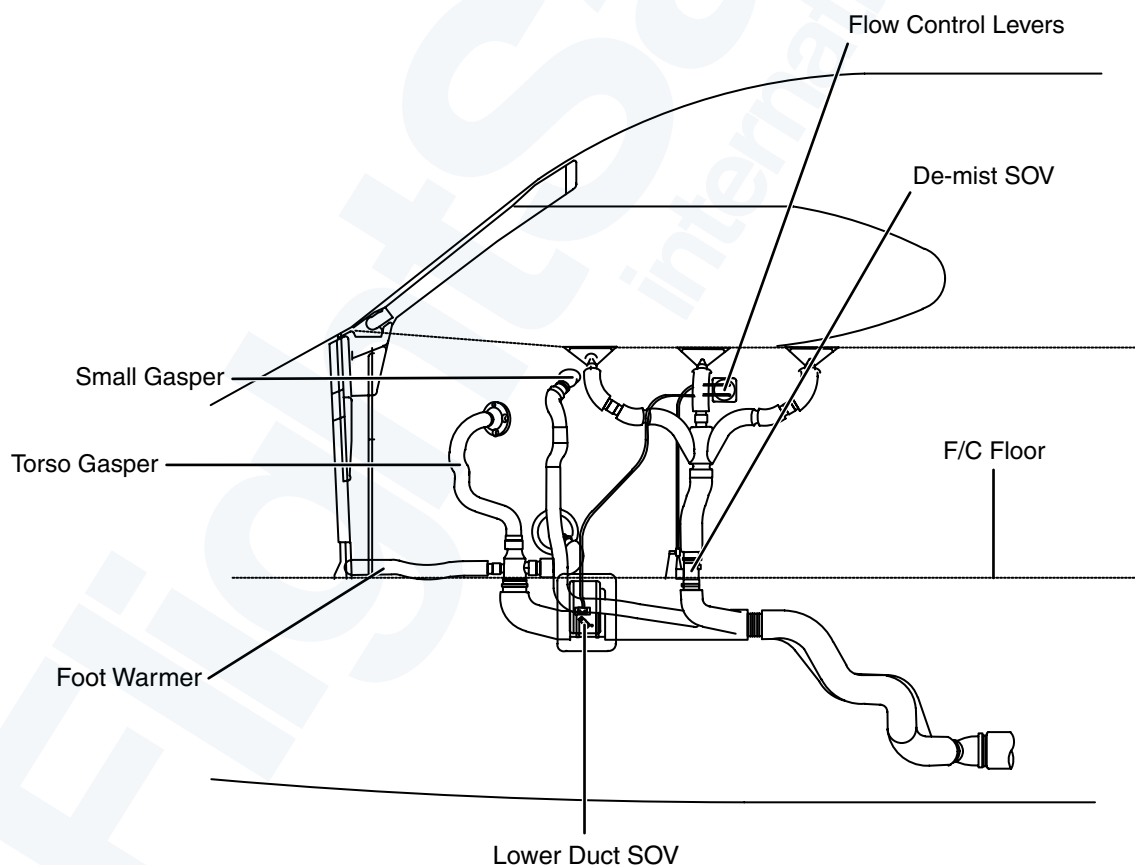


Figure 21-52. Flight Compartment Ducting

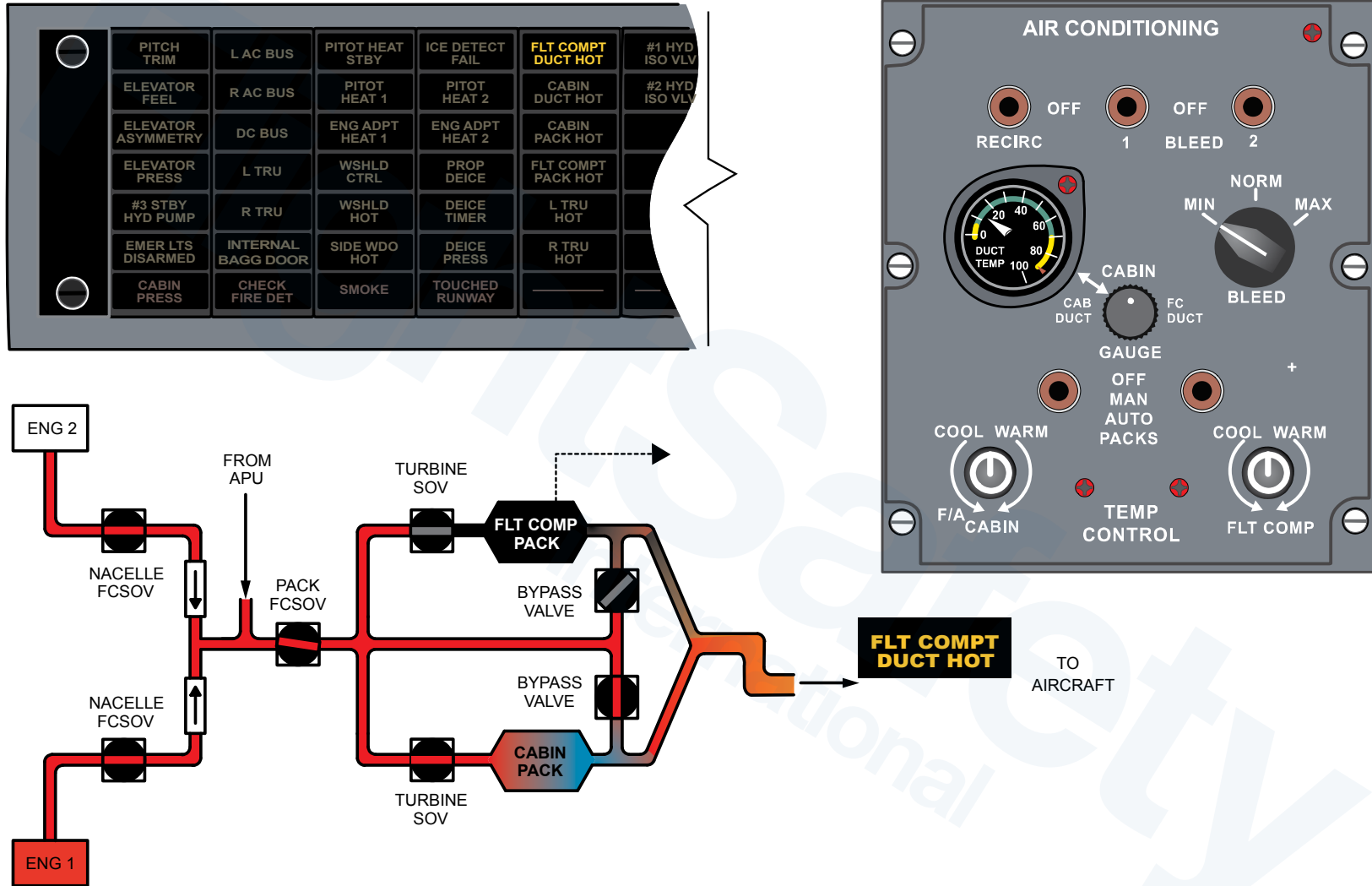
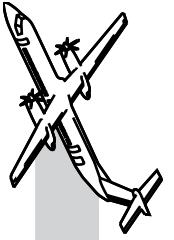
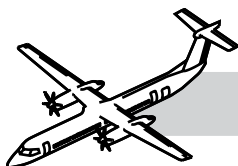


Figure 21-53. Flight Compartment Duct Hot Caution Light



FAULT DIAGNOSTICS

Refer to:

- Figure 21-53. Flight Compartment Duct Hot Caution Light.
- Figure 21-54. Flight Compartment Duct Hot Schematic.

Fault Scenario

Over-temperature is sensed during automatic operation

The FLT COMP DUCT HOT caution light illuminates during a flight. Maintenance interrogation of the CDS shows the message - FD OVRTMP SW.

The ECU receives a high F/C duct temperature 160°F (71°C) on connector J4 pins 2 and 3. The ECU will reduce the pack outlet temperature by modulating the ACM bypass valve towards close.

The ECU receives an over-temperature F/C duct temperature of 190°F (87.8°C) on connector J2 pin 59. The ECU will do the actions that follow:

- Close the turbine SOV
- Close ACM bypass SOV and
- Illuminate the FLT COMP DUCT HOT caution light.

NOTE

The FLT COMP DUCT HOT caution light will remain illuminated until the over-heat condition ends. A pack OFF/ON selection is required to RESET.

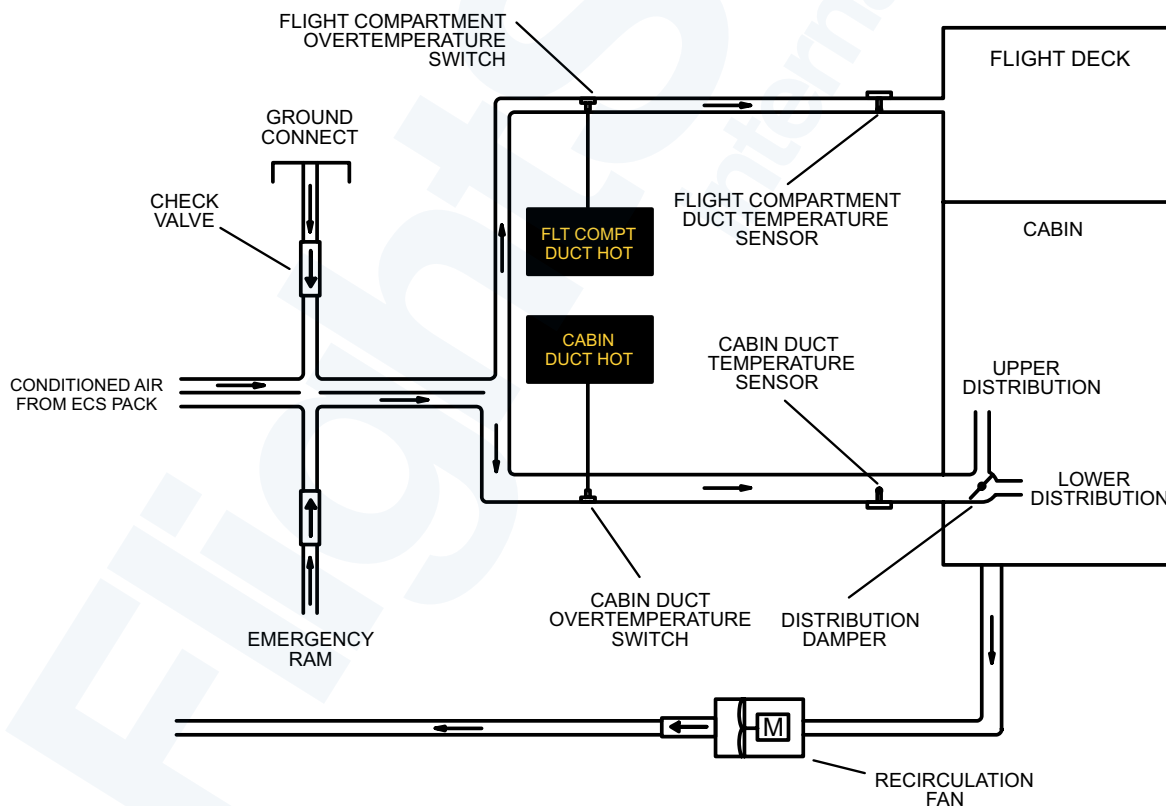
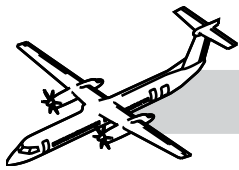


Figure 21-54. Flight Compartment Duct Hot Schematic



21-22-00 CABIN AND BAGGAGE AIR DISTRIBUTION

INTRODUCTION

Conditioned air is supplied to the cabin to maintain a comfortable environment for the passengers.

GENERAL

Refer to Figure 21-55. Cabin and Baggage Air Distributions.

The passenger cabin air distribution system divides the flow of conditioned air to the passenger cabin between the upper and lower duct systems. The air is ducted under the baggage compartment floor where it splits into an upper and lower supply duct for each side of the fuselage.

The ECU monitors temperature as part of the control loop.

SYSTEM DESCRIPTION

The cabin air system delivers air through a distribution damper to the aft flight attendant and galley gaspers.

The air is also routed to the:

- Cabin floor level vents
- Upwash and downwash vents and
- PSU gasper vents.

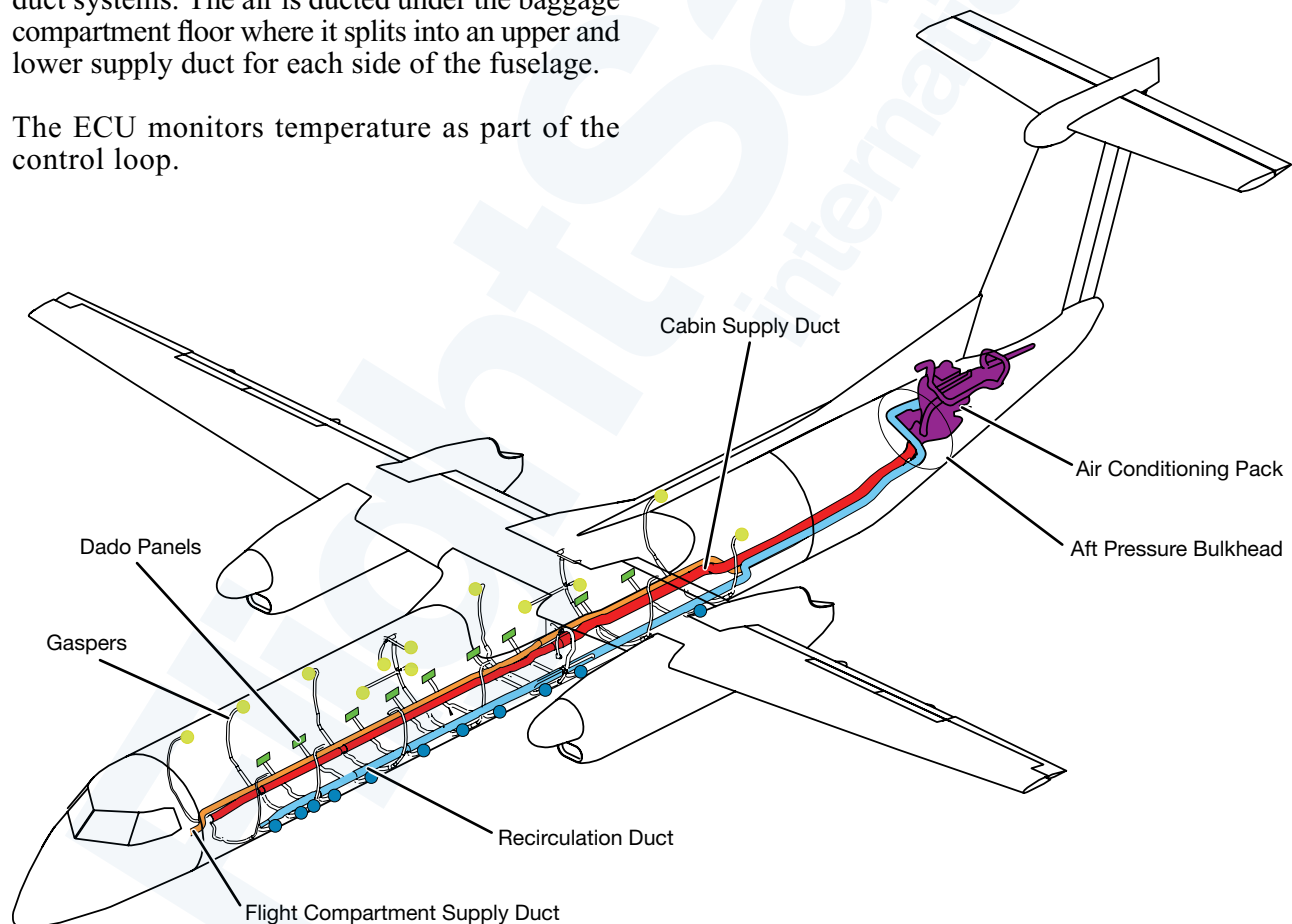
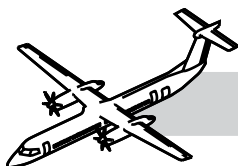


Figure 21-55. Cabin and Baggage Air Distributions



COMPONENT DESCRIPTION

Cabin Supply Ducts

Refer to Figure 21-56. Cabin Supply Ducts.

The cabin supply ducts consist of the following:

- Lower Cabin Supply Duct
- Distribution Damper
- Aft Galley Gasper
- Noise Treatment Duct
- Aft Flight Attendant Gasper
- Rear Pressure Bulkhead
- Upper Cabin Supply Duct
- Dado Riser
- PSU Gasper Riser
- Cabin Supply Duct.

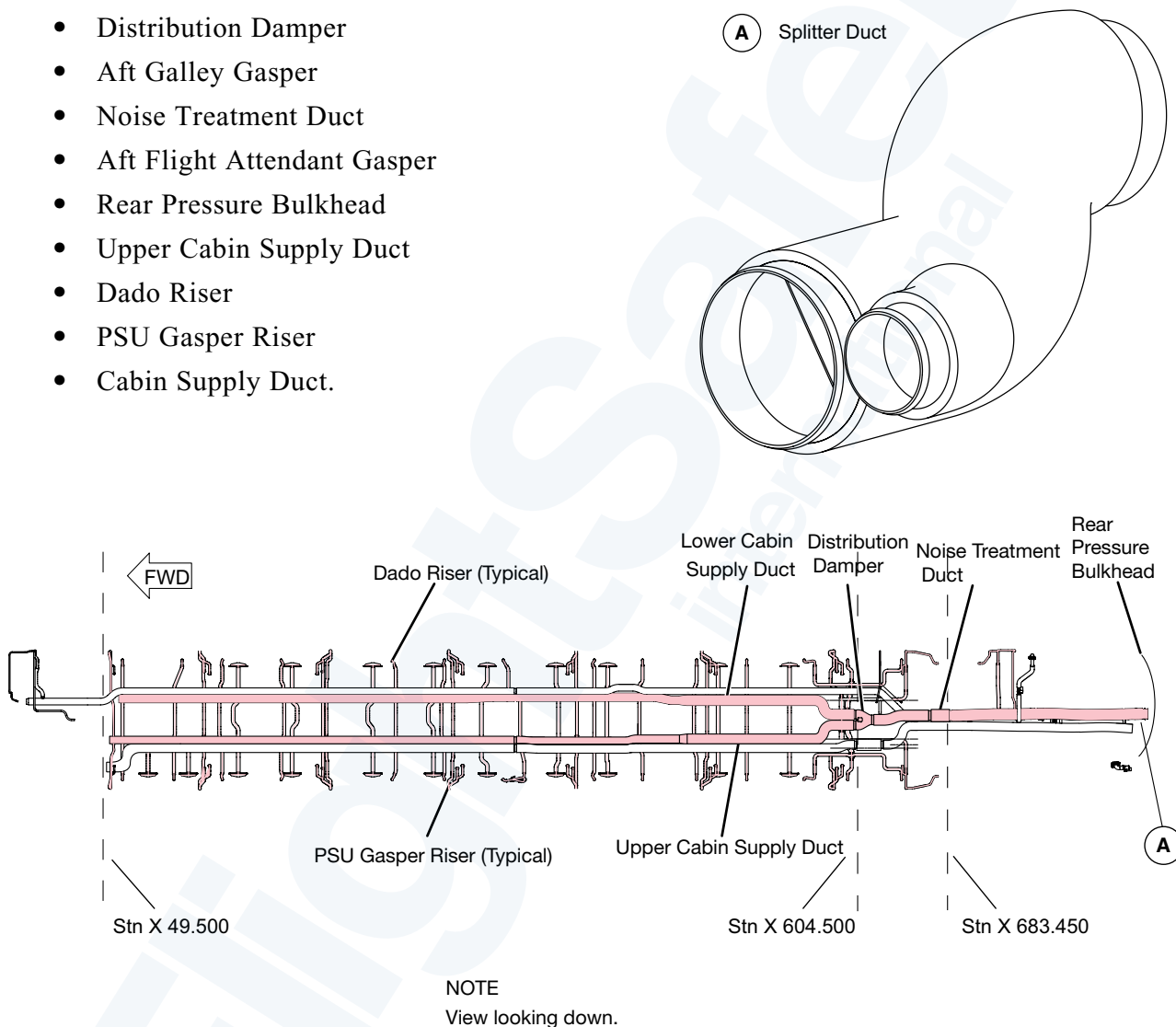
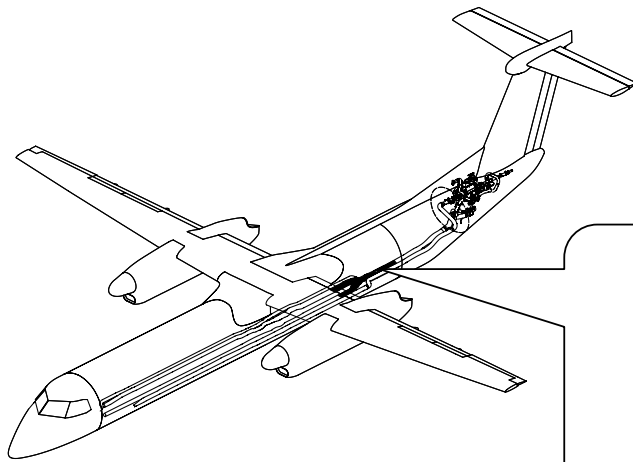
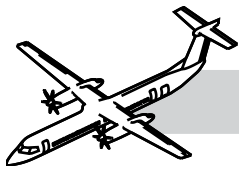
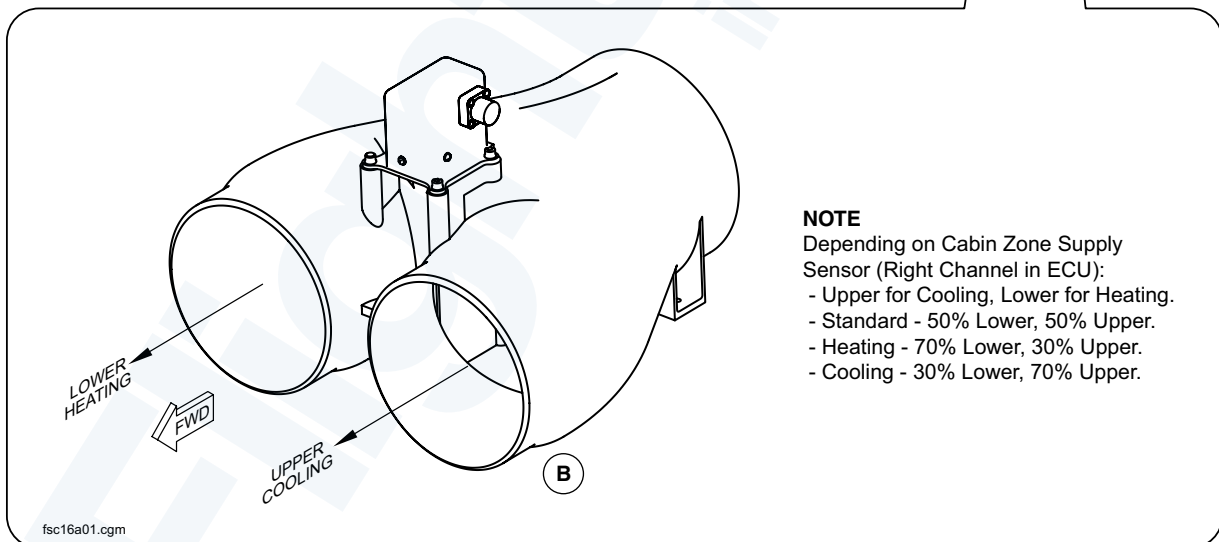
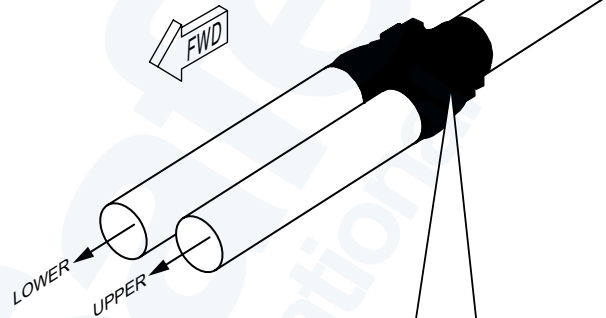


Figure 21-56. Cabin Supply Ducts



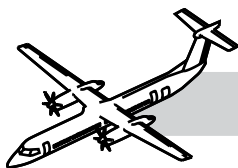
NOTE
If Right Channel or Damper Motor Fails - Valve Remains in Last Position.
ECU Logs Fail in CDS.

CABIN SUPPLY DUCT ASSEMBLY



NOTE
Depending on Cabin Zone Supply Sensor (Right Channel in ECU):
- Upper for Cooling, Lower for Heating.
- Standard - 50% Lower, 50% Upper.
- Heating - 70% Lower, 30% Upper.
- Cooling - 30% Lower, 70% Upper.

Figure 21-57. Distribution Damper Synoptic



DISTRIBUTION DAMPER

Refer to:

- Figure 21-57. Distribution Damper Synoptic.
- Figure 21-58. Distribution Damper.

The distribution damper, installed just aft of station X604.500, adjusts the amount of air being delivered to the upper and lower duct systems.

It contains a three-position electrically actuated valve. The left-hand side supplies the lower distribution system, for the heating mode. The right-hand side outlet supplies the upper distribution system, for the cooling mode.

The distribution damper function is controlled by the right digital channel of the ECU depending on the cabin zone supply temperature and cabin temperature request.

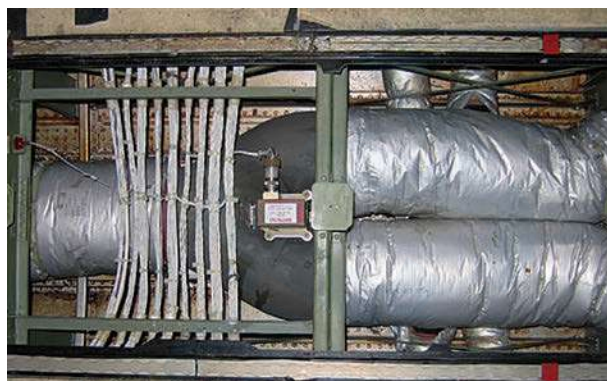
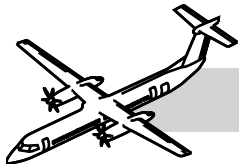


Figure 21-58. Distribution Damper



Distribution-Cabin Ducts

Refer to:

- Figure 21-59. Distribution Cabin Ducts (1 of 2).
- Figure 21-60. Distribution Cabin Ducts (2 of 2).

The conditioned air is distributed to the cabin through lower dado panel vents, overhead vents, passenger gaspers, galley gaspers and lavatory gaspers.

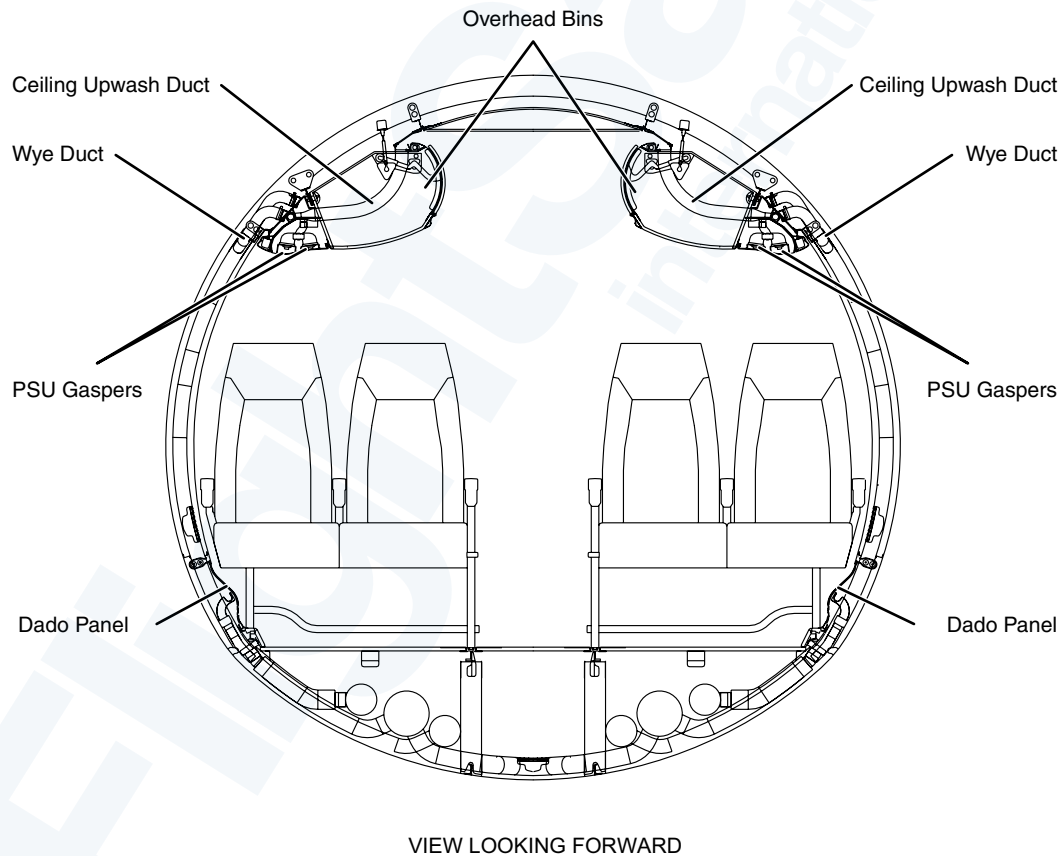
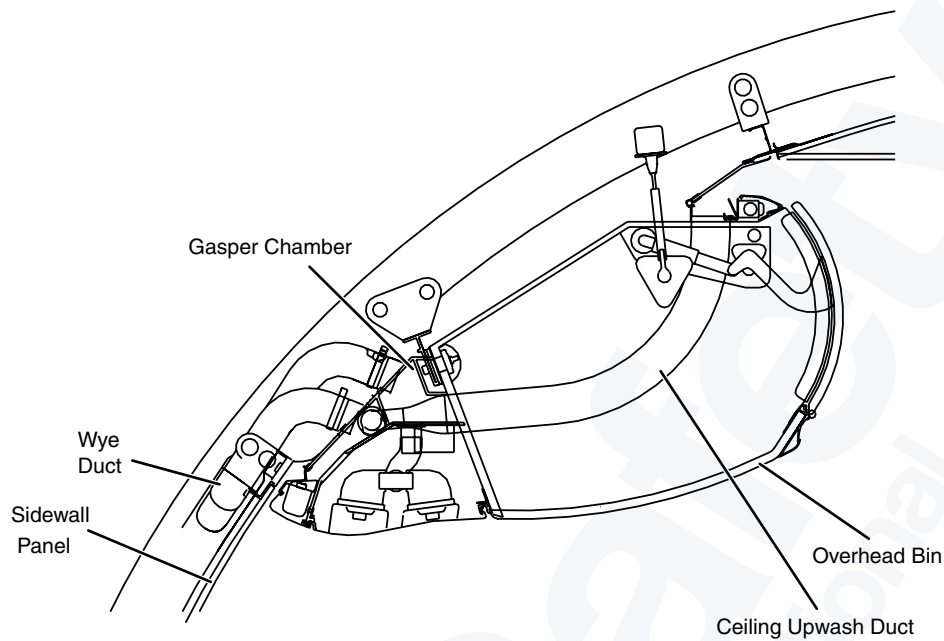
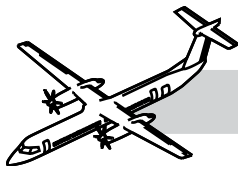
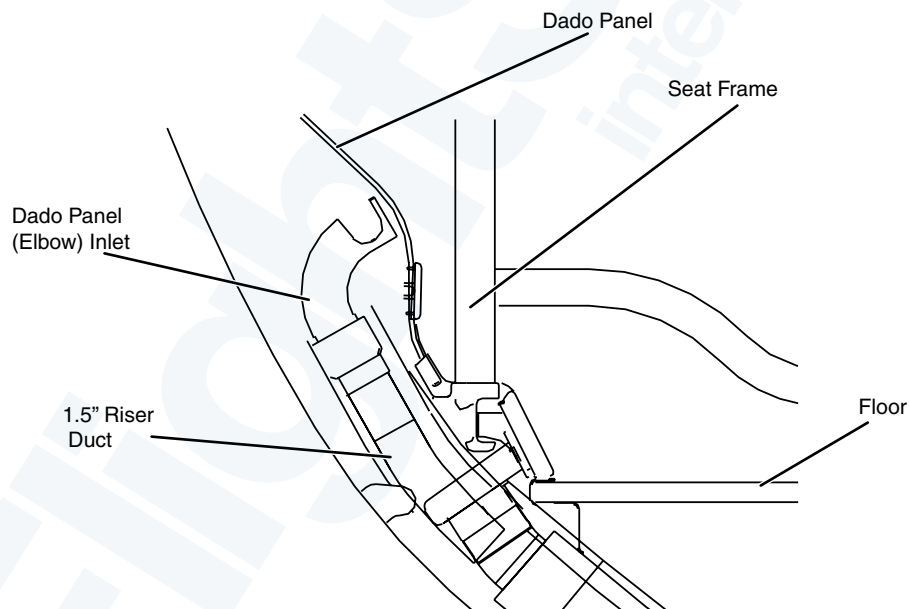


Figure 21-59. Distribution Cabin Ducts (1 of 2)



CEILING AIR DISTRIBUTION DETAIL



LOWER AIR DISTRIBUTION DETAIL

Figure 21-60. Distribution Cabin Ducts (2 of 2)

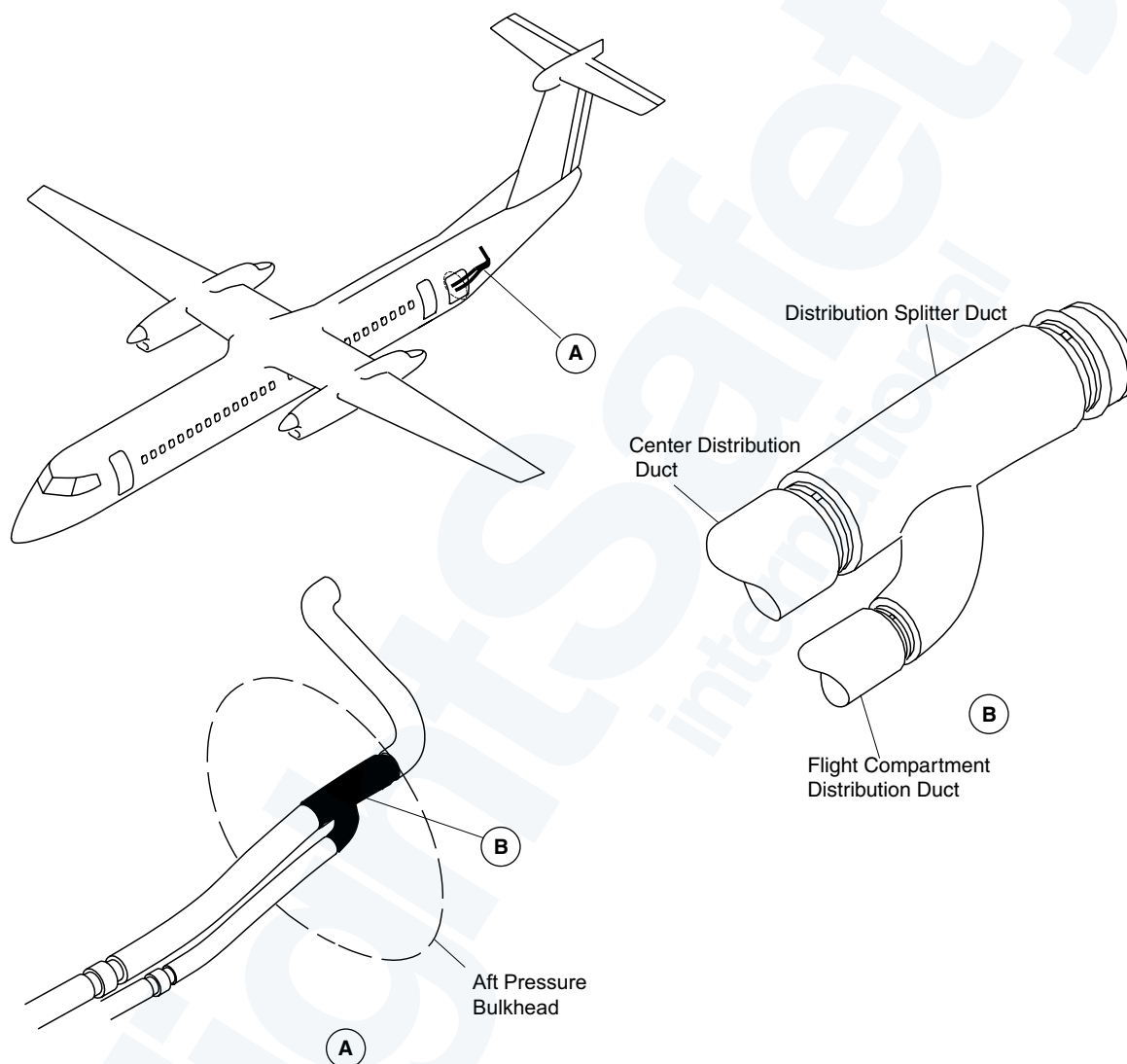
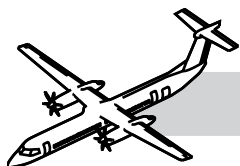
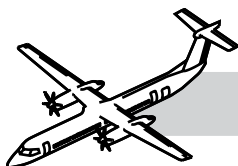


Figure 21-61. Splitter Duct - Distribution



Splitter Duct - Distribution

NOTES

Refer to Figure 21-61. Splitter Duct - Distribution.

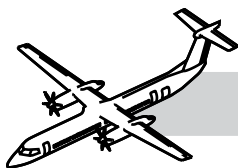
The flight compartment and cabin flows are separated by the splitter duct forward of the pressure bulkhead.

The splitter duct is a semi-rigid bifurcated duct made from silicone impregnated glass cloth. The left side of the bifurcation supplies flight compartment air while the right side supplies cabin air.



21 AIR CONDITIONING





Aft Baggage Compartment Vent Valves

NOTES

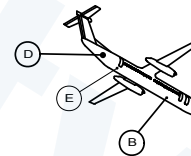
Refer to Figure 21-62. Aft Baggage Vent Valves.

The aft baggage compartment vent valves are installed in the aft baggage compartment.

One valve is installed at the inlet supply below the floor on the right hand fuselage. The other vent is installed at the outlet on the left hand fuselage.

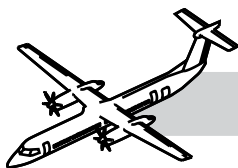
These valves allow the supply of conditioned air to the baggage compartment when the ECS packs are in operation.

When smoke is detected in the aft baggage compartment, the control amplifier will remove electrical power from the aft baggage vent inlet and vent outlet shutoff valves. This will minimize the air flow within the baggage compartment. Indication of vent valve status will be shown as VENT INLT and VENT OTLT lights on the Fire Protection Panel.



21-82

FOR TRAINING PURPOSES ONLY



OPERATION

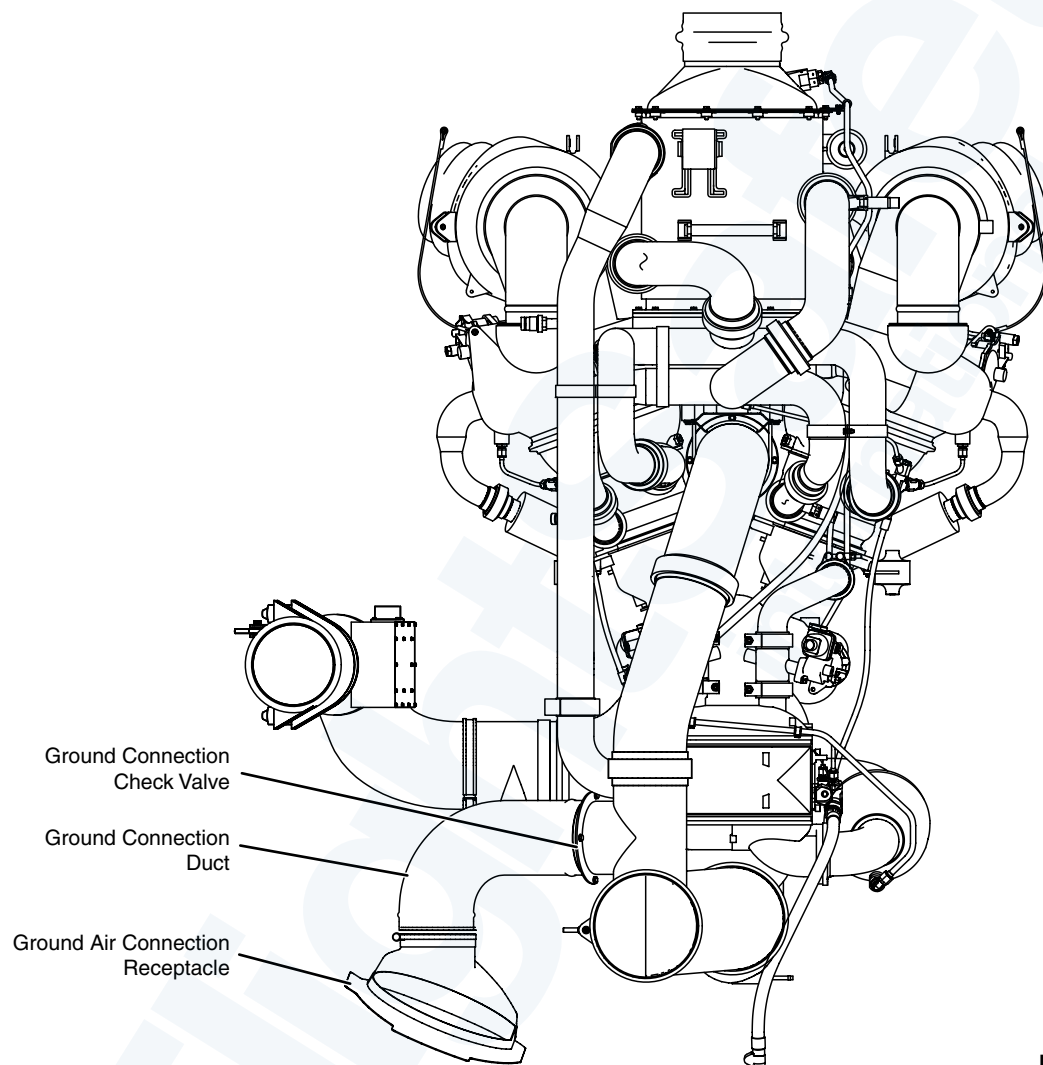
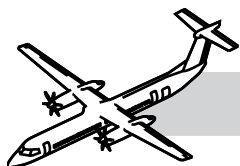
NOTES

Normal System Operation

Refer to Figure 21-63. Cabin Air Distribution Electrical Schematic.

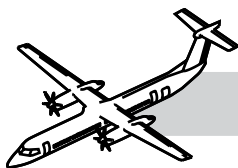
Operation of the switches that follow on the AIR CONDITIONING control panel operate the ECS:

- The BLEED switches are set to the 1 and/or 2 position.
- A PACKS switch is set to the AUTO position.
- The distribution damper is controlled by the ECU (connector J5 pins 29, 52 and 68). The damper controls the airflow in the positions that follow:
- Warm operation; 70% of the warm air goes to the lower vents and 30% goes to the overhead vents.
- Cooling operation; 30% of the airflow goes to the lower vents and the overhead receives 70%.
- At standard temperatures (55 to 95°F (12.8 to 35°C)) air is directed evenly to the overhead and lower vents.
- If the right digital channel in the ECU fails, the damper valve can not change position.



NOTE
View looking aft.

Figure 21-64. Ground Connection Check Valve and Duct



21-23-00 GROUND CONNECTION

INTRODUCTION

The ground connection provides a means of supplying conditioned air to the ECS distribution system.

GENERAL

An external source of conditioned air is attached to the ground connection and the air flow is sent to the air conditioning supply ducts.

The ground connection system has the following components:

- Ground connection
- Ground connection check valve and
- Ground connection ducts.

COMPONENT DESCRIPTION

Ground Connection

Refer to:

- Figure 21-64. Ground Connection Check Valve and Duct.
- Figure 21-65. Ground Connection - Detail.

The ground connection is a duct, accessed through a circular shaped, hinged, and sealed door, located at the rear fuselage right hand side.

The connection point has a cover to prevent contamination from entering the system.

The 5" diameter check valve prevents pressurized air from the air conditioning packs or the emergency ram air system from venting overboard. When there is no ground conditioned air pressure available, the check valve is held closed by a spring to prevent dirt from entering the system.

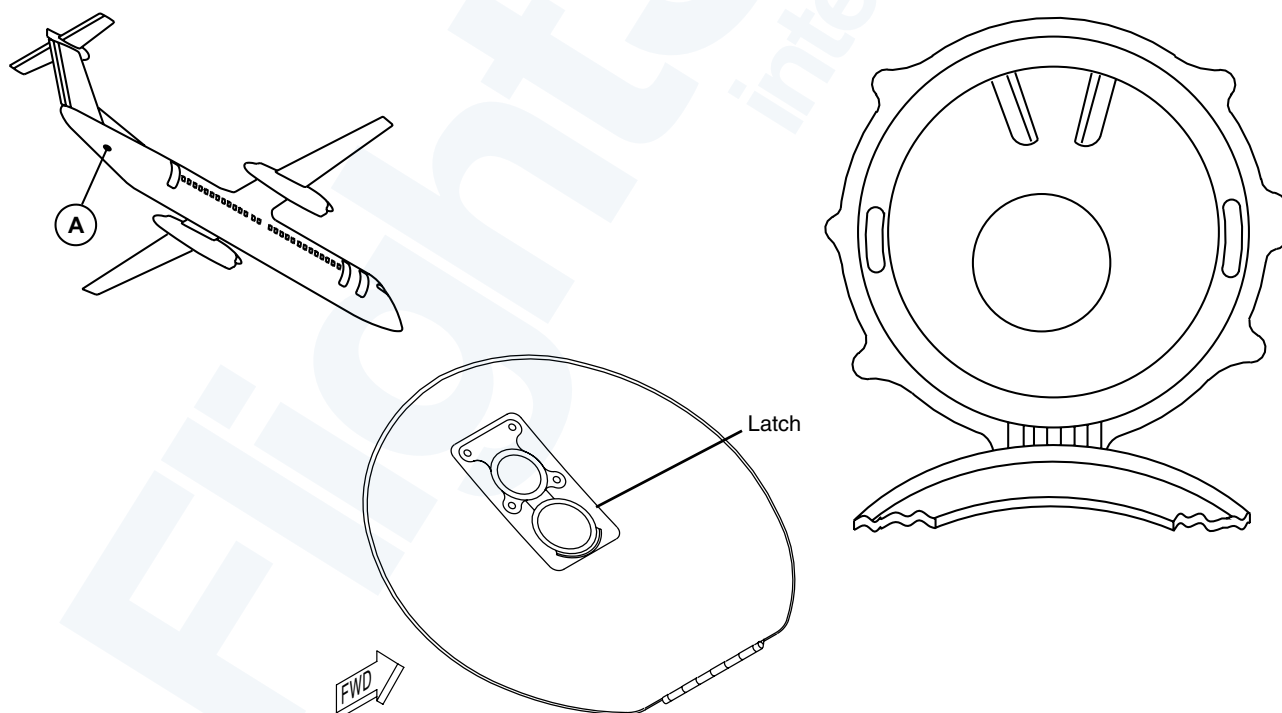
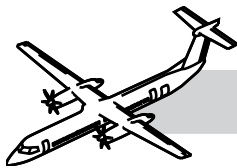


Figure 21-65. Ground Connection - Detail



21-24-00 RAM AIR SYSTEM

INTRODUCTION

The emergency ram air system supplies ambient air to the flight and cabin compartments in the event of a failure of the air conditioning packs.

SYSTEM DESCRIPTION

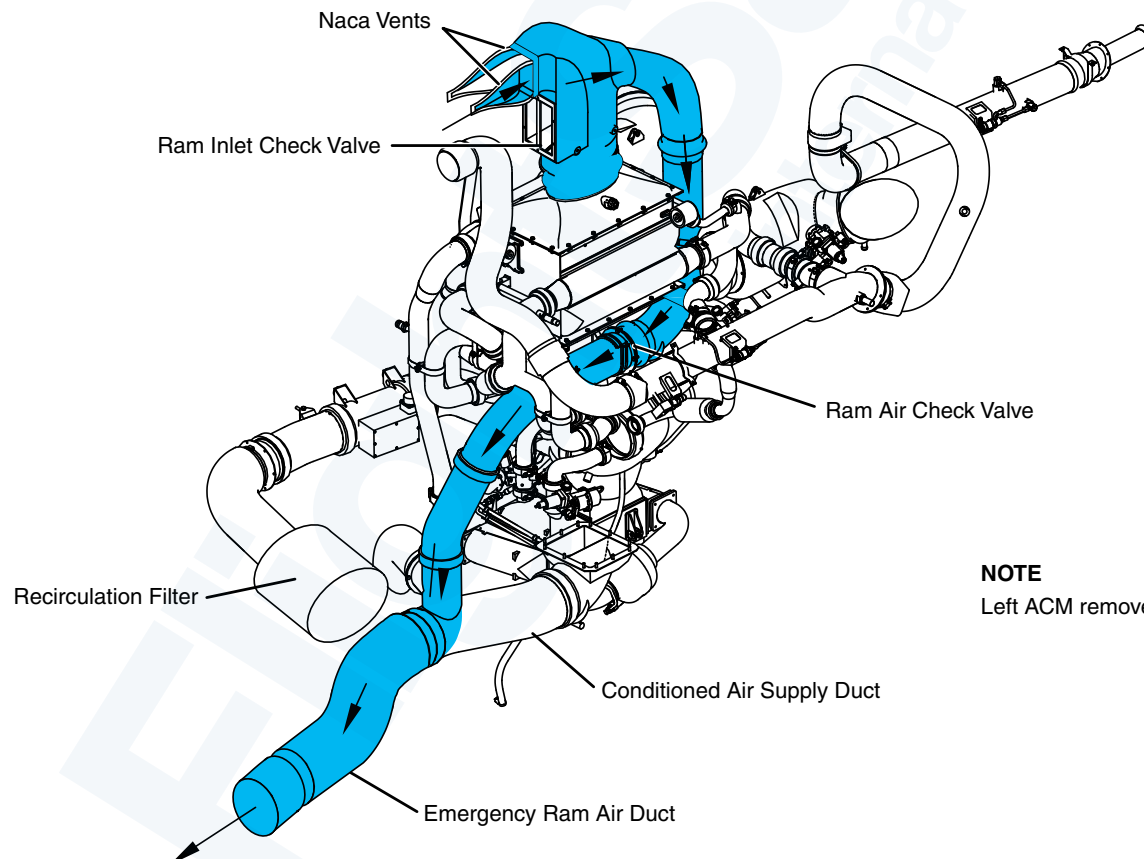
Refer to Figure 21-66. RAM Air System.

There are two functions of the ram air system:

- Supply cooling ram air to the dual heat exchanger and
- Supply emergency ram air to the cabin and flight compartments in the event of a failure of the air conditioning packs.

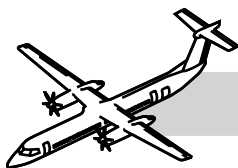
In flight, ram air enters the aircraft through the NACA vents at the base of the dorsal fin. It flows through the ram air ducting to the dual heat exchanger where it cools the bleed air entering the air conditioning packs.

If there is no air flow from the air conditioning packs, the pressure from the ram air is enough to overcome the spring in the emergency ram check valve. The check valve will open allowing cooling ram air to flow directly into the cabin and flight compartment supply ducts.

**NOTE**

Left ACM removed for clarity.

Figure 21-66. RAM Air System



Refer to Figure 21-67. RAM Inlet Check Valve.

During ground operations, when there is no ram effect, the airflow through the NACA is insufficient for cooling. The ACM fans continue to draw in air through the ram air inlet creating a negative pressure in the ram inlet. This negative pressure opens a rectangular-shaped ram inlet check valve to let in additional cooling air.

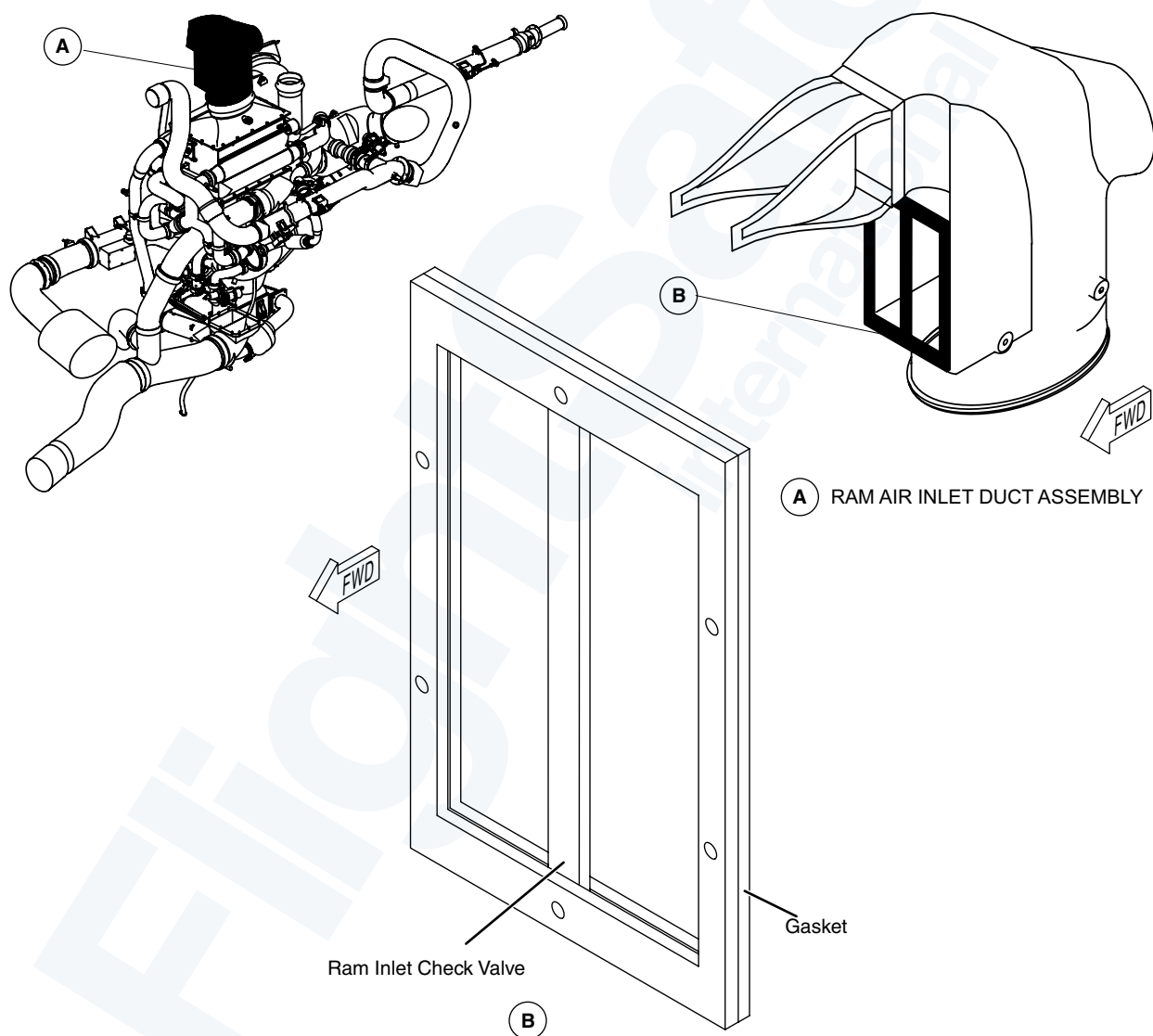
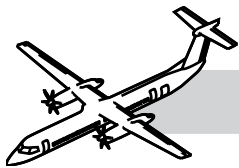


Figure 21-67. RAM Inlet Check Valve



COMPONENT DESCRIPTION

Refer to Figure 21-68. Emergency Ram Air Check Valve and Ducts.

Ram Air Check Valve

The emergency ram air check valve is a circular dual flapper valve installed between the upper and mid emergency ram air ducts.

The valve prevents loss of air through the ram ducting during pack operations. When the packs are off, the valve opens in flight, to provide emergency ram ventilation.

If the check valve can move freely by hand against the spring pressure, it is considered serviceable; otherwise it has to be replaced.

Ram Air Ducts

A series of four ducts in the aft fuselage carry air from the NACA vents to the flight and cabin compartments.

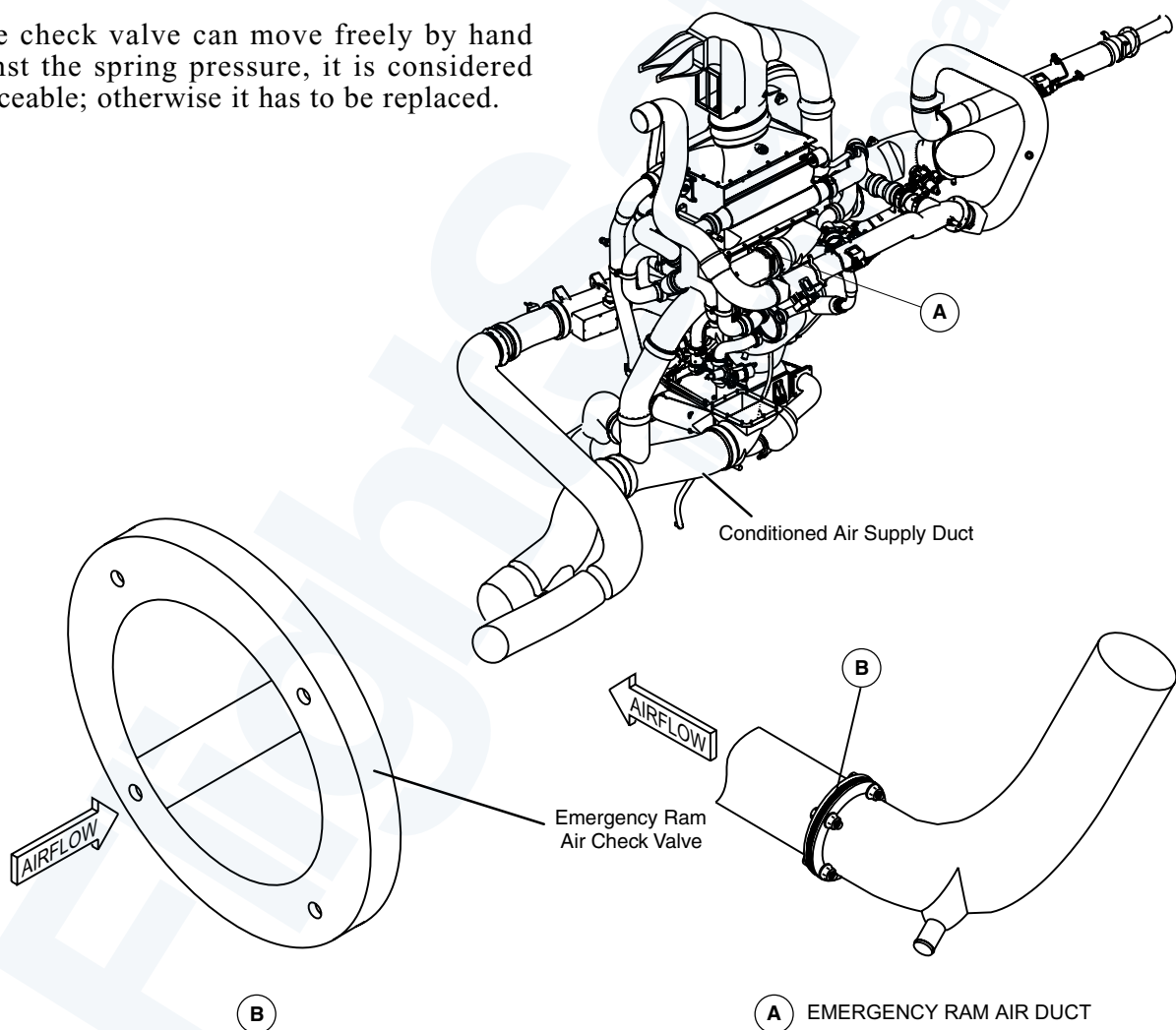
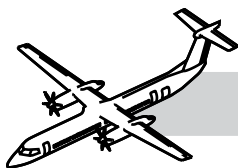


Figure 21-68. Emergency Ram Air Check Valve and Ducts



OPERATION

Refer to Figure 21-69. Emergency Operation.

Normal Operation

Ram air flows through the ducting, then through the dual heat exchangers to cool the engine bleed air in the Packs.

No Pack Operation

Ram air pressure opens the emergency ram check valve and the ram air flows into the cabin and flight compartment supply ducts.

Ground Operation

With no ram air flow, the ACM fans draw air through the ram air ducts.

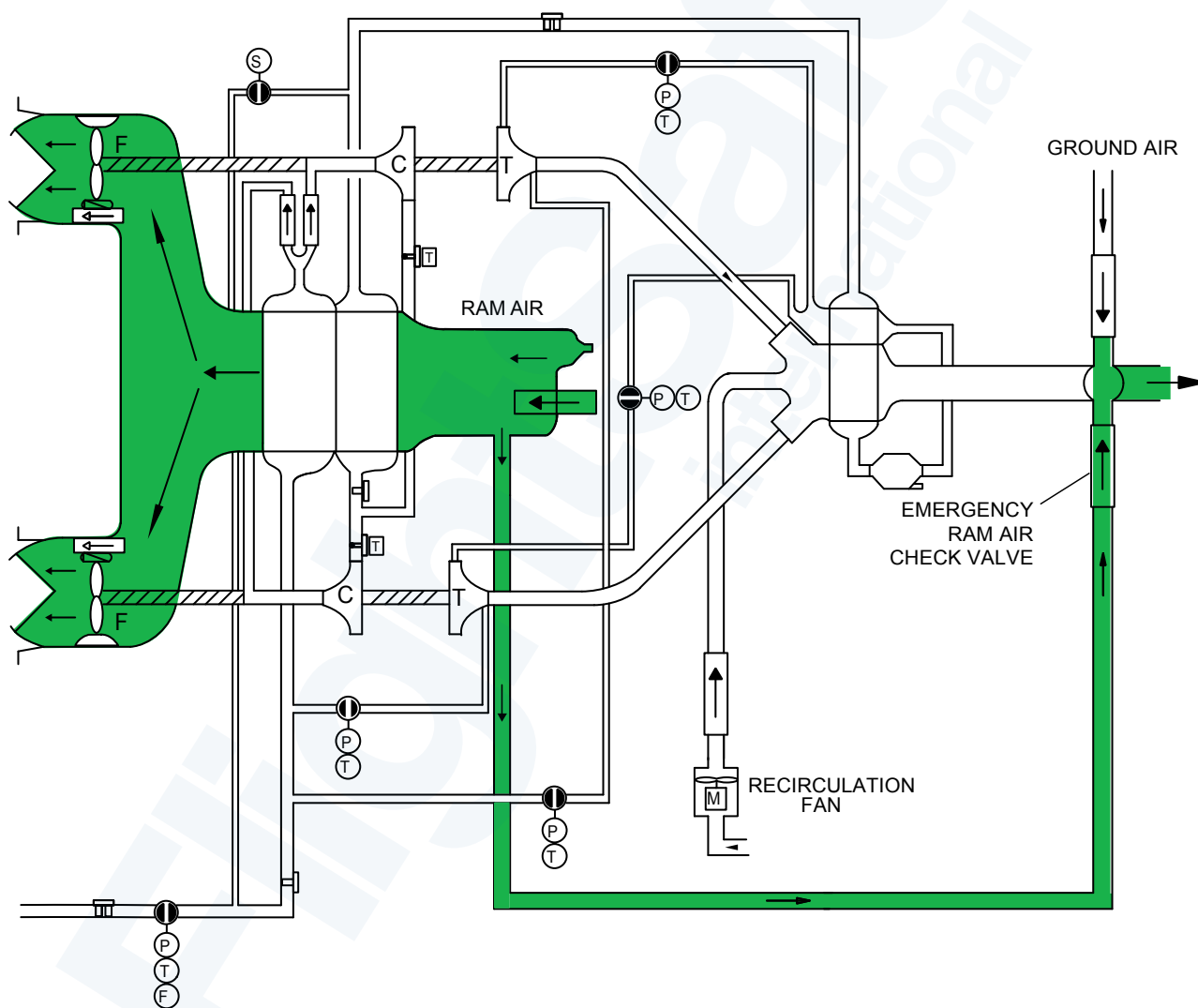


Figure 21-69. Emergency Operation

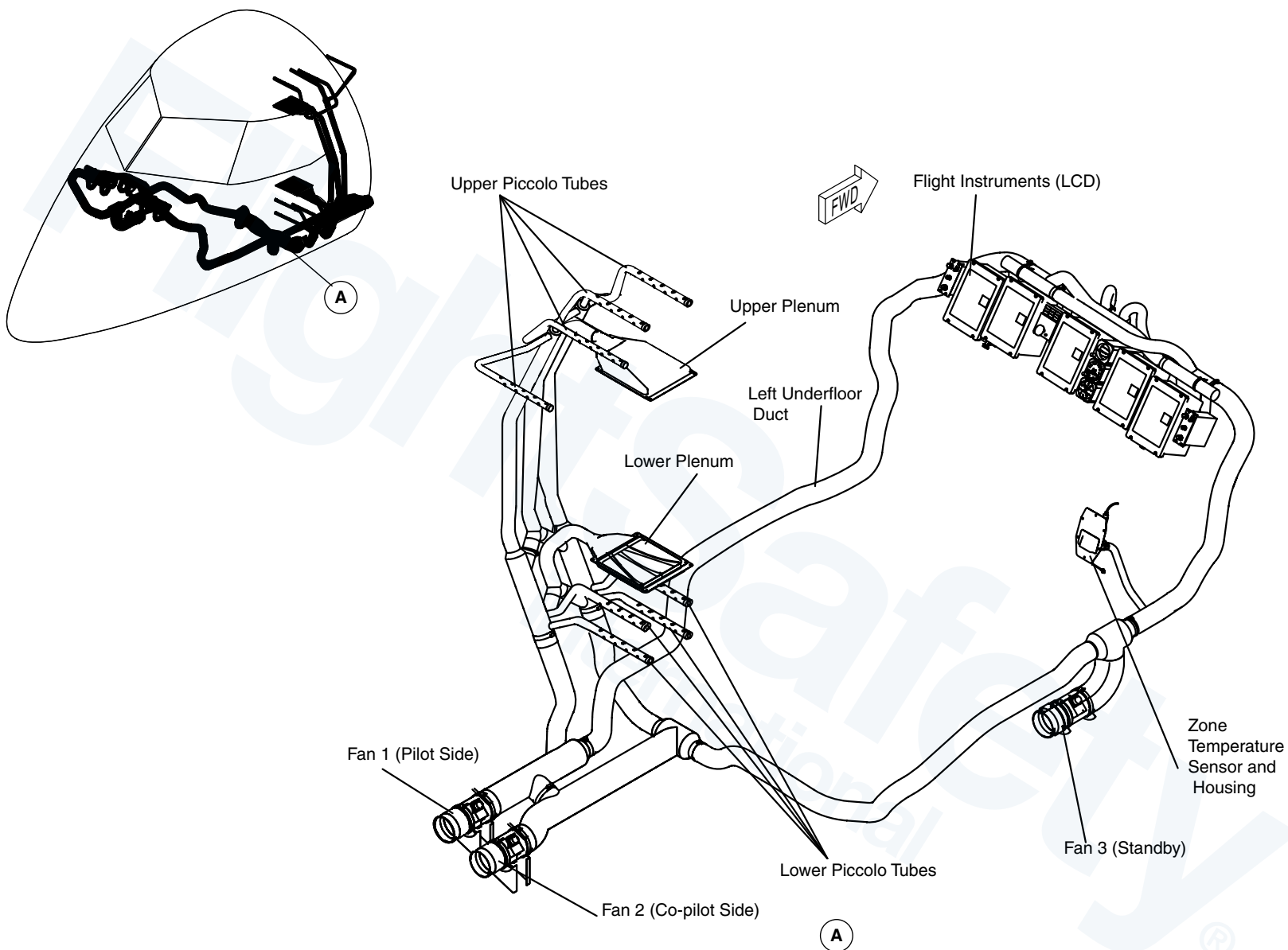
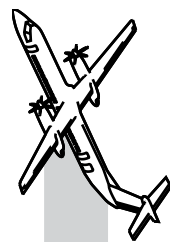
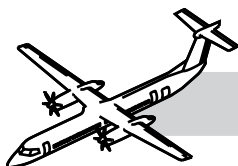


Figure 21-70. Avionics Equipment and Display Unit Cooling System Ducts



21-25-00 AVIONICS COOLING

NOTE

INTRODUCTION

The avionics cooling system has two independent duct networks and three fans. The system removes heat from the avionics rack, the wardrobe rack, and the five liquid crystal displays (LCDs).

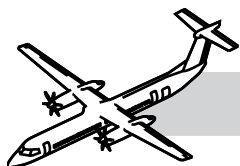
GENERAL

Refer to Figure 21-70. Avionics Equipment and Display Unit Cooling System Ducts.

The aircraft has an extraction type cooling system for the avionics and LCDs.

The cooling system has three identical fans. Each fan can supply half the required cooling flow.

The electrical and ducting systems have been designed so that single failures do not result in the loss of the DUs.



SYSTEM DESCRIPTION

Refer to Figure 21-71. Avionics and LCD Cooling System.

The avionics cooling system in the flight compartment has dual redundant duct assemblies. Each duct assembly has a cooling fan and branches of smaller ducts that are connected to the 5 LCDs, avionics rack and the wardrobe rack.

There are three fans: Fan 1, Fan 2 and Fan 3, that supply airflow in the two avionics cooling ducts.

The avionics cooling fan has an electronically-controlled two speed brushless DC motor.

Each fan has a built-in electronic LSWD that sends a signal to the IOM when the fan is rotating slower than 5000 rpm.

A fan speed control unit is installed on the side wall of each of the three avionics cooling fans.

One of the duct assemblies is routed to the instrument panel from the pilot side and the other assembly is routed from the co-pilot side.

The duct assembly on the co-pilot side also has the standby fan No.3 attached. The hot air is exhausted under the floor behind the flight compartment.

Fan 1 and fan 2 are normally running whenever the electrical power is applied to the aircraft DC main bus.

If either of these fans fails, the standby fan 3 automatically starts operating.

The operational time of the equipment without forced air-cooling is limited to 60 minutes because of the location and the high-density design of the LCD displays.

Operation of the fans is inhibited on the ground when the flight compartment temperature is below 41°F (5°C).

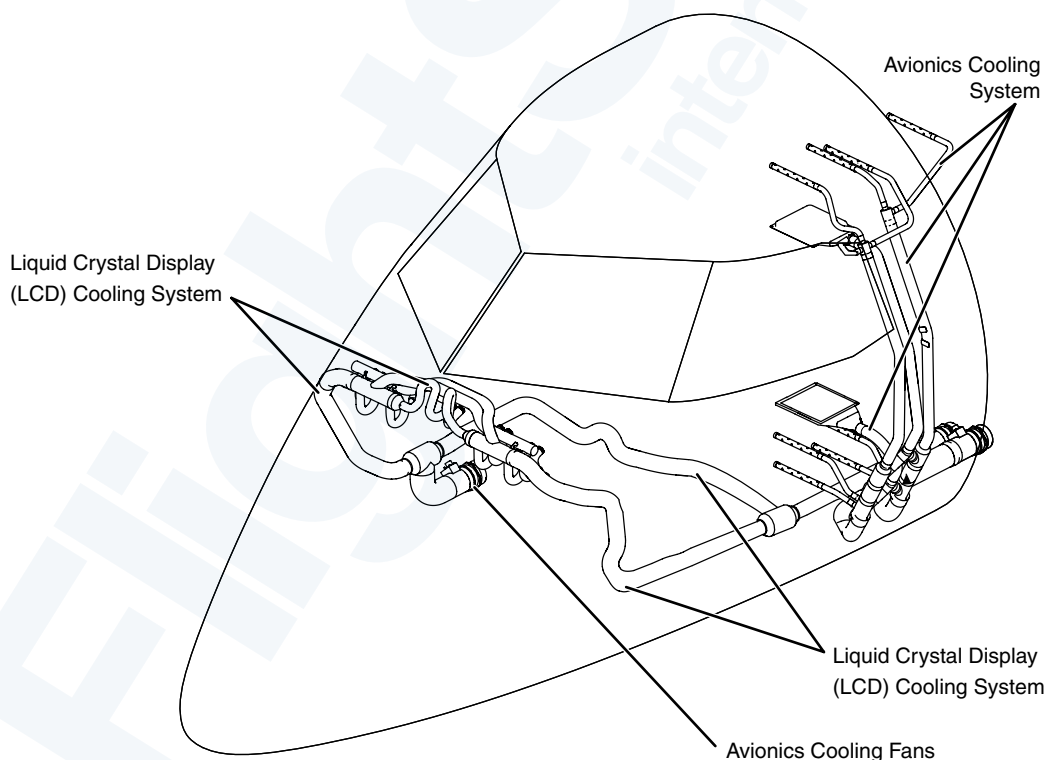
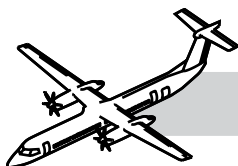


Figure 21-71. Avionics and LCD Cooling System



COMPONENT DESCRIPTION

Avionic Cooling Fans

Refer to Figure 21-72. Cooling Fans.

Two cooling fans are installed in the fuselage aft of the pilot bulkhead and one is installed on the right side of the flight compartment below the floor.

The avionics cooling fans draw warm air away from the display units and the Avionics rack. They also supply a constant flow of air across the flight compartment zone temperature sensor. Only two of the three fans are required to be operational for dispatch.

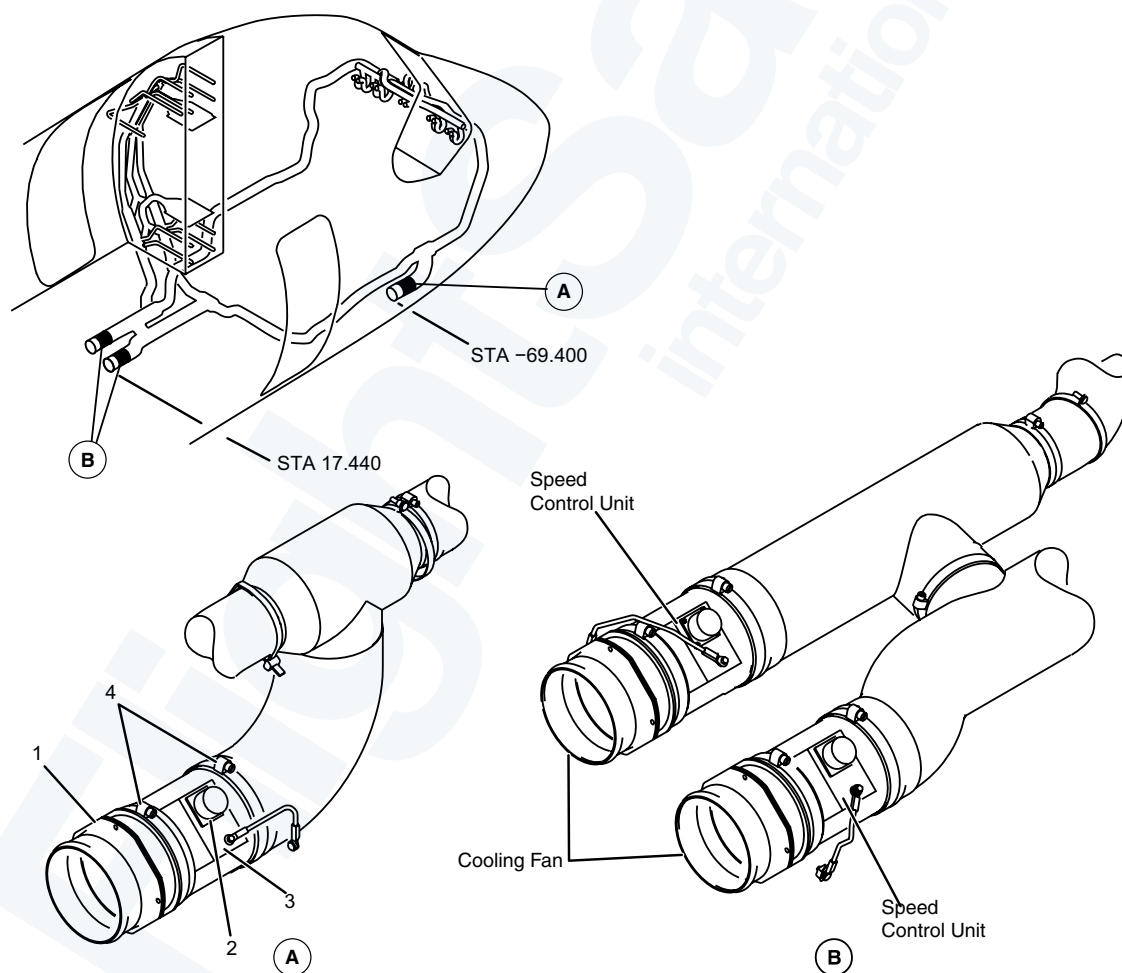


Figure 21-72. Cooling Fans

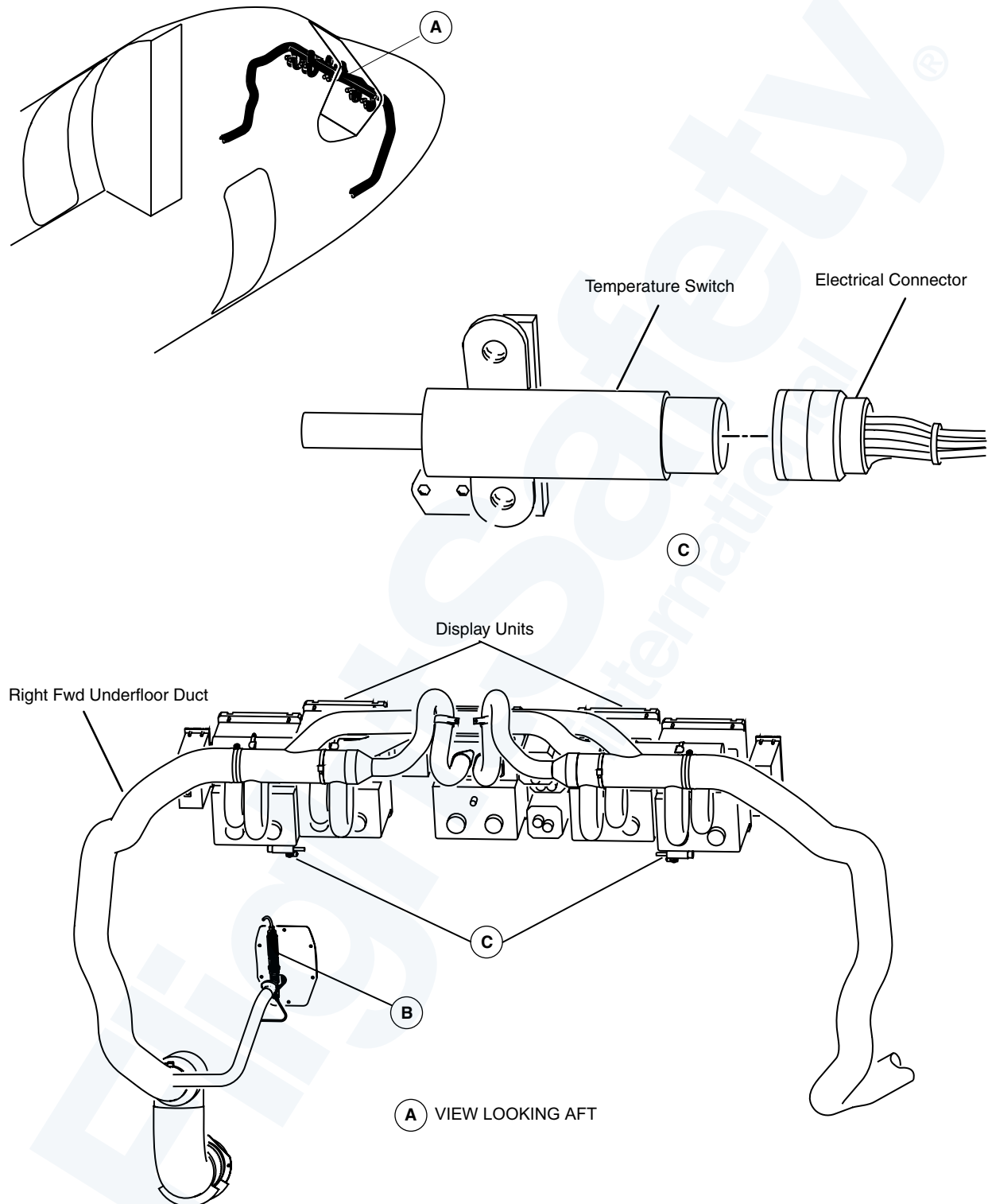
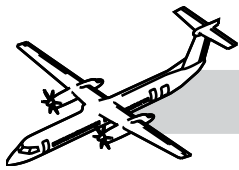
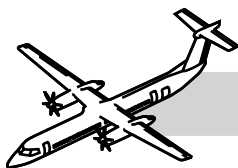


Figure 21-73. Zone Temperature Switches



Zone Temperature Switches

NOTES

Refer to Figure 21-73. Zone Temperature Switches.

There are two zone temperature switches located under the LCDs. When the flight compartment temperature is:

- Below 41°F (5°C) - Fan operation is inhibited if the aircraft is on the ground
- Below 55°F (10°C) - Fault monitoring is inhibited
- Above 86°F (30°C) - Fan will go high speed if the aircraft is post Modsum 4-126287.

NOTE

The cooling fans will not operate if the flight compartment temperature is less than 46°F (8°C).

The failure monitoring system will not operate if the flight compartment temperature is less than 55°F (13°C).

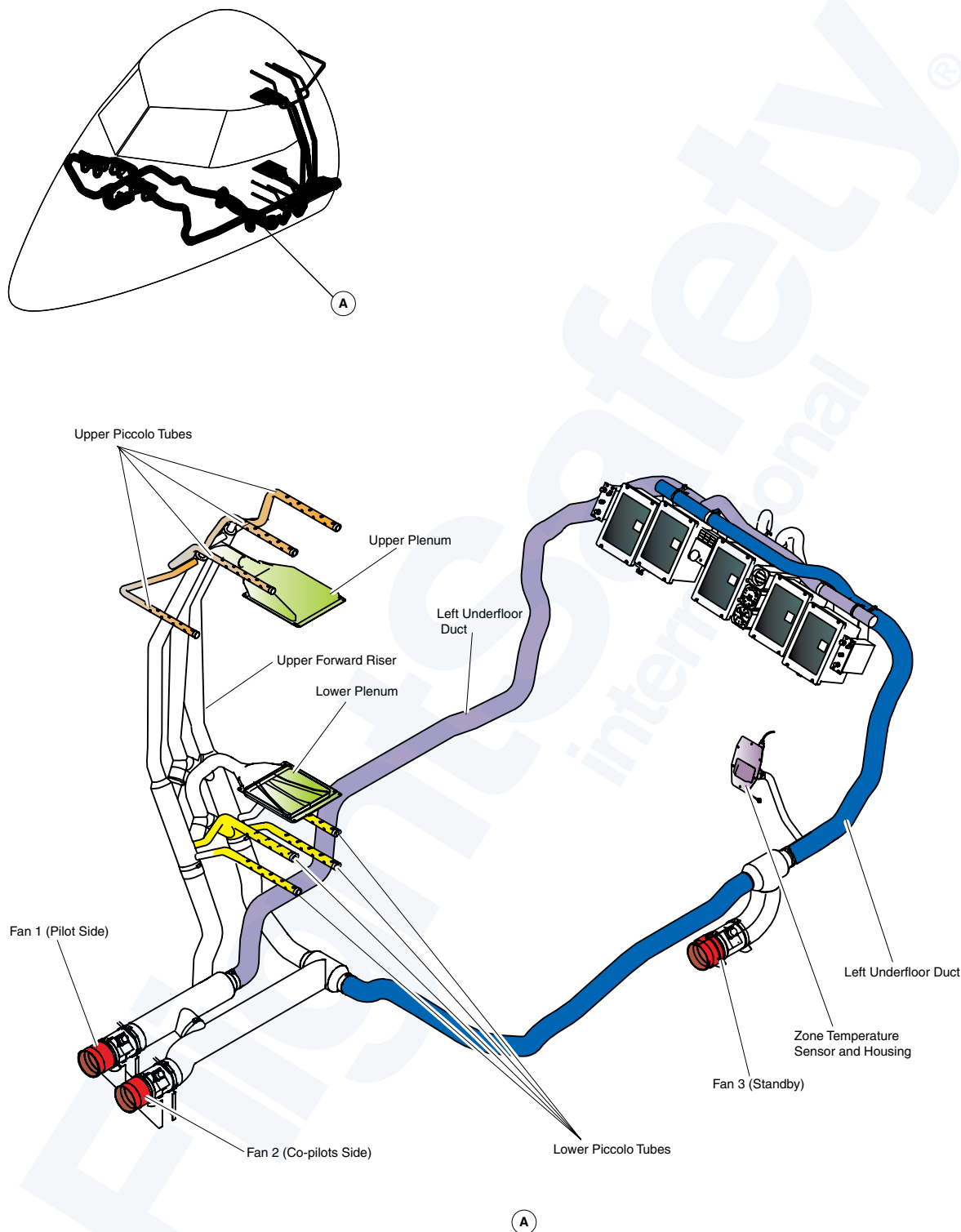
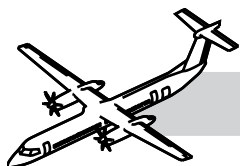
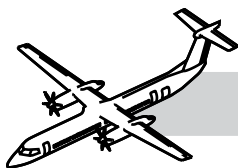


Figure 21-74. Avionics and LCD Cooling System Ducts



Avionics and LCD Cooling System Ducts

Refer to:

- Figure 21-74. Avionics and LCD Cooling System Ducts.
- Figure 21-75. Front Fuselage LCD Cooling Filters.

The dual redundant duct assemblies are as follows:

- Lower forward riser duct
- Upper forward riser duct
- Lower aft riser duct
- Upper aft riser duct
- Lower Y-ducts
- Lower plenum
- Upper Y-duct
- Upper plenum
- Left underfloor duct
- Left double tee fitting
- Left manifold duct
- Right underfloor duct
- Tee fitting and Elbow
- Right double tee fitting
- Right manifold duct.

The duct assemblies used above in the avionics cooling system are rigid, semi-rigid and flexible type ducting.

One assembly is routed from fan 1 below the cabin floor to the instrument panel on the pilot side, and behind the avionics rack and wardrobe shelf.

The second duct assembly is routed from fan 2 below the cabin floor to the instrument panel on the co-pilot side.

The duct assembly on the co-pilot side contains the standby fan located below the flight compartment floor.

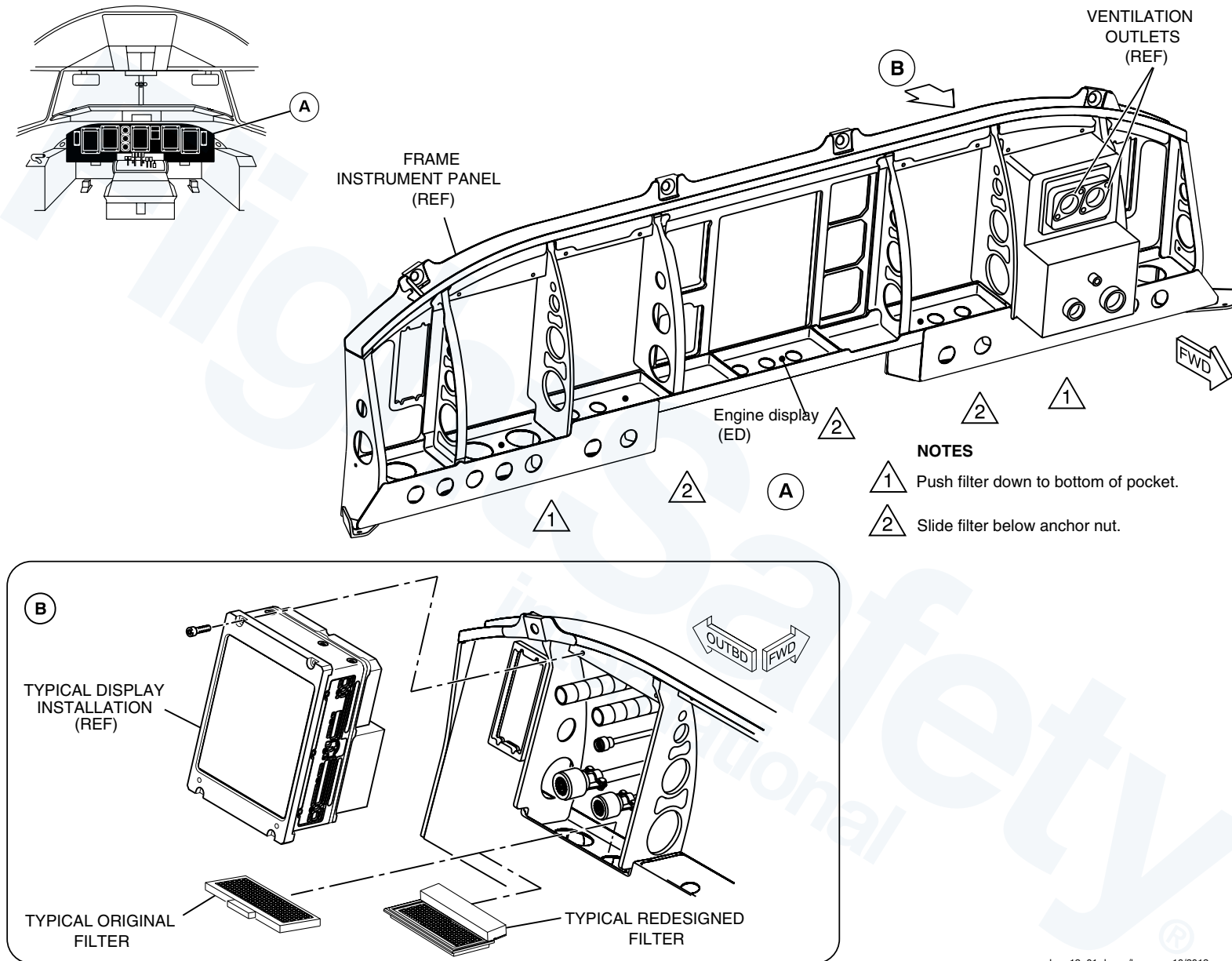
Each duct assembly alone can supply enough cooling for continuous operation of the DUs.

The hot air is exhausted into the exit ducts that direct the air to the rear of the baggage compartment liner.

Display Unit Filter Maintenance Practices

The filter at the bottom of each LCD is used to ensure the cooling air going into the LCDs is clean. They need to be replaced at 1000 FH or every 12 months whichever comes first. Here are some helpful hints:

- Do not disconnect the electrical connector
- Disconnect the cooling air hoses so the display units can be pulled out sufficiently for access to the filter
- PFD, MFD and ED use different size filters. Check mod status and refer to IPC
- Display Unit filters are not cleanable, they must be replaced with new ones
- Filters are different sizes and are installed with different orientation depending on whether the original or redesigned type is used. Adhere to AMM Task 21-25-00-900-801 to ensure the applicable filter type is installed properly. Incorrect installation can lead to premature Display Unit failure.



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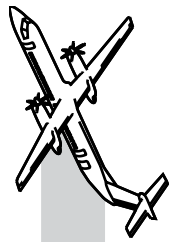
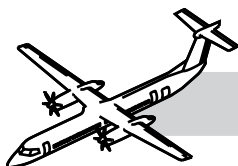


Figure 21-75. Front Fuselage LCD Cooling Filters



Avionics Cooling Fans Operational Check

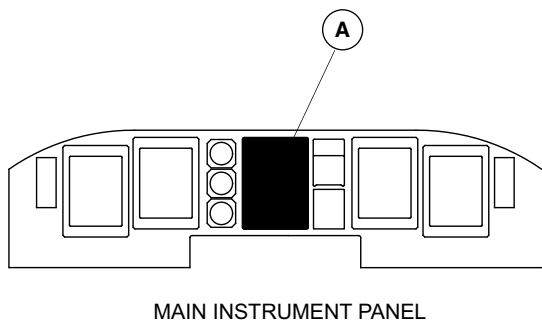
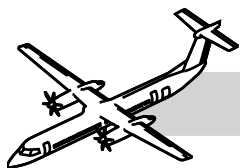
NOTE

A maintenance task is used for operational check of the three avionics cooling fans as well as the two temperature switches. The proper operating speeds of each avionics cooling fan are tested by:

- Pulling and resetting their PRI and SEC circuit breakers
- Applying and removing electrical power to DC main buses
- Applying and removing heat to the left and right temperature switches (if aircraft has modsum 4-126287 or 4-215000).

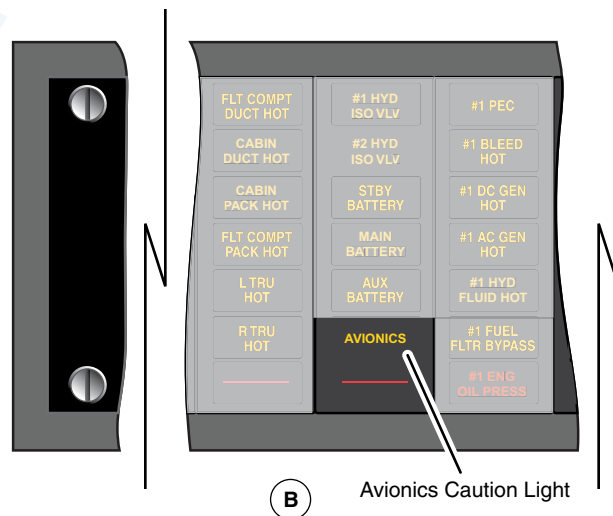
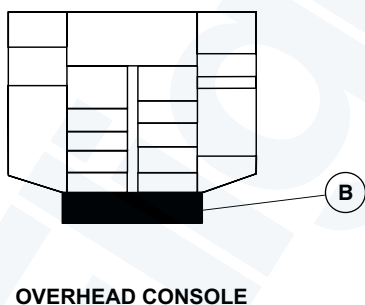
NOTE

This test must be done with the flight compartment temperature higher than 13°C as the avionics cooling fan failure monitoring system will not operate.



A

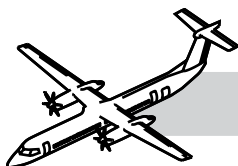
Advisory Message



B

Avionics Caution Light

Figure 21-76. Avionics Cooling Fans Malfunctions Indications /Advisory Message



CONTROLS AND INDICATIONS

NOTES

Refer to Figure 21-76. Avionics Cooling Fans Malfunctions Indications /Advisory Message.

There are no manual controls for the avionic cooling fans. The avionics cooling system status is indicated on the ED and/or CAWP.

One Fan Failure: Single fan failures are logged within the CDS and are not shown to the flight crew.

Two Fans Failure (Air mode): FANS FAIL advisory message is displayed on the ED.

Two Fans Failure (Ground mode): If speed is less than 50 Kts, the FANS FAIL advisory message is displayed on the ED and the IOM turns on the AVIONIC caution light.

Interfaces

Low Speed Warning Detector

Each fan has a built-in electronic LSWD which sends a signal to the IOM when the fan is rotating slower than 5000 rpm.

The IOM uses the low speed warning signal to record and display faults. Both IOM 1 and IOM 2 monitor all three fan fail speeds.

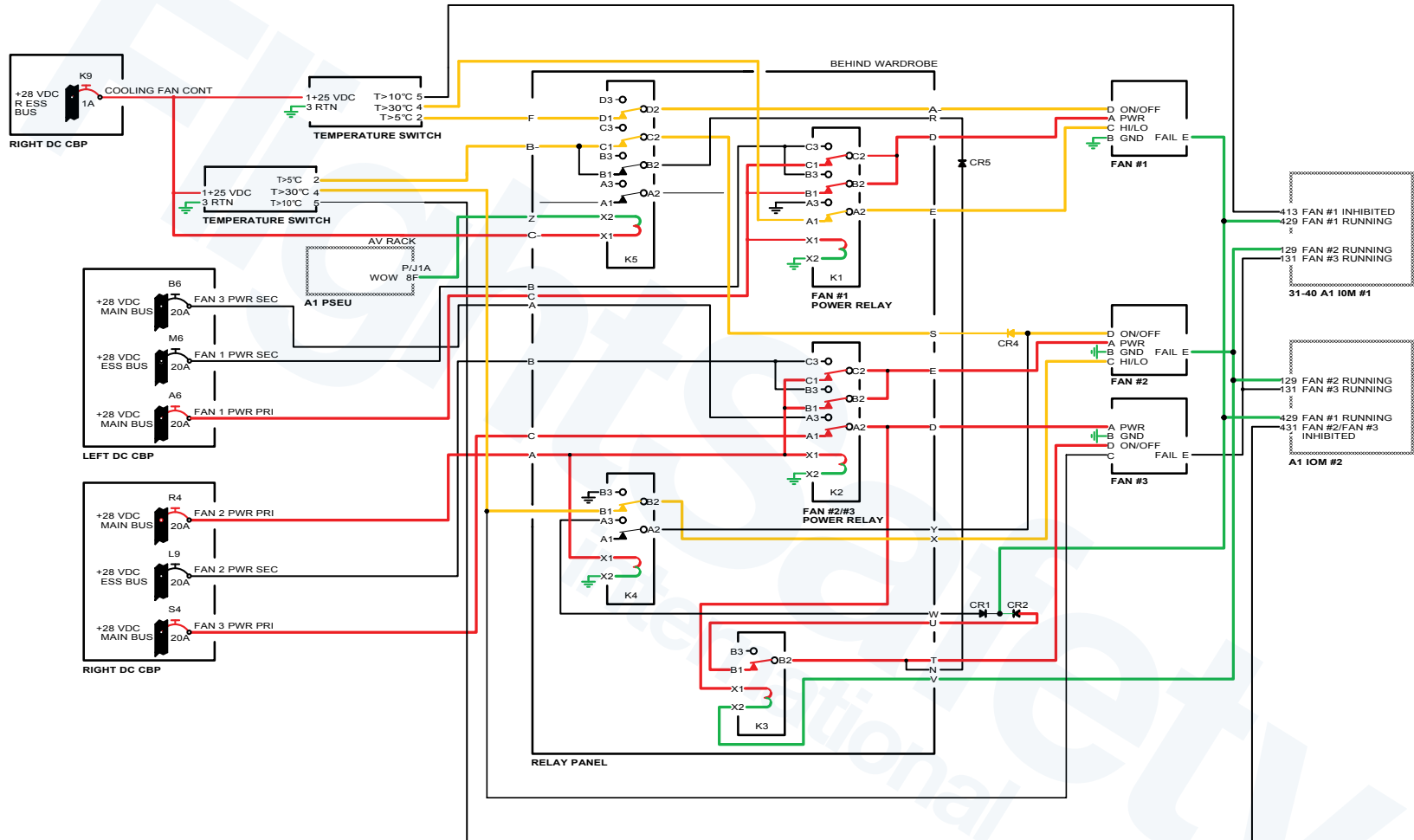
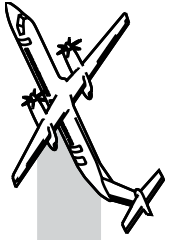
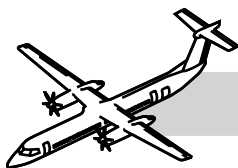


Figure 21-77. Avionics Cooling Normal Operation (Post Modsum)



OPERATION

Normal System Operation

Refer to Figure 21-77. Avionics Cooling Normal Operation (Post Modsum).

On initial start-up, electrical power is supplied to all three fans and all three fans will run. If there is no fan fail signal from the LSWD, fan No.3 will shut down and fan No.1 and fan No.2 will continue to operate. If either fans No.1 or No.2 fails, fan No.3 comes on automatically.

Only two of the three fans are required to be operational for dispatch. When DC generated power is available and the aircraft is in flight, only two avionics cooling fans are operating.

The operational time of the equipment without forced air cooling is limited to 60 minutes. When the internal temperature increases, the displays' brightness will be reduced to 60%. If there is the possibility of a misleading or total loss of display due to internal temperature increases, the display unit will shut down.

In the remote case of both main DC buses are not available in flight and the aircraft is only powered by the batteries, fan No.1 or No.2 will operate at low speed. Fan No.3 will not be available if both main buses are not powered. This operational mode is capable of supplying the minimum air flow required for the LCDs to operate at reduced brightness but below the automatic shutoff threshold.

The avionics cooling system has two fan zone temperature switches which operate at 5°C, 10°C and 30°C. On aircraft before Modsum 4-126287, the 30°C discrete is not installed.

- The 5°C, ON/OFF discrete to the fan is on pin 2 of the switch. When the temperature rises above 5°C, an open circuit causes the fan to operate. If the ON/OFF output is grounded, the fan stops operation.
- The 10°C, inhibit discrete to the IOM is on pin 5 of the switch. When the temperature rises above 10°C the Inhibit

output is open circuit which causes the IOM to record if FAN FAIL output. If the Inhibit output is grounded, the IOM does not record FAN FAIL outputs.

- For aircraft manufactured before Modsum 4-126287 - the fans always run at high speed during normal operation.
- For aircraft with Modsum 4-126287 - when the left hand zone temperature switch senses 30°C or higher, the HI/LO output is open circuit which causes fan No.1 to operate at high speed. If the HI/LO output is grounded, the fan operates at low speed. Fan No.2 speed is affected by the right hand zone temperature switch.
- The 5°C and 10°C discrettes are inhibited in flight mode and the 30°C discrete will cause the fan to go high speed if the main buses are powered.

Each fan uses a LSWD which sends a FAN FAIL output on pin E of the fan to the IOMs. This output is grounded if the fan is in normal operation.

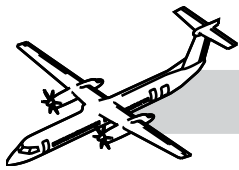
Crew Reports

Avionics fans appear to start late as none of them could be heard running until 10 or 20 minutes after the DC external power was applied. Subsequent operation of the cooling fans was found normal. What is your assessment?

Fault Diagnostics

In conjunction with the aircraft centralized diagnostics system, maintenance can use one of the following troubleshooting tools:

- Bombardier's ECS HEX Decoder
- Fault Isolation Manual (FIM).
- Aircraft Diagnostics Solutions (ADS)



Central Diagnostics System

The aircraft Central Diagnostics System (CDS) has 7 pages of information dedicated for the pneumatics and air conditioning systems.

Page 1 shows the first 12 present faults detected by the left channel of the ECSECU. Any additional faults are not shown. The faults are displayed with fault code in hexadecimal followed by a 15 character description.

Page 2 shows the present faults of the right hand channel. Present faults are faults currently stored in memory. When maintenance personnel clear the present faults, they will be moved to their corresponding fault history page.

Page 3 lists the fault history for the left channel. **Page 4** lists the fault history for the right channel. The 3 digit number is the time in hours since the last reset. If it is more than 999, then the 3 digits will change to “OLD”. A maximum of 12 historical faults can be displayed per channel.

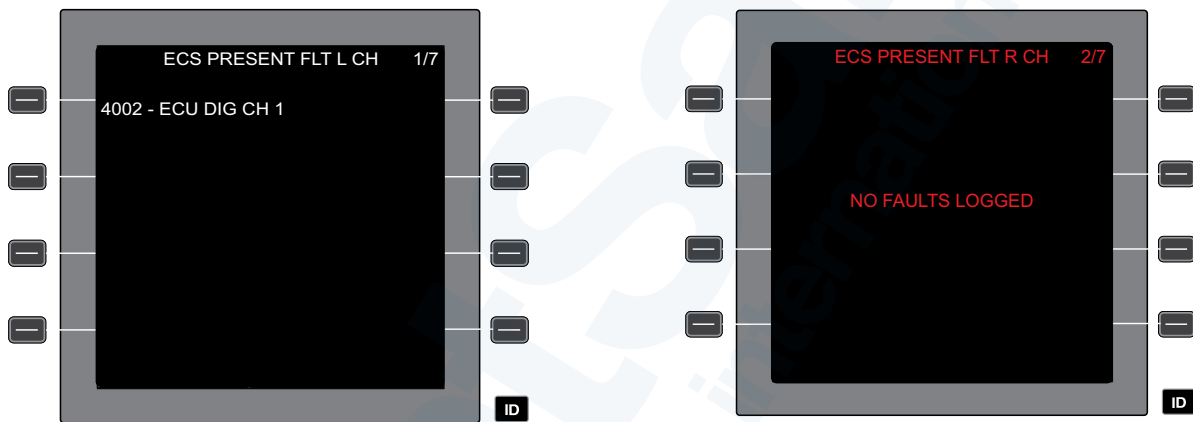


Figure 21-78. ECSECU - Present Faults of the Left Channel

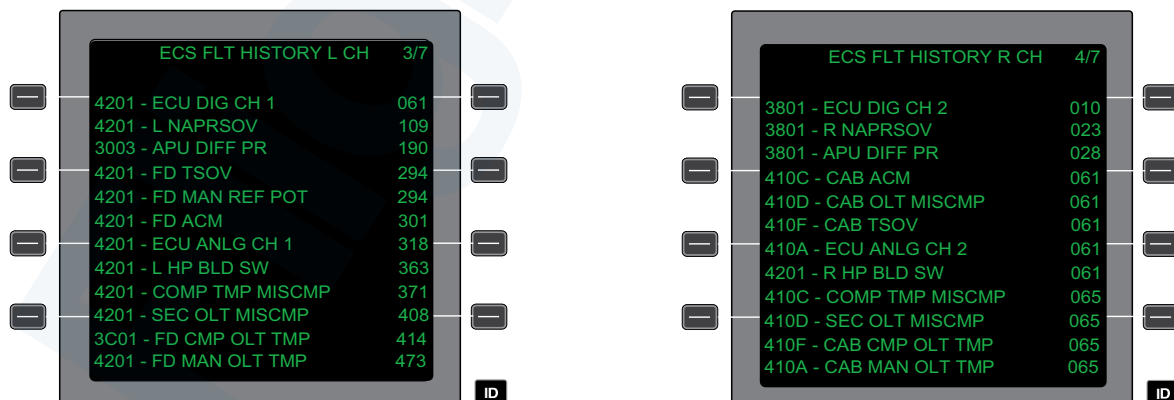
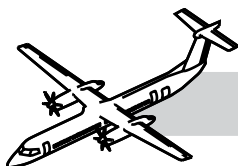


Figure 21-79. ECSECU - Present Faults of the Right Channel



Page 5 shows the software version currently loaded on the two channels of the ECSECU.

Page 6 shows the diagnostics data collected by the left digital channel. They are recorded in hexadecimal. **Page 7** shows the hexadecimal data for the right digital channel. The ECS data sent to the CDS is a “snapshot” of the data contained in the ECU. The first 3 characters are for identification and the remaining 4 are for signal information. They represent ECS temperatures, pressures and valve commands. In order to refresh the data, the ECS menu must be exited and re-entered. There are 36 hexadecimal codes for each channel.

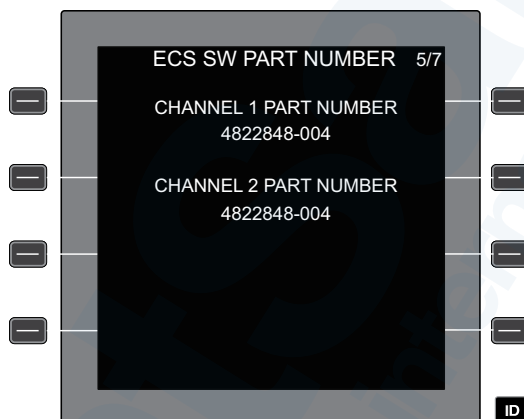


Figure 21-80. Software Version on Both Channels of the ECSECU

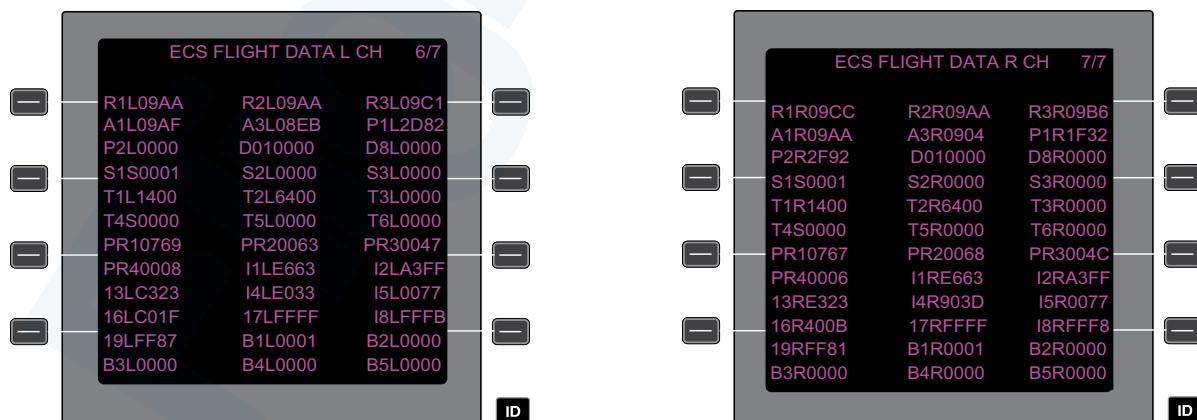


Figure 21-81. Diagnostic (Left) and Hexidecimal (Right) Data Display

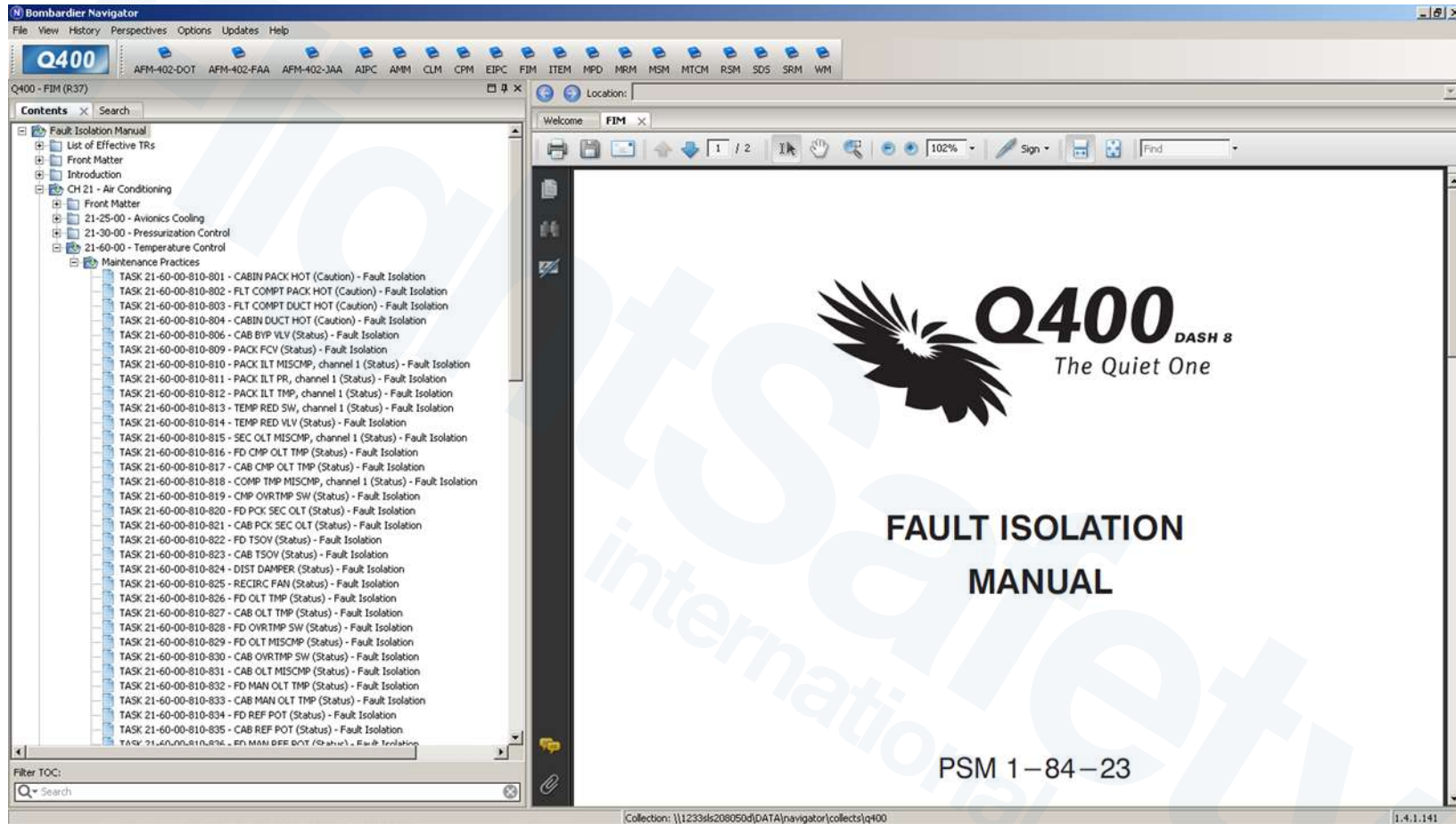
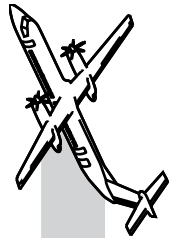
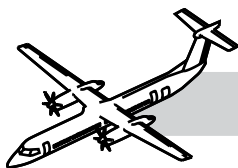


Figure 21-82. Bombardier Navigator - FIM



The Fault Isolation Manual (FIM)

The Fault Isolation Manual (FIM) provides maintenance personnel with instructions to identify, isolate, and correct faults. To avoid or reduce dispatch delays, the FIM provides the Original Equipment Manufacturer's (OEM) recommended procedures to quickly and effectively isolate discrepancies in the Q400 aircraft.

The FIM has fault isolation tasks to identify single point discrepancies.

Fault Isolation Process

The fault isolation process has three parts.

- A flight compartment discrepancy, BITE indication, or cabin discrepancy is viewed
- Find the fault isolation task in the Master Fault Table.
- Do the fault isolation task.

Fault Tables

Alert Message List

The alert message list is an alphanumeric list of all flight compartment discrepancies that are sensed by the aircraft monitoring systems. Alert messages are shown or heard in the flight compartment (e.g. CABIN PRESS or cabin pressure control panel FAULT light illuminates).

Observed Fault List

The observed fault list is an alphanumeric list of the flight compartment discrepancies that the pilots can see. An observed fault does not have a related alert indication. Examples of observed faults are caution or advisory lights do not come into view or audio message is not heard or a control panel selection does not give the correct result.

Status Message List

The status message list is an alphanumeric list of all the system BITE indications that can be seen or heard while the aircraft is on the ground.

These BITE indications are made available on systems such as Cabin Pressure Control System, Environmental Control System etc.

An alert message is related to one or more status messages. Therefore, start from the alert fault isolation task to find the correct status task.

Class 2 status messages do not have a related alert message. Therefore, for the status messages that are in the Class 2 Fault List, start from the status fault isolation task.

Cabin Fault List

The cabin fault list is an alphanumeric list of cabin discrepancies that are seen in the cabin and affect passenger comfort. An example of cabin fault is potable water system.

Aircraft Diagnostics Solutions

Refer to Figure 21-83. Aircraft Diagnostic Solutions.

The Aircraft Diagnostics Solutions application is accessible on the right hand side of the home page under "My Services and Tools" after you log into the iflybombardier.com web site. You can open a new ADS troubleshooting session in 3 different ways:

- **Images**

Click on the relevant area on the cockpit image, e.g. Caution and Warning Panel for DUCT HOT caution light (refer to Figure 21-84. ADS Session via Images.)

- **Keywords**

Enter a keyword, e.g. Bleed Hot (refer to Figure 21-85. ADS Session via Keywords.)

- **ATA**

Select an ATA chapter, e.g. ATA 21 (refer to Figure 21-86. ADS Session via ATA.)

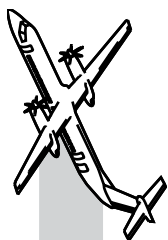


Figure 21-83. Aircraft Diagnostic Solutions

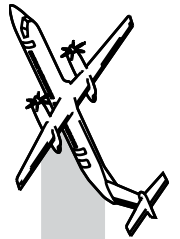
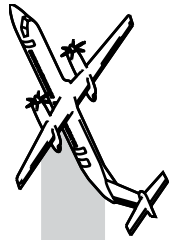


Figure 21-84. ADS Session via Images



Aircraft Diagnostics Solutions V4.4.0.0124 - Windows Internet Explorer provided by FlightSafety International

http://ads.casebases.com/ads/main.jsp#cq400_H0_

File Edit View Favorites Tools Help Links

Aircraft Diagnostics Solutions V4.4.0.0124

Aircraft Diagnostics Solutions Plus powered by CaseBank

You are logged in as John Chan

Home D8 - Q400 Untitled

1. Enter Problem using Keywords 2. Answer Questions & View Solutions 3. Exit Session

Switch to Images | Switch to ATA

Problem Keywords: Bleed Hot (Optional) Select ATA: All Categories

- #1 BLEED HOT : #1 BLEED HOT illuminated
- #2 BLEED HOT : #2 BLEED HOT illuminated
- #1 BLEED HOT : Both #1 and #2 BLEED HOT illuminated
- #2 BLEED HOT : Both #1 and #2 BLEED HOT illuminated
- BLEED - Message on ED/MFD : BLEED message not shown when BLEED switch selected
- WSHLD HOT : Illuminated
- Evidence of hot brakes : Yes, a brake is hot
- Evidence of hot brakes : Yes, two brakes on one side are hot
- Pneumatic system (bleed air) problems : No or low bleed air from one engine
- Pneumatic system (bleed air) problems : Bleed air fluctuation or surging at low power

Selected Symptoms (0)

Continue >

Figure 21-85. ADS Session via Keywords

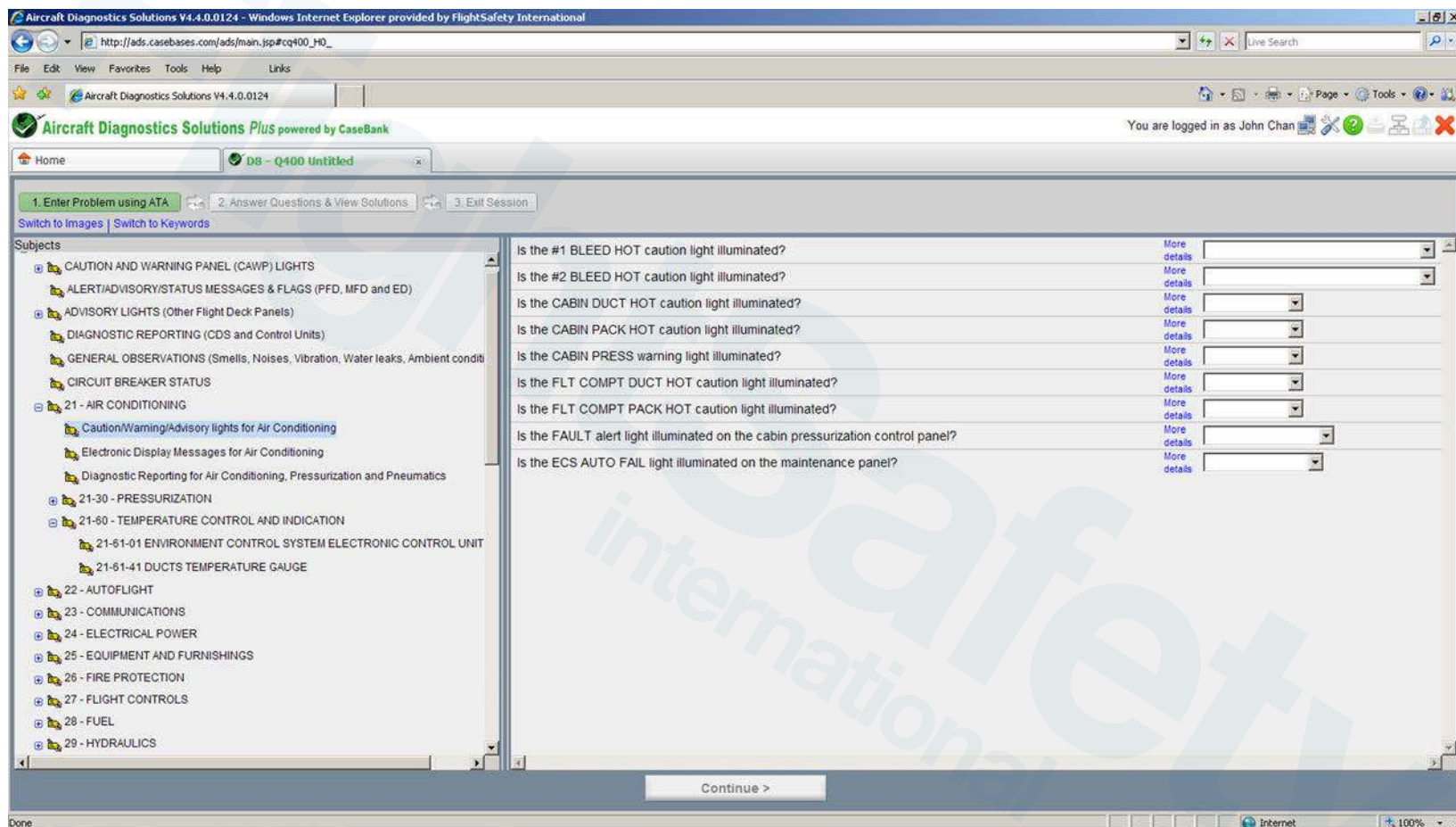
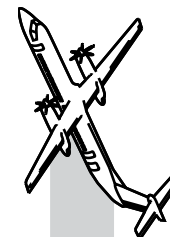
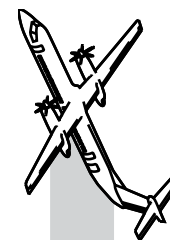


Figure 21-86. ADS Session via ATA





Q400 ECS System Hex decoder

File Print About

Enter Hex | Control Panel | Pack page | Bleeds Page | Fault Codes | Hex List | Components

DATE:

AIRCRAFT S/N:

READING #:

NOTE:

Enter Ambient Temperature:

(Enter from Cockpit Panel Gauge:)
Cabin Temperature:

Cabin Duct Temperature:

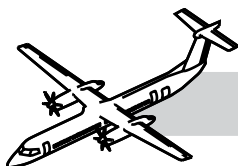
FC Duct Temperature:

Decode

R1L	07A2	R2L	07D9	R3L	07A2	R1R	07A8	R2R	07BF	R3R	07A8
A1L	07C2	A3L	081B	P1L	35C1	A1R	0788	A3R	07B2	P1R	2881
P2L	0006	D01	0001	D8L	0000	P2R	0007	D01	0002	D8R	0000
S1S	0001	S2L	0000	S3L	0000	S1S	0001	S2R	0001	S3R	0000
T1L	0000	T2L	6400	T3L	0000	T1R	0000	T2R	6400	T3R	0000
T4L	0000	T5L	0000	T6L	0000	T4R	0000	T5R	0000	T6R	0000
PR1	07B0	PR2	0045	PR3	006F	PR1	07B2	PR2	0040	PR3	0071
PR4	0014	I1L	EEE3	I2L	A3FF	PR4	001D	I1R	EAE3	I2R	A344
I3L	E323	I4L	A41D	I5L	0077	I3R	C323	I4R	D413	I5R	0077
I6L	400B	I7L	FFFE	I8L	FFFA	I6R	C01F	I7R	FFFE	I8R	FFF9
I9L	FF85	B1L	0001	R4L	0000	I9R	FF8B	B1R	0001	R4R	0000
B3L	0000	B4L	0000	B5L	0000	B3R	0000	B4R	0000	B5R	0000

FOR TRAINING PURPOSES ONLY

Figure 21-87. Enter Hex Tab Screen



Bombardier's ECS HEX Decoder

Bombardier has an ECS troubleshooting tool that you can download from their iflybombardier.com web site. It is called the ECS HEX Decoder tool. You can locate the zipped version of the tool under the Resources tab.

To download a copy of the tool, follow the path below:

- ▶ iflybombardier.com
 - ▶ Resources
 - ▶ Service Documents
 - ▶ Training Material
 - ▶ ECS Decoder V9

To start using the tool, you need to do the following:

- On the initial Enter Hex Tab screen enter date, aircraft registration, ambient temperature and hex codes recorded on pages 6 & 7 of the CDS ECS section. Preferably 3 readings should be taken:
 - When the air-conditioning system is OFF and cooler down
 - When both engines are running with bleeds ON and within a few minutes after hot temperature was selected
 - After cold temperature selection has just been made
- Enter cabin and duct temperatures indicated on the cockpit gauge
- Enter fault codes or any relevant information in the NOTE text box
- Select "File" - "Save as" from menu to save the reading to a file in a folder of your choice.
- Click the "Decode" button to decode and enable the top toolbar tabs.

Refer to:

- Figure 21-87. Enter Hex Tab Screen.
- Figure 21-88. Pack Page Tab Screen.
- Figure 21-89. Control Panel Screen.
- Figure 21-90. Bleeds Page Tab Screen.
- Figure 21-91. Fault Codes Tab Screen.
- Figure 21-92. Hex List Tab Screen.
- Figure 21-93. Components Tab Screen.

For sample screens of the Bombardier ECS Decoder.

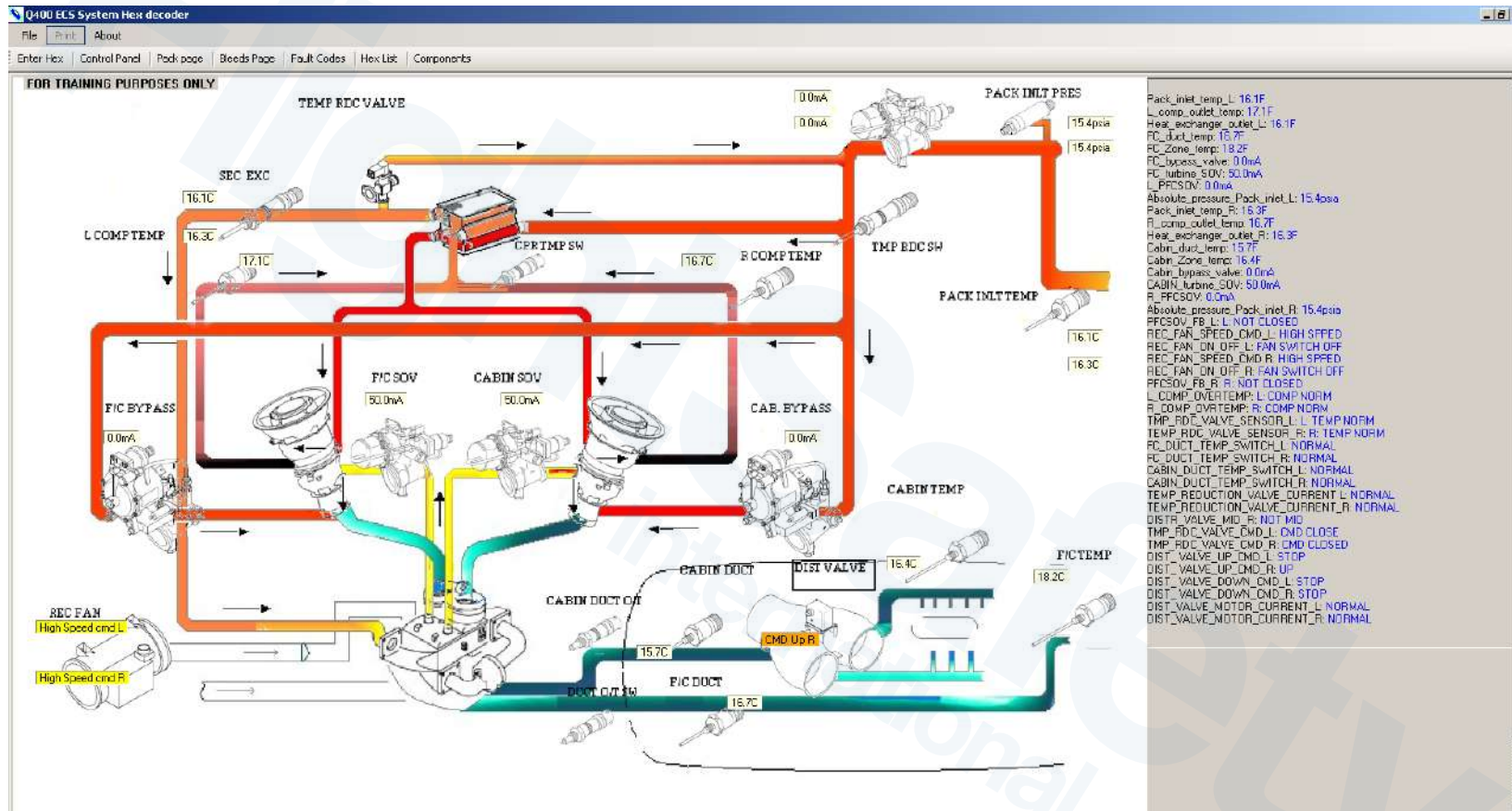
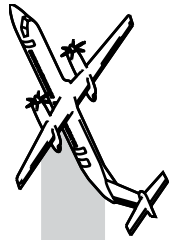


Figure 21-88. Pack Page Tab Screen

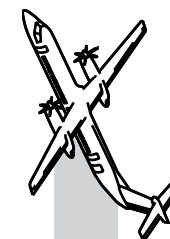
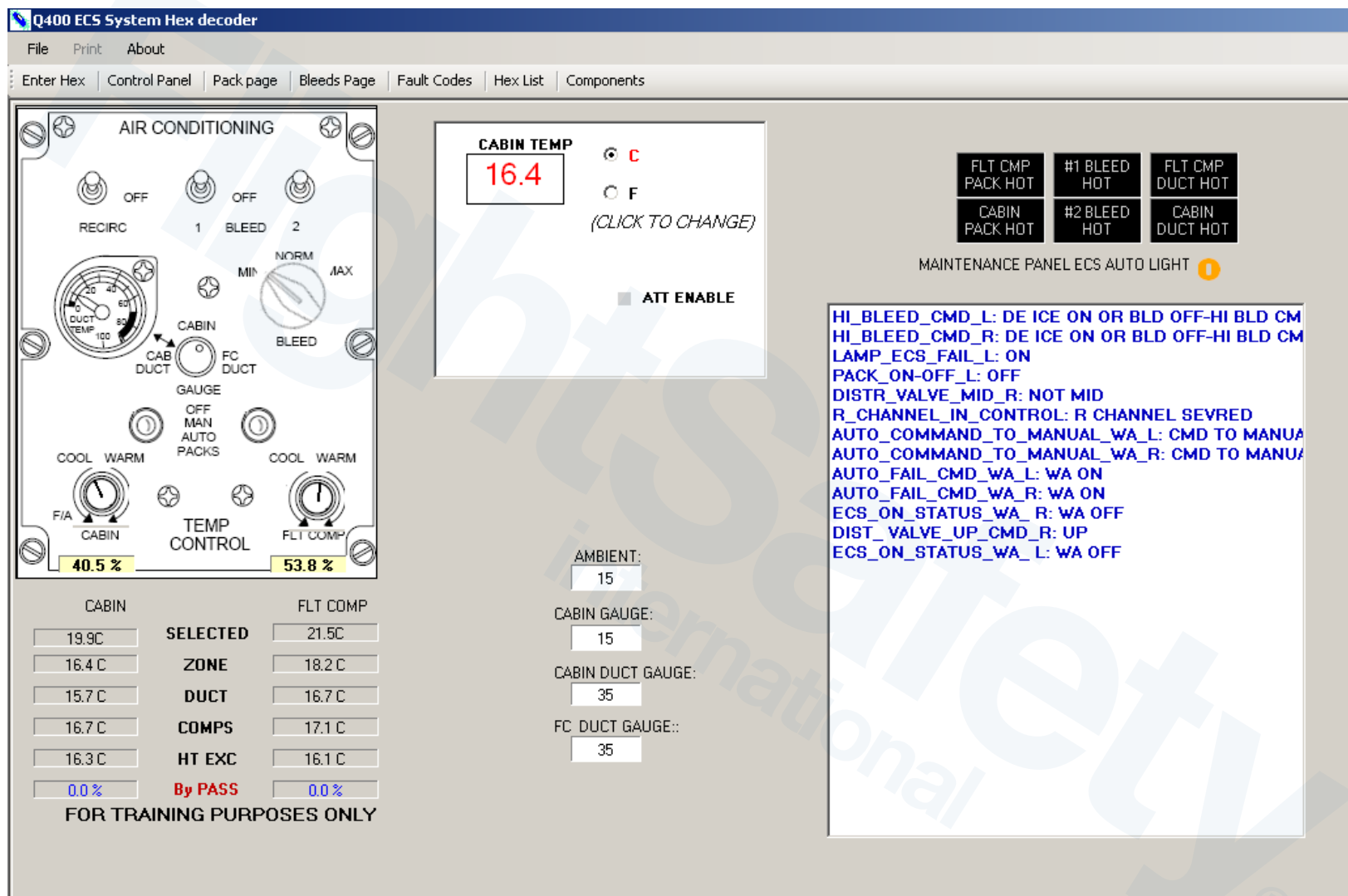


Figure 21-89. Control Panel Screen

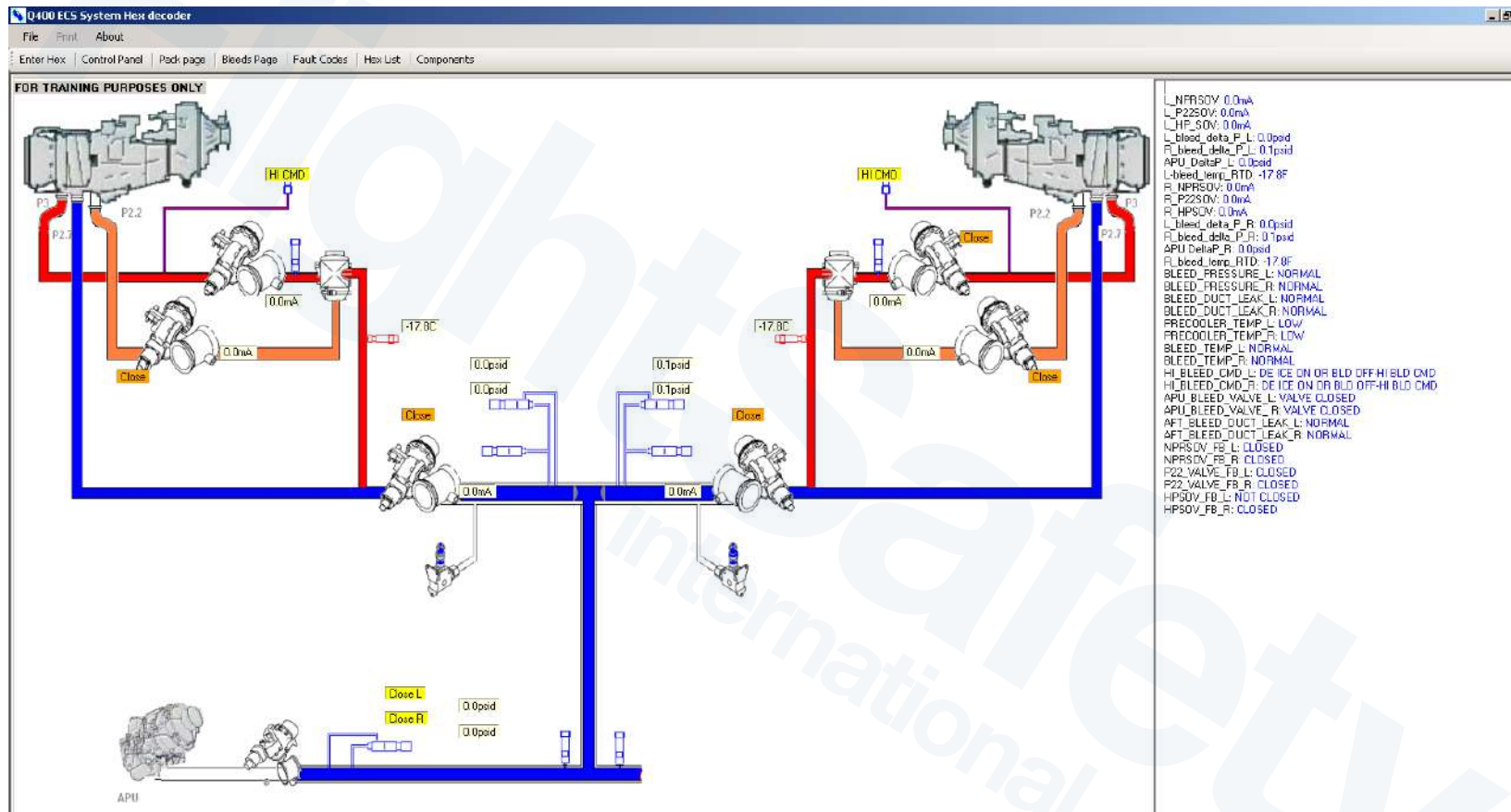


Figure 21-90. Bleeds Page Tab Screen

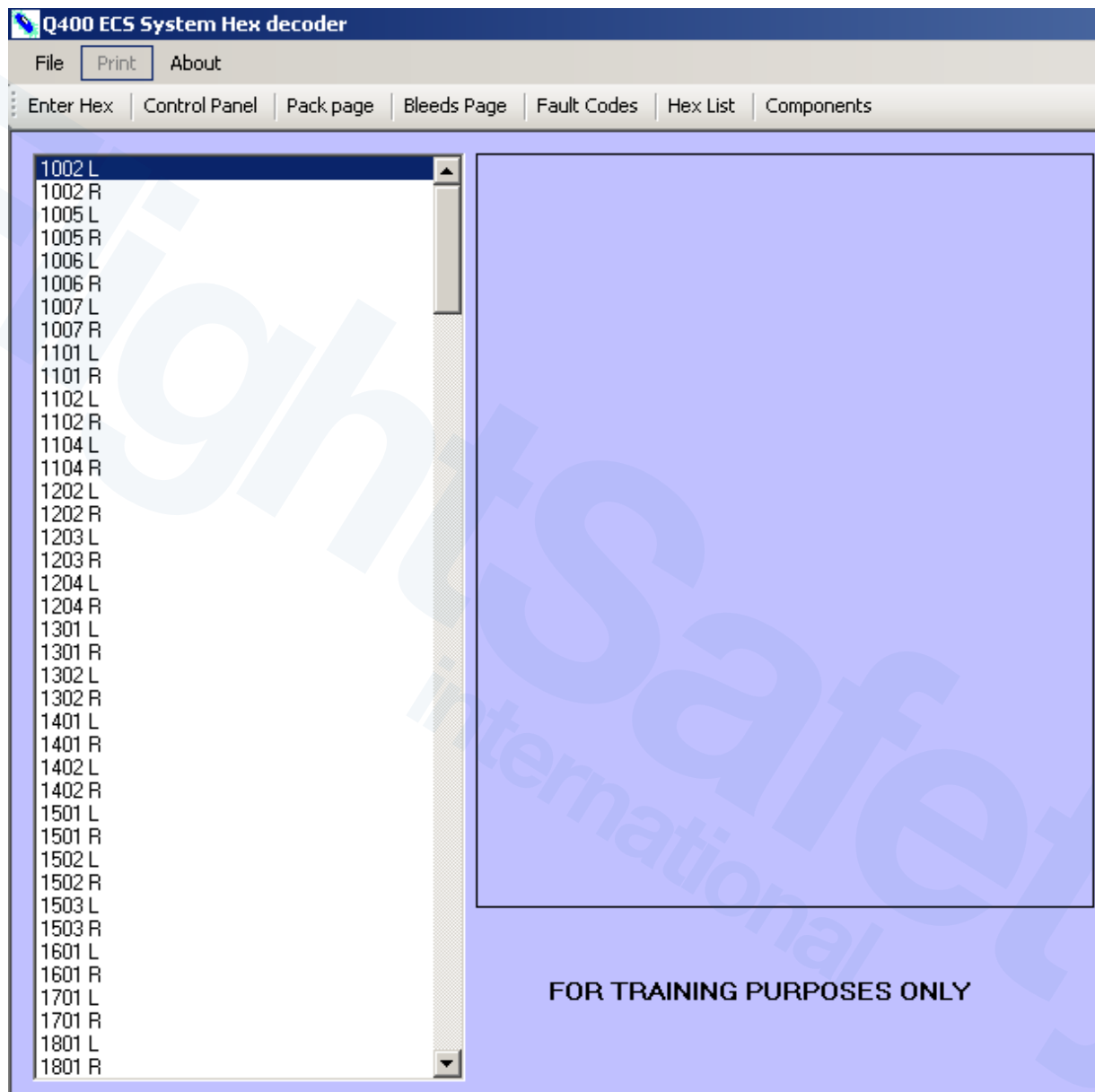
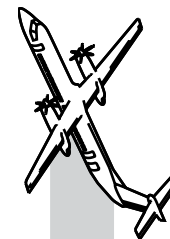


Figure 21-91. Fault Codes Tab Screen



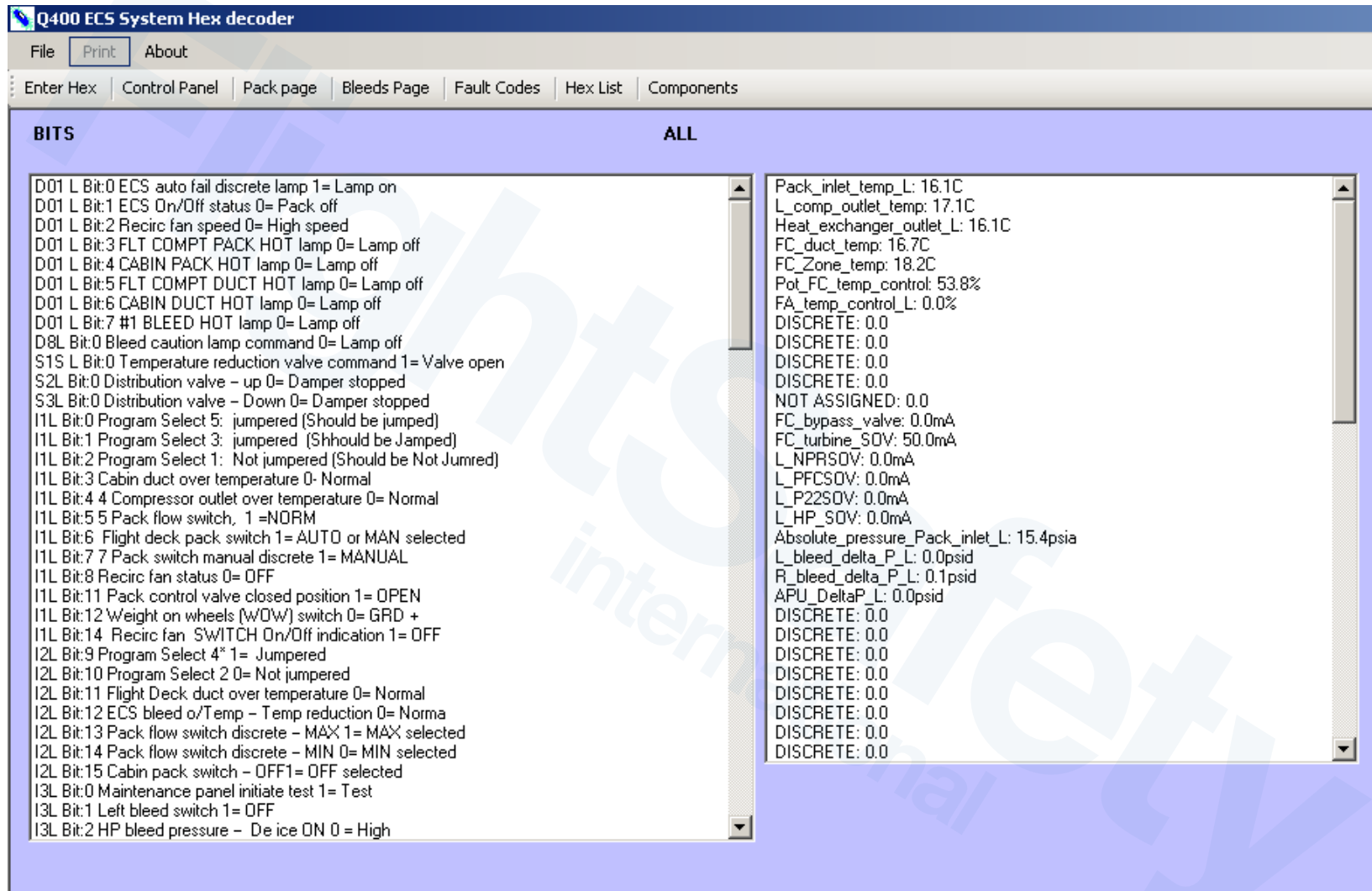
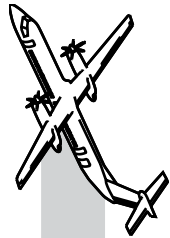


Figure 21-92. Hex List Tab Screen

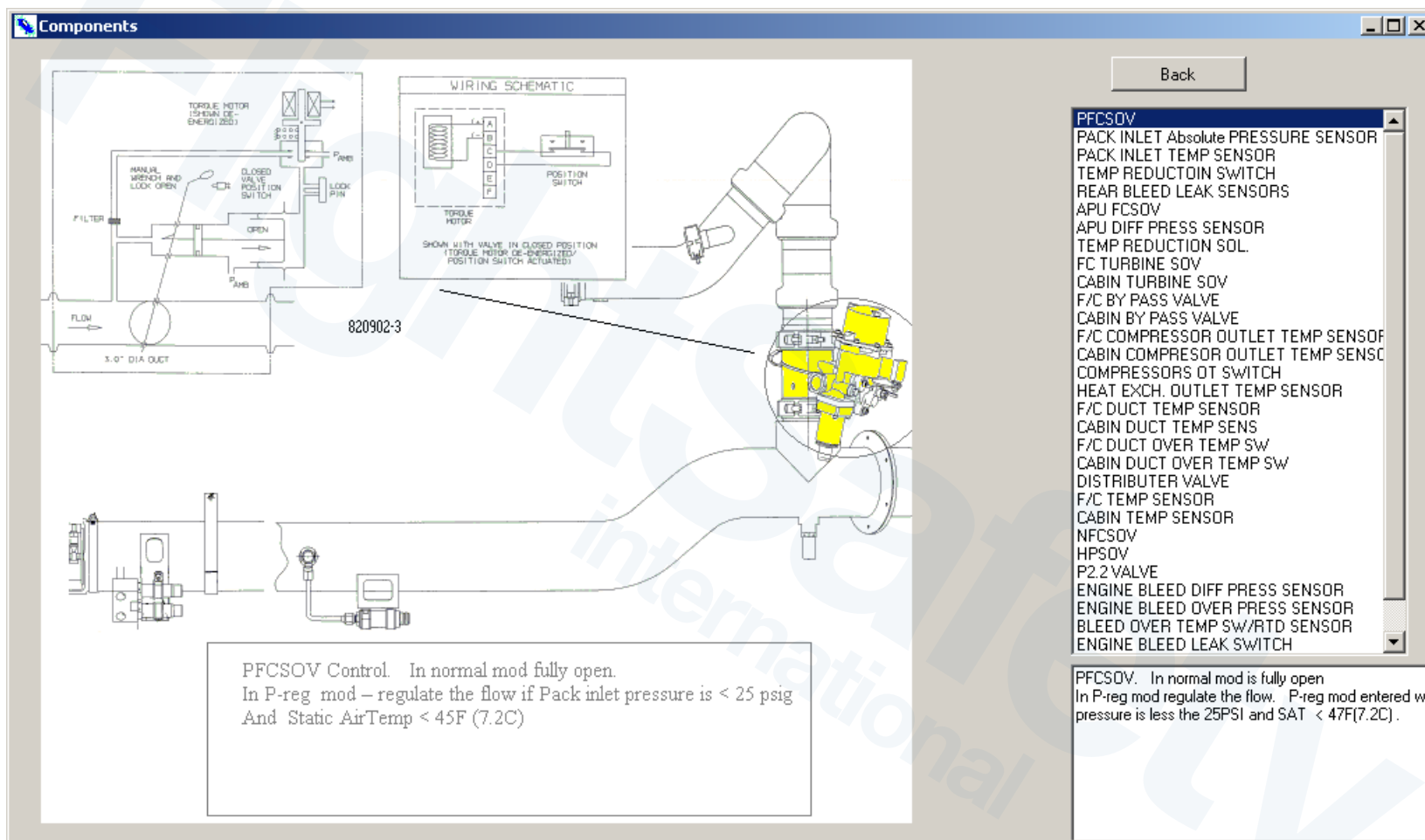


Figure 21-93. Components Tab Screen

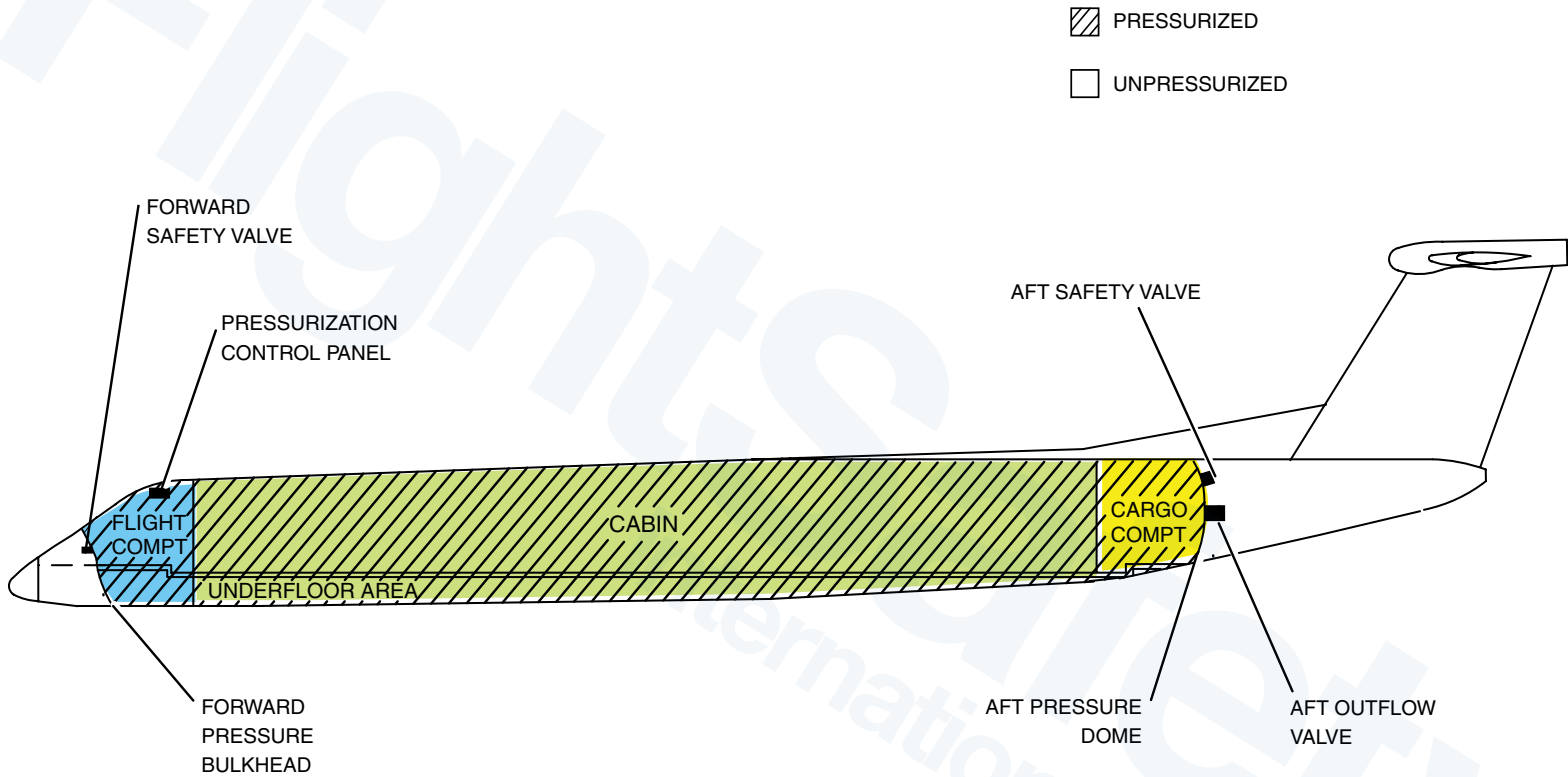
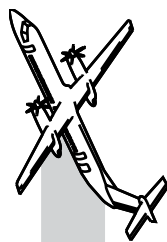
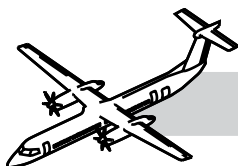


Figure 21-94. Fuselage Pressurized Areas



21-31-00 PRESSURIZATION CONTROL

NOTES

INTRODUCTION

The pressurization system controls the cabin altitude and the cabin rate of change.

GENERAL

Refer to Figure 21-94. Fuselage Pressurized Areas.

The pressurized area of the fuselage extends from the front pressure bulkhead to the rear dome. This area includes the flight compartment, passenger compartment, under floor space and baggage compartments.

The CABIN PRESSURE control panel controls the system in automatic and manual modes. The Cabin Indication Module shows the data to monitor the system.

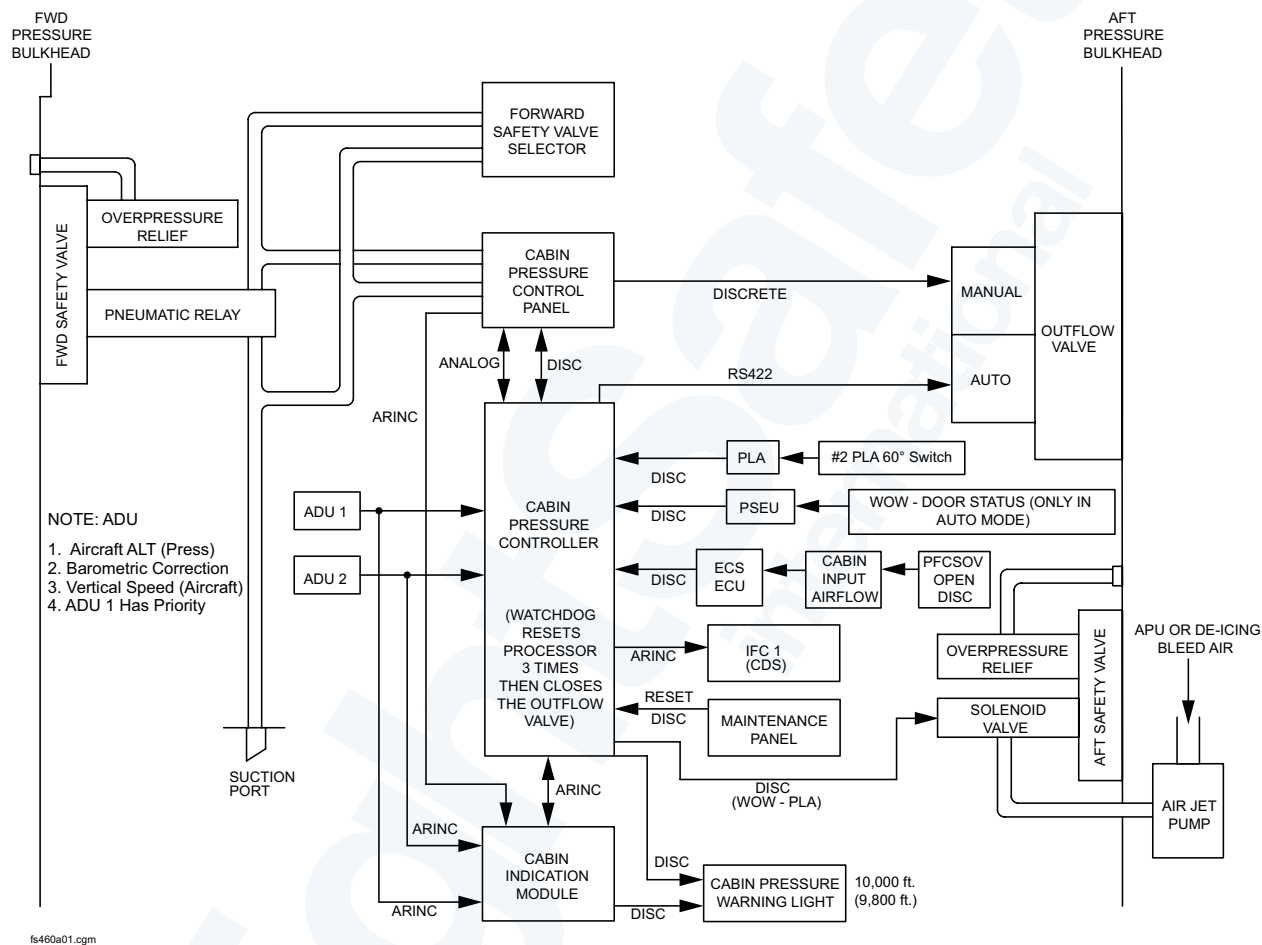
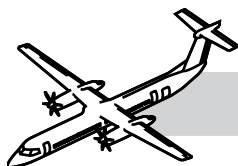
If the cabin altitude is too high, the cabin pressure controller and the cabin indication module send the warning signal to the aircraft systems.

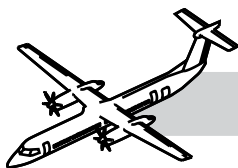
Electrical power supply for pressurization control is supplied by the 28 VDC left main bus through a CABIN PRESS AUTO CONT circuit breaker.

The aft outflow valve is used for automatic and manual control of the pressurization system.

The forward safety valve is used for emergency operation and for smoke removal from the flight compartment. It also protects the aircraft against excessive positive and negative air pressures.

The aft safety valve protects the aircraft against excessive positive and negative pressures and assists in cabin pressure equalization after aircraft lands.





SYSTEM DESCRIPTION

Refer to:

- Figure 21-95. Pressurization Control Synoptic.
- Figure 21-96. Typical Flight Profile.

The system operates in the modes that follow:

- Normal/Automatic - Software Profile
- On Ground - Pressurize 400' below take off field elevation during take-off roll
- Take-off - Maintain software schedule with reference altitude (ADU inputs)
- Landing - 400' below selected landing altitude
- Emergency/Smoke Removal
- Pressure Dump
- Manual

Flight compartment indications are:

- Cabin altitude
- Cabin altitude rate of change
- Differential pressure
- Cabin altitude warning
- Cabin Pressure Control System fault annunciation

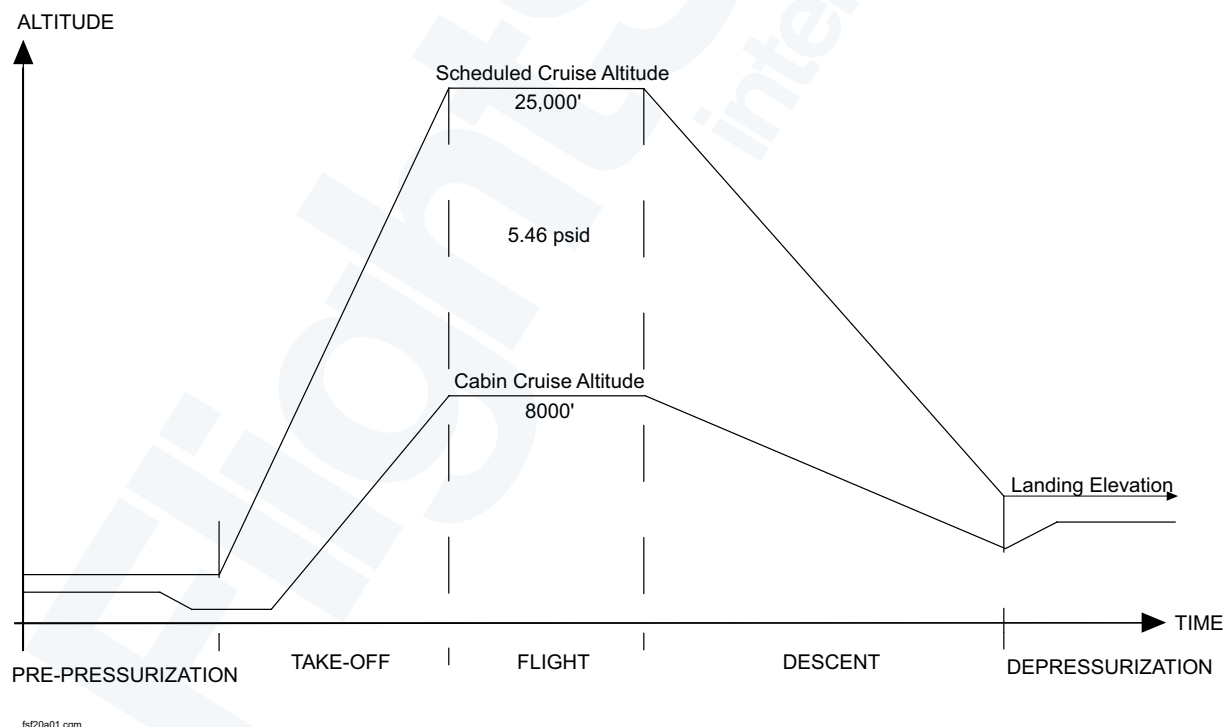


Figure 21-96. Typical Flight Profile

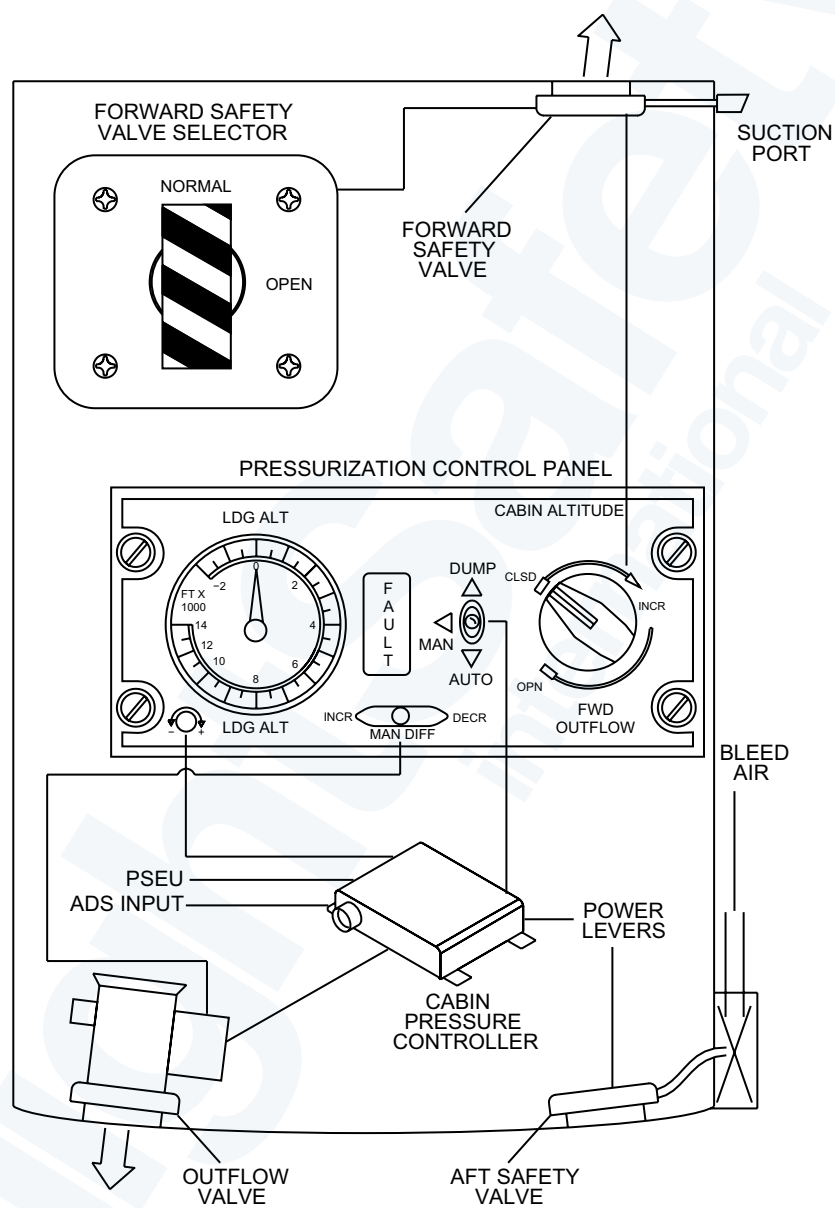
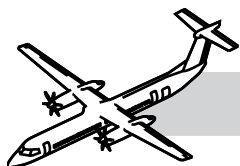
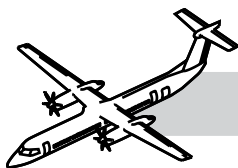


Figure 21-97. Pressurization Synoptic



Refer to Figure 21-97. Pressurization Synoptic.

NOTES

The pressurization control system is electrically operated. The pressurized area is supplied with a flow of conditioned air from the ECS.

Modulating the aft outflow valve controls the fuselage pressure. If the external ambient pressure differs from the fuselage pressure by more than a preset amount, safety valves will open for positive or negative pressure relief.

When electrical power is first supplied to the system, a full electrical test of the system is done. The FAULT alert light, on the CABIN PRESSURE control panel illuminates, during the POST. The system operation is automatic with the data programmed into the controller.

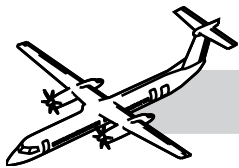
The aft outflow valve can be manually controlled from the CABIN PRESSURE control panel on the flight compartment overhead panel.

The aft safety valve protects against cabin overpressure and negative pressure and supplies venting when WOW.

A manually operated forward safety valve in the front pressure bulkhead is used for backup pressurization control or for smoke removal. The forward safety valve is operated by the FWD OUTFLOW rotary knob on the CABIN PRESSURE control panel or by the forward safety valve selector on the co-pilot's side console.

The CABIN PRESSURE control panel and the cabin pressure controller control the system.

Operation of the outflow valve is automatically controlled by the cabin pressure controller according to the flight profile and landing elevation set by the aircrew. The valve can also be operated in the manual mode in the event of failure of the automatic mode of control.



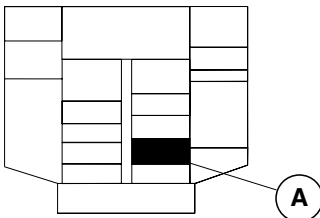
COMPONENT DESCRIPTION

CABIN PRESSURE Control Panel

Refer to Figure 21-98. CABIN PRESSURE Control Panel (CPCP).

The CABIN PRESSURE control panel, on the overhead panel contains the required indicators and switches to manage the cabin pressure.

For details, refer to Controls and Indications CPCP later in this chapter.



OVERHEAD CONSOLE

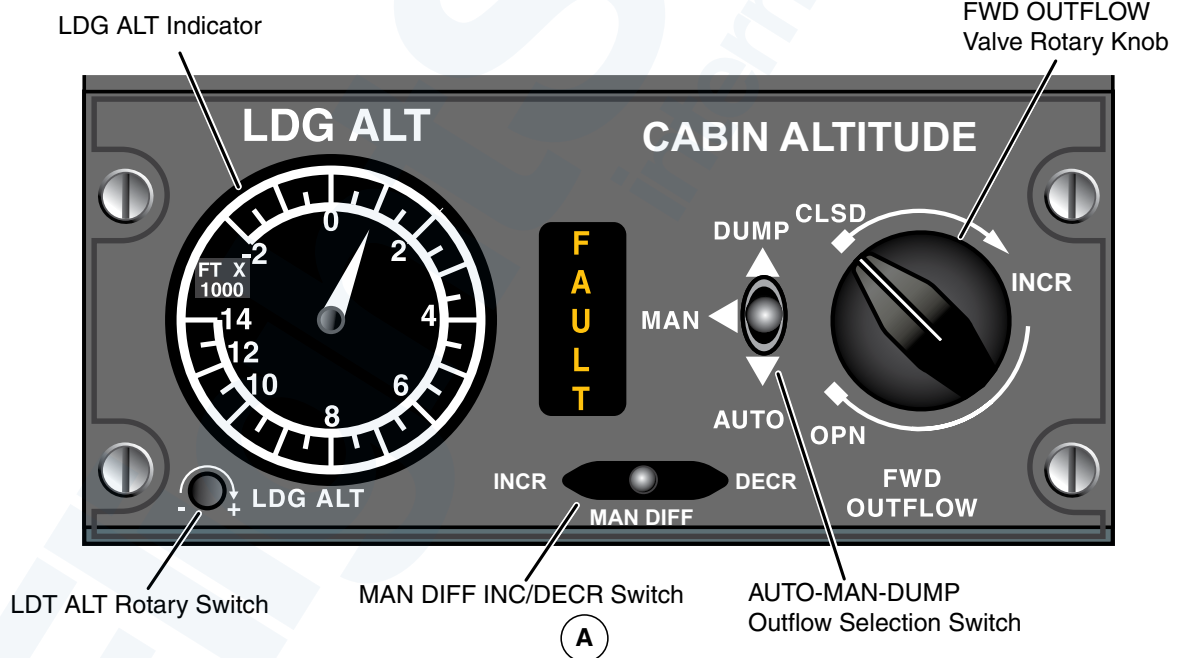
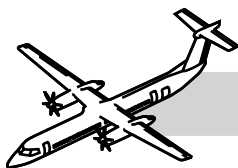


Figure 21-98. CABIN PRESSURE Control Panel (CPCP)



Cabin Pressure Controller

Refer to Figure 21-99. Cabin Pressure Controller (CPC).

The CPC is located on the bulkhead forward of the wardrobe. It is a microprocessor based unit that controls the cabin pressure schedule based on the aircraft flight profile.

The CPC:

- Provides the calculation of the control logic and cabin pressurization schedule according to aircraft flight profile.
- Provides control of the cabin pressure by driving the auto mode actuator of the aft outflow valve.
- Provides an excessive cabin altitude warning signal to the CAWP. (9800 ft).
- Monitors the Cabin Indication Module, Aft Safety Valve and the Outflow Valve for fault detection.
- Initiates a system level power up BIT when on the ground.
- Provides an interface to the CDS.

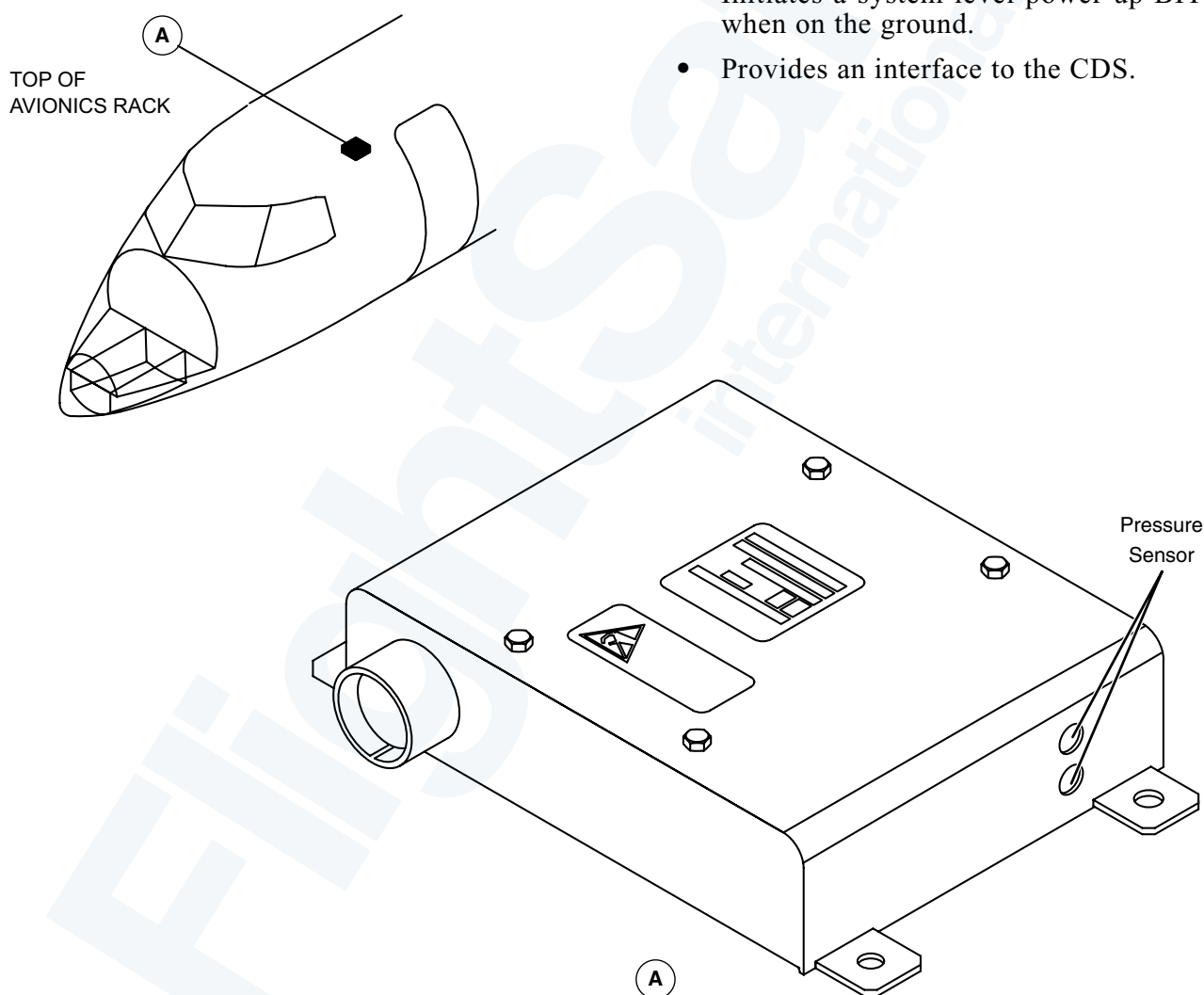
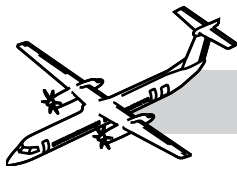


Figure 21-99. Cabin Pressure Controller (CPC)

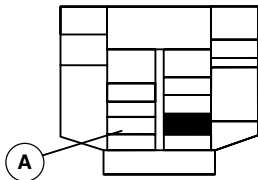


Cabin Indication Module (CIM)

Refer to Figure 21-100. Cabin Indication Module (CIM).

The cabin indication module located in the flight compartment on the overhead panel contains a differential pressure indicator, a cabin altitude indicator, a cabin rate of change indicator, and a pressure sensor.

The CIM interfaces with the CPC and ADUs for differential, altitude and rate indications. For details, refer to Controls and Indications CIM later in this chapter.

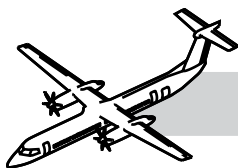


OVERHEAD CONSOLE



(A)

Figure 21-100. Cabin Indication Module (CIM)



Outflow Valve

Refer to:

- Figure 21-101. Aft Outflow Valve.
- Figure 21-102. Outflow Valve Synoptic.

The 6 in. aft outflow valve is located on the rear pressure bulkhead.

The electrical actuator has two DC motors, a gearbox and a control box.

The gearbox is arranged to allow either actuator to drive the valve butterfly between limits, regardless of the position of the other actuator.

Auto mode actuator has a brushless 28 VDC motor. Manual mode has a 28 VDC motor with brushes.

The function of the aft outflow valve is to modulate and control the cabin pressure in the dump, auto and manual modes.



Figure 21-101. Aft Outflow Valve

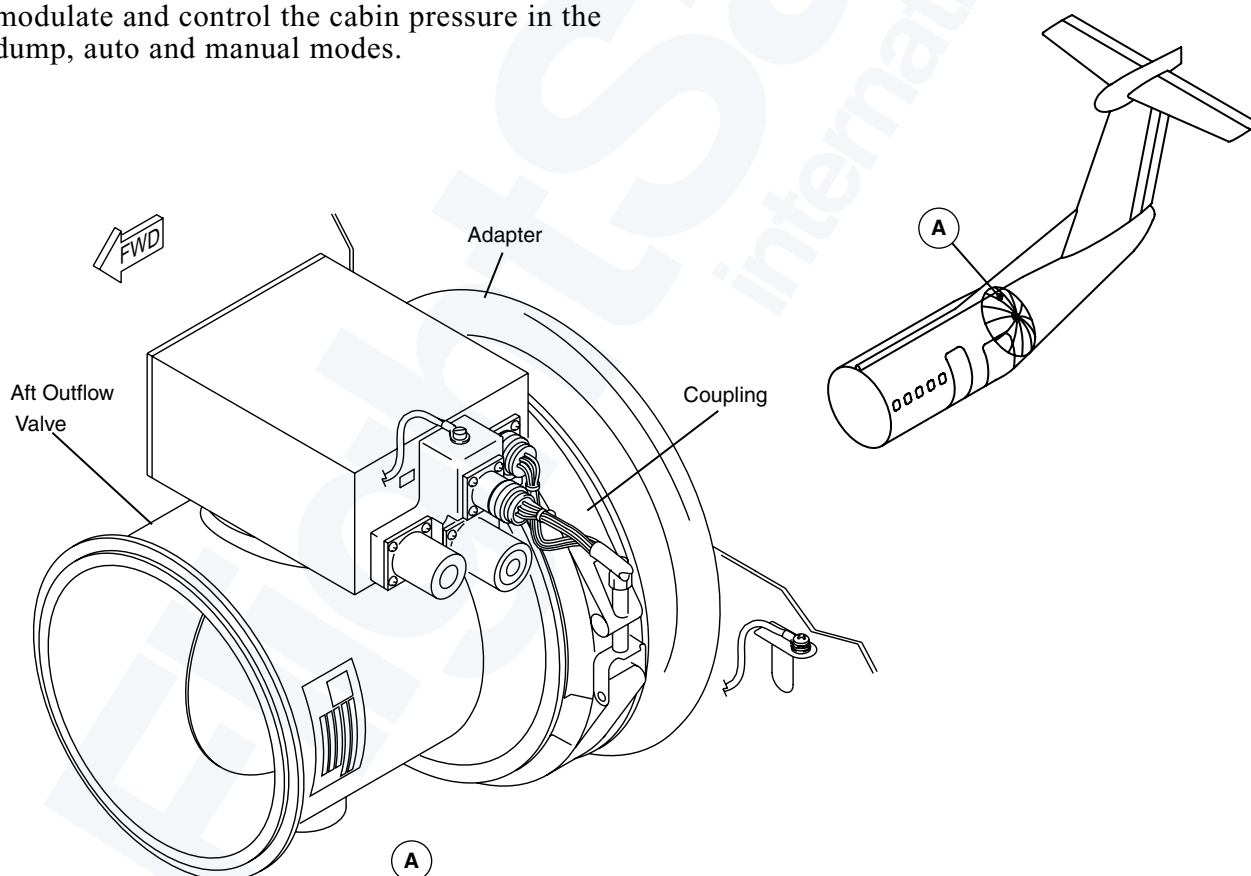


Figure 21-102. Outflow Valve Synoptic

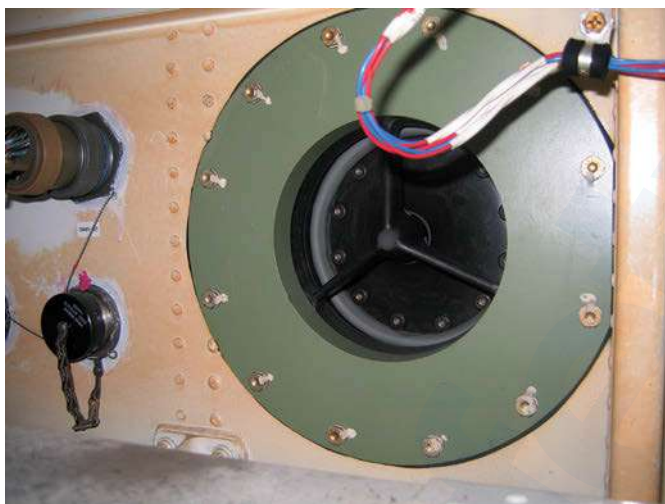
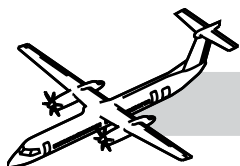


Figure 21-103. Forward Safety Valve

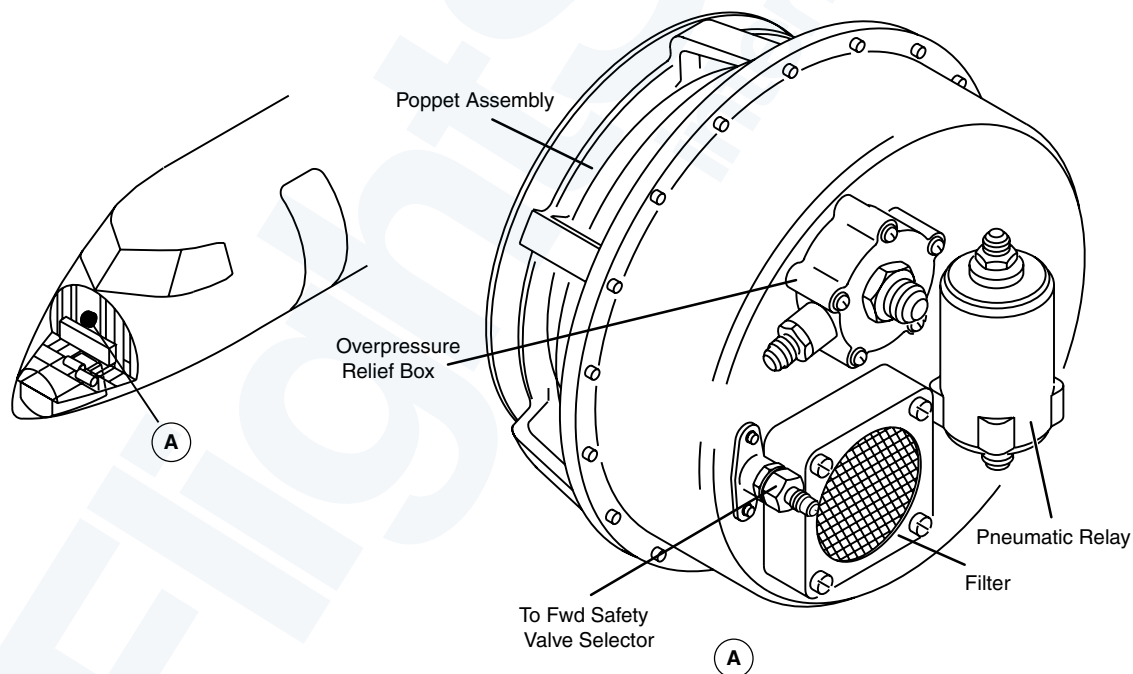
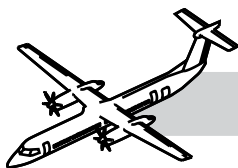


Figure 21-104. Forward Safety Valve Synoptic



Forward Safety Valve

NOTES

Refer to:

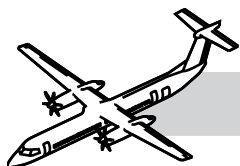
- Figure 21-103. Forward Safety Valve.
- Figure 21-104. Forward Safety Valve Synoptic.

The forward safety valve body is located in the nose compartment and is installed on the pressure bulkhead.

Its cover includes an overpressure relief box and a filter box, a poppet/diaphragm assembly, a deflector, a spring and a pneumatic relay that includes a manometric capsule and a check valve. The valve is of mainly composite construction.

The forward safety valve:

- Provides overpressure and negative pressure relief.
- Controls the discharge airflow for flight deck smoke evacuation or RAM air. Controlled by the FWD OUTFLOW knob of the CABIN PRESSURE control panel or the FWD safety valve selector on the co-pilot side panel.
- The filter is non cleanable and has to be replaced periodically as per maintenance requirement
- The valve is held by a quick attachment detachment clamp and a spring pin ensure the valve is installed in one fixed position
- There are two ways to do an operational check for the forward safety valve:
 - Connect a Vacuum Pump (GSB100006) to the suction port to create suction pressure for the forward safety valve and test operation of the valve by rotating the forward safety valve selector
 - Run both engines and pressurize the aircraft via MAN mode until CIM registers 2 psi differential pressure. Turn the FWD OUTFLOW selector to the INCR position and see the valve opens.



Forward Safety Valve Selector

Refer to Figure 21-105. Forward Safety Valve Selector.

The forward safety valve selector is located on the co-pilot side console.

Opening the forward safety valve provides smoke evacuation or ram air flow.

The selector fully opens the forward safety valve to release all cabin pressure. The selector is a shut-off valve.

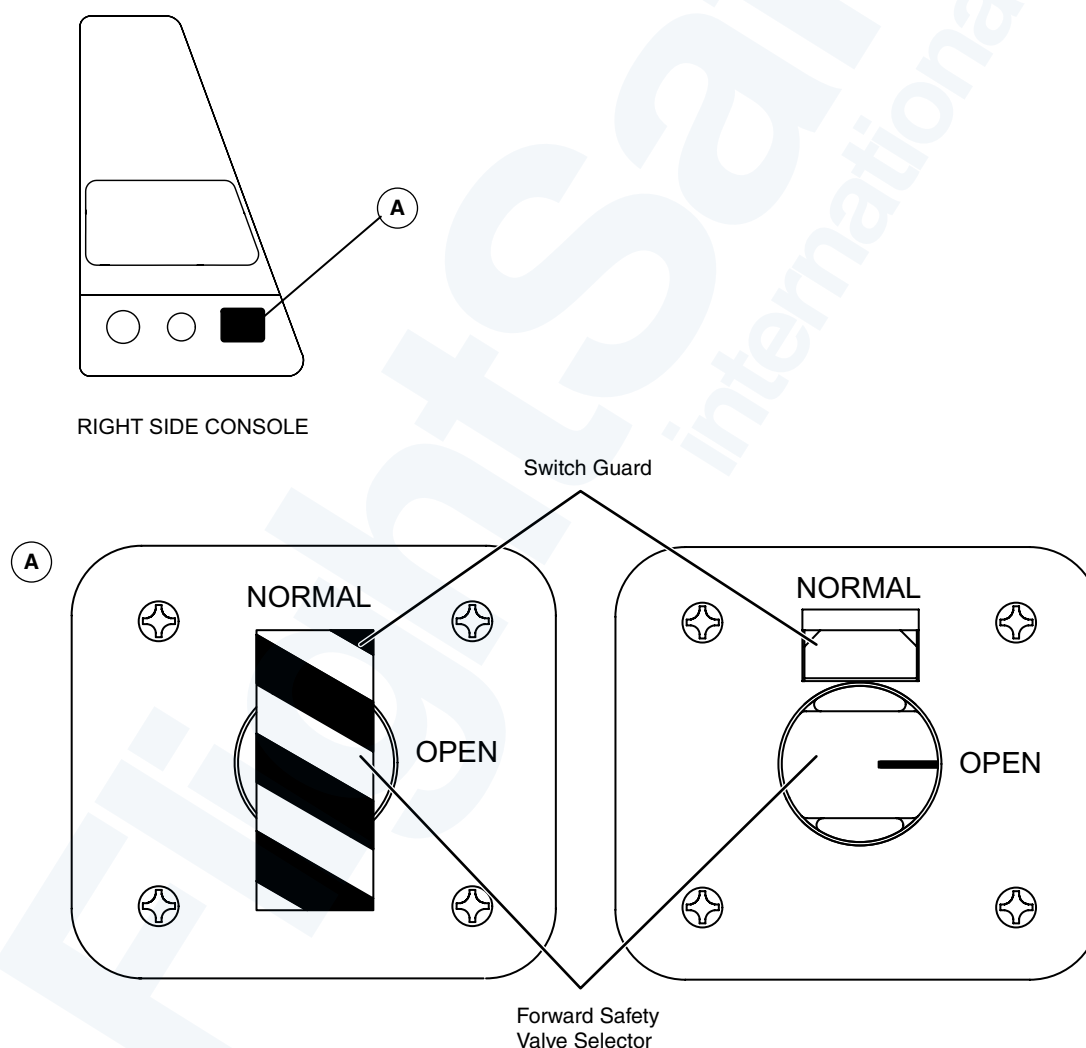
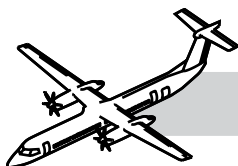


Figure 21-105. Forward Safety Valve Selector



Aft Safety Valve, Air Jet Pump & Safety Valve Filter

Refer to Figure 21-106. Aft Safety Valve.

Aft Safety Valve

The aft safety valve, located on the aft pressure bulkhead, has a solenoid valve and opens on the ground with the No.2 power lever below 60 degrees, if there is a bleed air source.

The aft safety valve also ensures protection for overpressure and negative pressure relief.

It is non cleanable and has to be replaced periodically. The filter on the aft safety valve is identical to the forward safety valve and is secured by four screws and a plate.

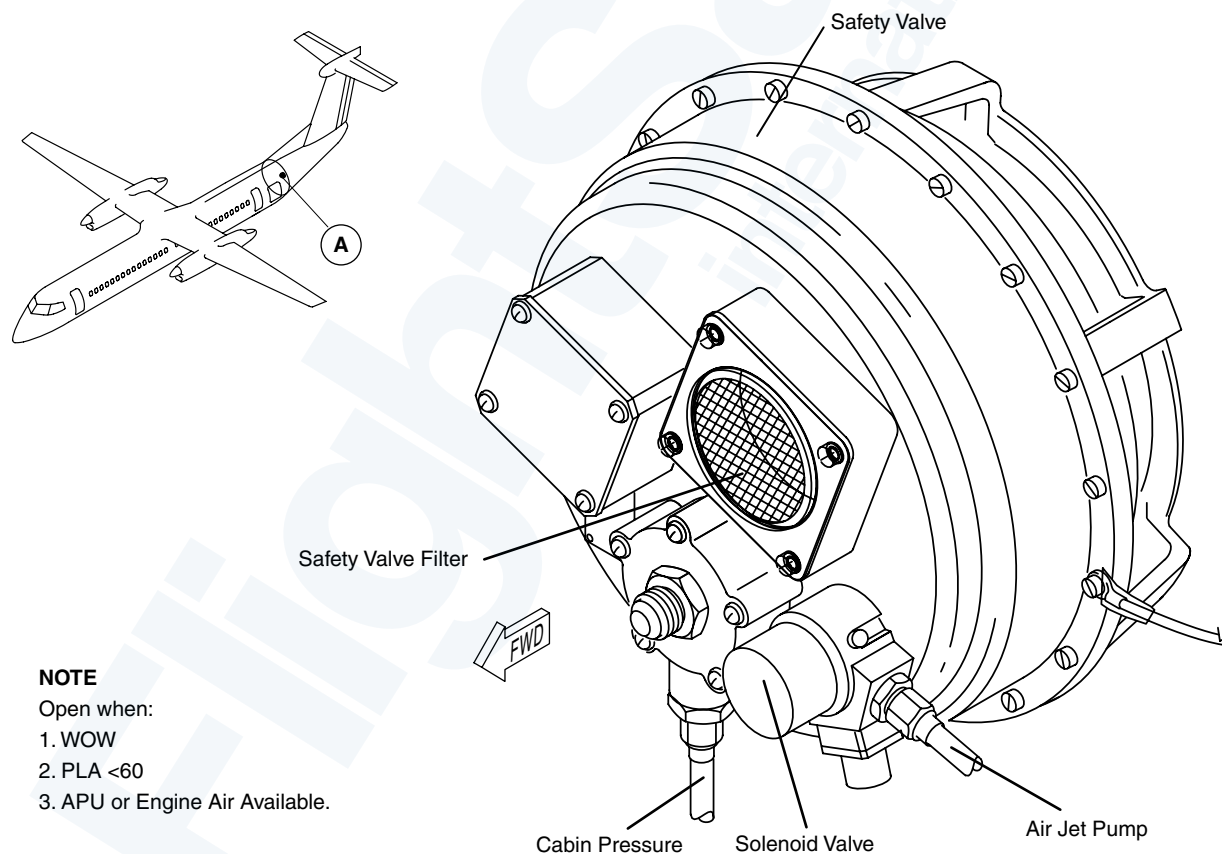
Air Jet Pump

The air jet pump, located on the aft safety valve, has a body and three orifices. The pump uses bleed air to cause a negative pressure on the solenoid valve to open the valve.

The pump fully opens the aft safety valve on the ground.

Safety Valves Filter

The filters, installed on the forward and aft safety valve, connect the cabin air to the safety valves. The filters feed the slave chamber of the safety valve with cabin pressure.

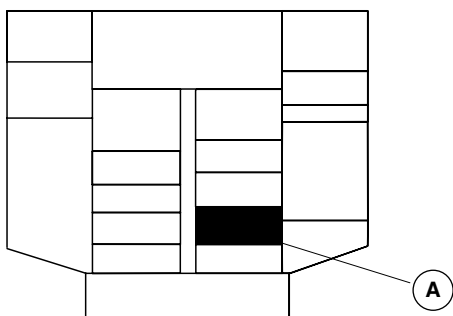
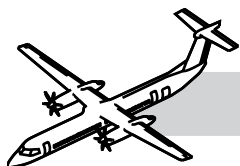


NOTE

Open when:

1. WOW
2. PLA <60
3. APU or Engine Air Available.

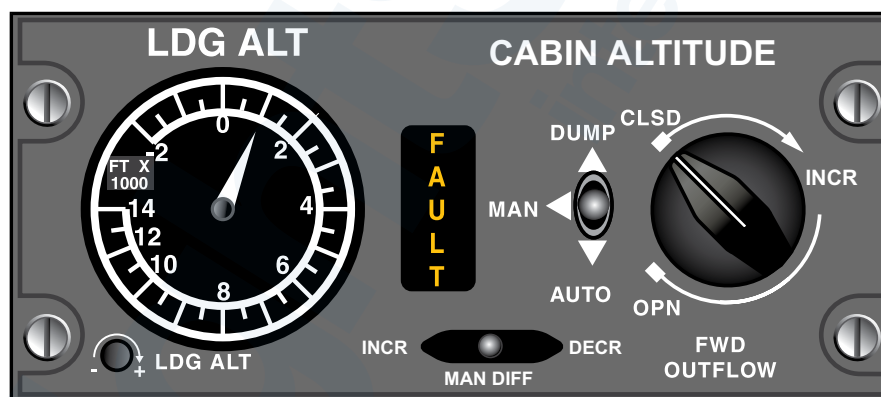
Figure 21-106. Aft Safety Valve



OVERHEAD CONSOLE

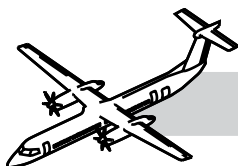
NOTE

-1,500 ft/min ↔ +2,500 ft/min
Rate of Change (AUTO Mode).



A

Figure 21-107. Cabin Pressurization Control Panel (CPCP)



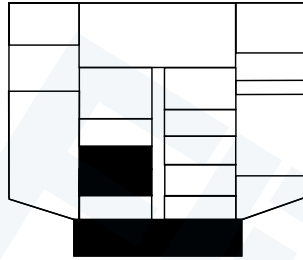
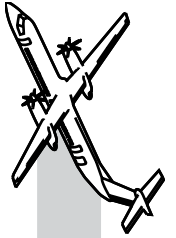
CONTROLS AND INDICATIONS

Cabin Pressurization Control Panel (CPCP)

Pressurization system control in auto, manual and dump modes.

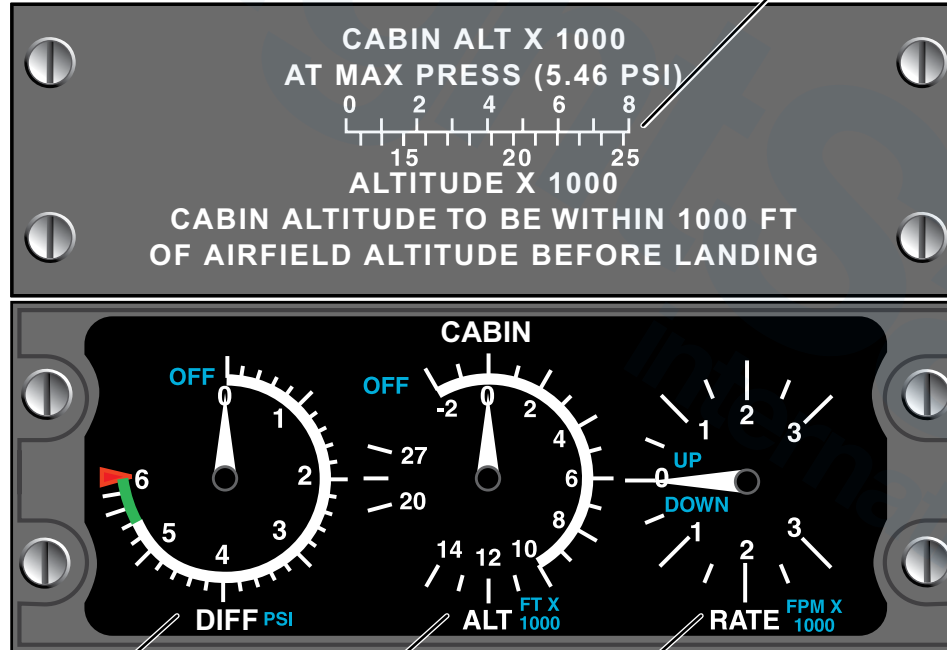
Refer to Figure 21-107. Cabin Pressurization Control Panel (CPCP).

1. LDG ALT Indicator
 - Displays the landing altitude selected with the LDG ALT rotary knob. Landing altitude selected provides CPC input to pressurize 400' below during landing.
2. FAULT ANNUNCIATOR Light (amber)
 - Is illuminated on CPCP during the POST mode, i.e. on initial aircraft power up or whenever the Cabin Pressure Controller circuit breaker is reset
 - Is illuminated on CPCP continuously when the cabin pressure control system detects a fault.
3. FWD Outflow Valve Rotary knob (rotary action)
 - CLSD - Closes the forward safety valve (if the forward safety valve selector is in the NORM position.)
 - OPN - Opens the forward safety valve
 - Usually set fully counterclockwise (CLSD) when in AUTO mode
 - In case of SMOKE evacuation, a manual controller drives the pneumatic relay of the Forward Outflow (Safety) Valve to position the valve. The FWD Safety Valve selector is provided to Open/Close the FWD Safety Valve.
4. LDG ALT Rotary Switch (rotary action)
 - Selects the field elevation for landing as shown on the LDG ALT indicator.
5. MAN DIFF Switch (three position, spring loaded to the center OFF position)
 - DECR - Decreases differential pressure when AUTO/MAN/DUMP switch is in the MAN position
 - INCR - Increases differential pressure when AUTO/MAN/DUMP switch is in the MAN position
 - Drives outflow valve open/closed for manual pressure control.
6. OUTFLOW VALVE Selection Switch (three position)
 - DUMP - Releases pressure by fully opening the outflow valve. Outflow valve is driven full open by the CPC.
 - MAN - Turns off the cabin pressure controller – enables manual control of the outflow valve and the cabin differential pressure using the MAN DIFF switch. The CPCP drives the electrical Outflow Valve through a toggle switch.
 - AUTO - Cabin altitude is controlled by the CPC using the outflow valve which modulates to maintain a pre-programmed cabin altitude schedule. The CPC controls the cabin pressure based on a software schedule that is referenced to aircraft altitude as sensed by the ADU.



OVERHEAD CONSOLE

Manual Cabin Alt/Differential
Pressure Schedule

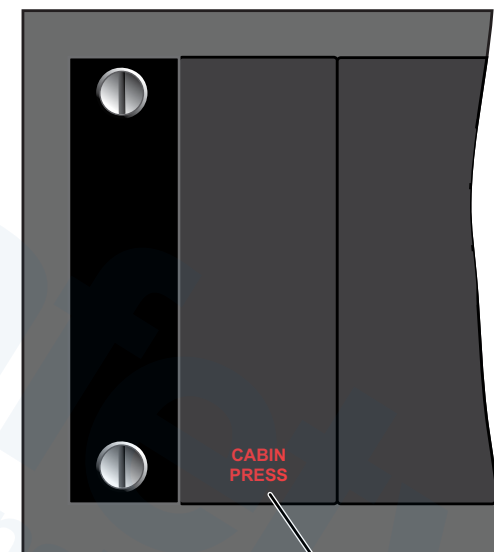


Diff Pressure
Indicator

Cabin Altitude
Indicator

Cabin Altitude Rate of
Climb Indicator

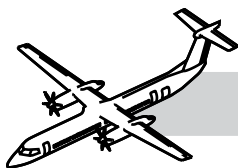
CABIN INDICATION MODULE



Cabin Press
Warning Light

CAUTION WARNING PANEL

Figure 21-108. Pressurization Indicator Panel



Cabin Indication Module (CIM)

The cabin indication module displays:

- Differential Pressure
- Cabin Altitude
- Rate of Change.

Refer to Figure 21-108. Pressurization Indicator Panel.

1. Manual cabin Aft / Differential pressure schedule
 - Used as a guide to set cabin differential pressure when using the MAN DIFF switch in MAN mode.
2. DIFF PRESSURE Indicator
 - Shows the differential pressure between cabin and aircraft flight altitude pressure in pounds per square inch.
 - The indicator gives a differential pressure from -0.5 to 5.8 ± 0.15 psi (-3.45 to 40 ± 1.03 kPa) between the cabin pressure and the outside static pressure.
3. CABIN ALT Indicator
 - Shows the cabin altitude in thousands of feet.
 - An internal absolute cabin pressure sensor performs cabin altitude indication. This sensor's information is transmitted on an ARINC 429 data bus to the CPC.
4. CABIN ALTITUDE RATE OF CLIMB Indicator
 - Shows cabin altitude rate of climb or descent in thousands of feet per minute.
 - Shows cabin altitude rate of climb or descent in thousands of feet per minute.
5. CABIN PRESSURE WARNING
 - The cabin indication module shows data to monitor the system. When the cabin altitude reaches 9,800 ft., an internal switch applies a ground signal to the CABIN PRESS warning light. If the

cabin altitude continues to rise and reaches 10,000 ft., a switch integral to the cabin pressure controller applies 28 VDC to the CABIN PRESS warning light causing it to come on.

Interfaces

The interfaces for the pressurization system are:

- Power Lever Angle (60° Micro Switch) right lever only.
- PSEU (WOW)
- ECS Controller
- IFC 1 (CDS)
- ADU 1 and ADU 2
- Caution and Warning System (CABIN PRESS)
- 28 VDC Left Main Bus (Cabin Press Auto Cont circuit breaker).

OPERATION

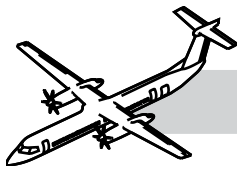
Normal System Operation

The aircraft cabin pressurization system is controlled automatically and/or manually. The CPCP and the CIM allow the crew to have control and indication of the cabin pressurization. The CPC programming maintains the cabin pressure during automatic mode. The crew maintains the cabin differential pressure during manual mode.

Automatic Mode

The CPCS will operate in automatic mode when:

- Left main bus energized and circuit breaker C4, CABIN PRESS AUTO CONT provides 28 VDC to the CPC connector J1 pin 1 and to the CPCP connector J2 pin E.
- Left essential bus is energized and circuit breaker L5, CABIN PRESS MAN provides 28 VDC to the aft safety valve connector P5 pin A.
- PSEU WOFFW open circuit discrete is sensed on CPC connector J1 pin 11.



- PSEU door closed discrete sensed on CPC connector J1 pin 5.
- Right power lever > 60° switch plus WOW sensed on CPC connector J1 pin 26.
- ECS ECU PFCSOV (valve open) discrete from connector 2100-P/ J2 pin 49 sensed on CPC connector J1 pin 41.
- AUTO selection on CPCP provides a discrete from CPCP connector J2 pin L to CPC connector J1 pin 40. This causes the CPC to provide an enable discrete from CPC connector P1 pin 17 to the aft outflow valve connector J3 pin A.
- AUTO selection also provides a 28 VDC AUTO discrete from CPCP connector J2 pin D to aft outflow valve connector P3 pin G.

The CPCS starts the flight sequence mode with WOFFW and No.2 power lever > 60°.

The cabin pressure is programmed to be a maximum of 8,000 ft (2,438 m) at a flight altitude of 25,000 ft (7,620 m), maximum differential pressure of 5.46 psi (377 mb).

A cabin sensor in the CPC measures the cabin pressurization.

Before flight the crew selects the DUMP/MAN/AUTO switch to the AUTO position and selects the LDG ALT on the landing altitude indicator.

During flight the automatic mode:

1. Calculates the cabin pressure rate of change in accordance with the landing altitude selection and software schedule.
2. Maintains the lowest possible cabin altitude within the maximum differential pressure (5.46 psi) range.
3. Sets the minimal rate of decent at 300 fpm or 0.159 psi per minute (10.99 mb/min) and modulates the rate based on the aircraft decent rate. During a high rate of decent, the CPC uses ADU vertical speed data and remaining flight time to the set landing altitude to calculate the increased rate (maximum cabin rate of decent is 2,000 fpm).

4. Sets the cabin pressure to land at 400 ft below the selected landing altitude.
5. Controls the after landing de-pressurization mode.

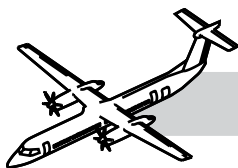
Manual Mode

The CPCS will operate in manual mode when:

- Left main bus is energized and circuit breaker C4, CABIN PRESS AUTO CONT provides 28 VDC to CPC connector J1 pin 1 and CPCP connector J2 pin E.
- Left essential bus is energized and circuit breaker L5, CABIN PRESS MAN provides 28 VDC to aft safety valve connector P5 pin A.
- The MAN in selection on CPCP provides a discrete from CPCP connector J2 pin L to the CPC connector J1 pin 40. This causes the CPC to provide a disable discrete from CPC connector P1 pin 17 to the aft outflow valve connector P3 pin A. The AUTO discrete from CPCP connector J2 pin D to the aft outflow valve connector J3 pin G is removed.
- 28 VDC manual control discrettes on CPCP connector J2 pins G (close) and F (open) to the aft outflow valve connector J4 pins A (close) and C (open).

The manual mode is used to control the aft outflow valve when the AUTO-MAN-DUMP switch is set to MAN. The MAN DIFF switch is set to the DECR position to open the aft outflow valve which causes the cabin differential pressure to decrease.

The differential pressure is shown on the CIM DIFF indicator. When the toggle switch is set to the INCR position, the outflow valve closes and causes the cabin differential to increase and the cabin altitude to decrease.



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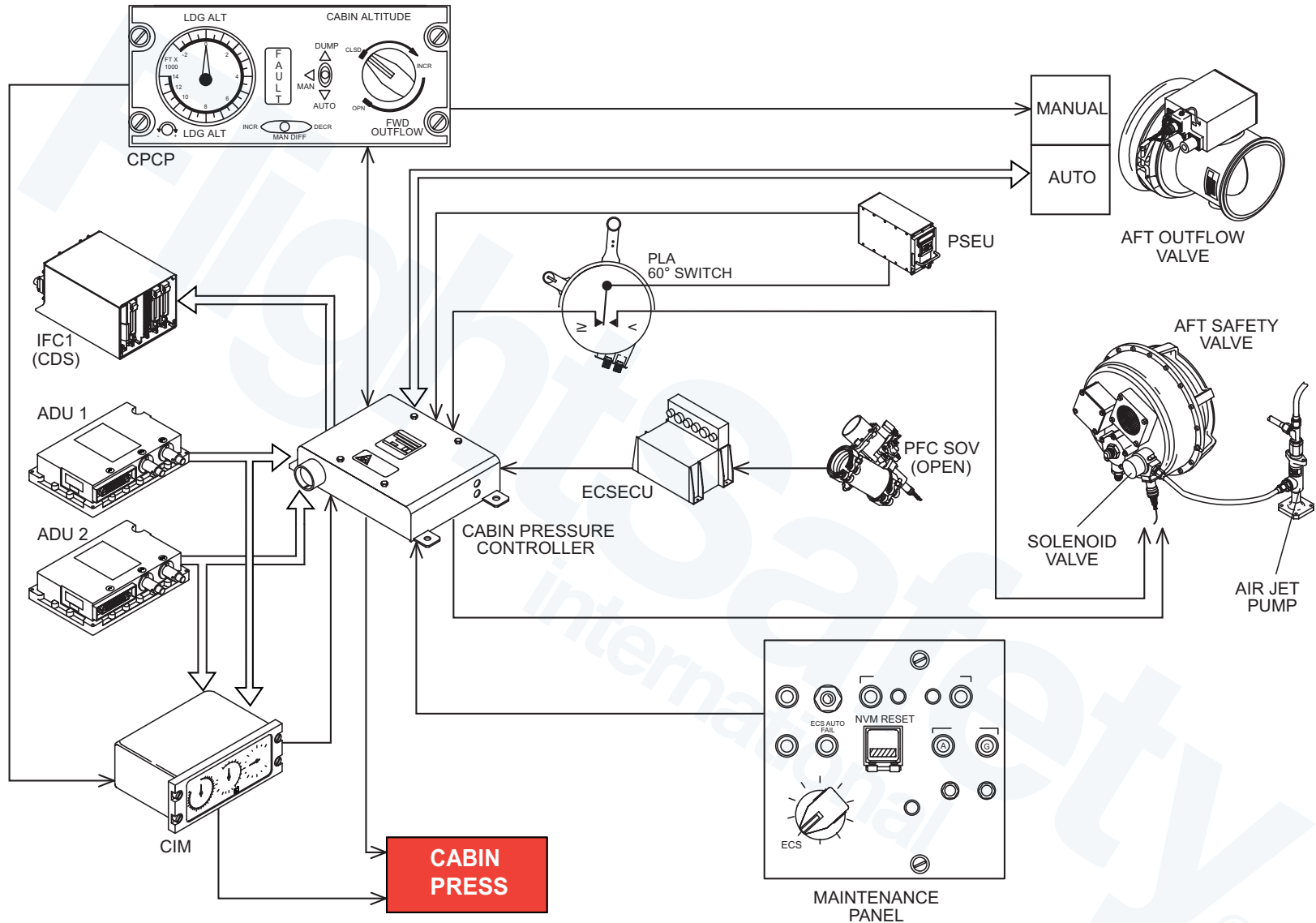
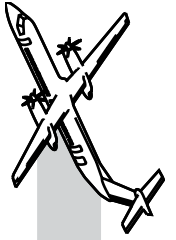
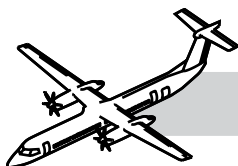


Figure 21-109. CPCS Diagram



Ground Mode (Manual or Automatic)

NOTES

Refer to Figure 21-109. CPCS Diagram.

During Ground mode, the Aft Safety valve stays open if:

- 28 VDC from circuit breaker L5 is available.
- Bleed air is supplied to the Aft Safety valve Air Jet pump.
- PSEU WOW discrete is sent through the PLA connector 7600 P8 pin 20 60 degrees switch to the Aft Safety valve solenoid connector P5 pin B.

Take-Off Mode

During the take-off ground mode the CPC will modulate the aft outflow valve to pressurize the cabin to 400 ft (121.9 m) below ambient pressure.

Landing Mode

During the landing mode depressurization is automatic:

If the set field altitude is higher than actual field altitude, the aircraft will land un-pressurized.

- If the field altitude is set less than landing field altitude, the aircraft will land pressurized. The WOW signal from PSEU connector 3261-P/J1A pin 6K is received by the CPC on connector J1 pin 11. The CPC sends an enable signal to the Aft Outflow valve on connector P/J3 pin A, which causes the cabin to depressurize at a calculated rate for one minute to reach the landing field altitude.

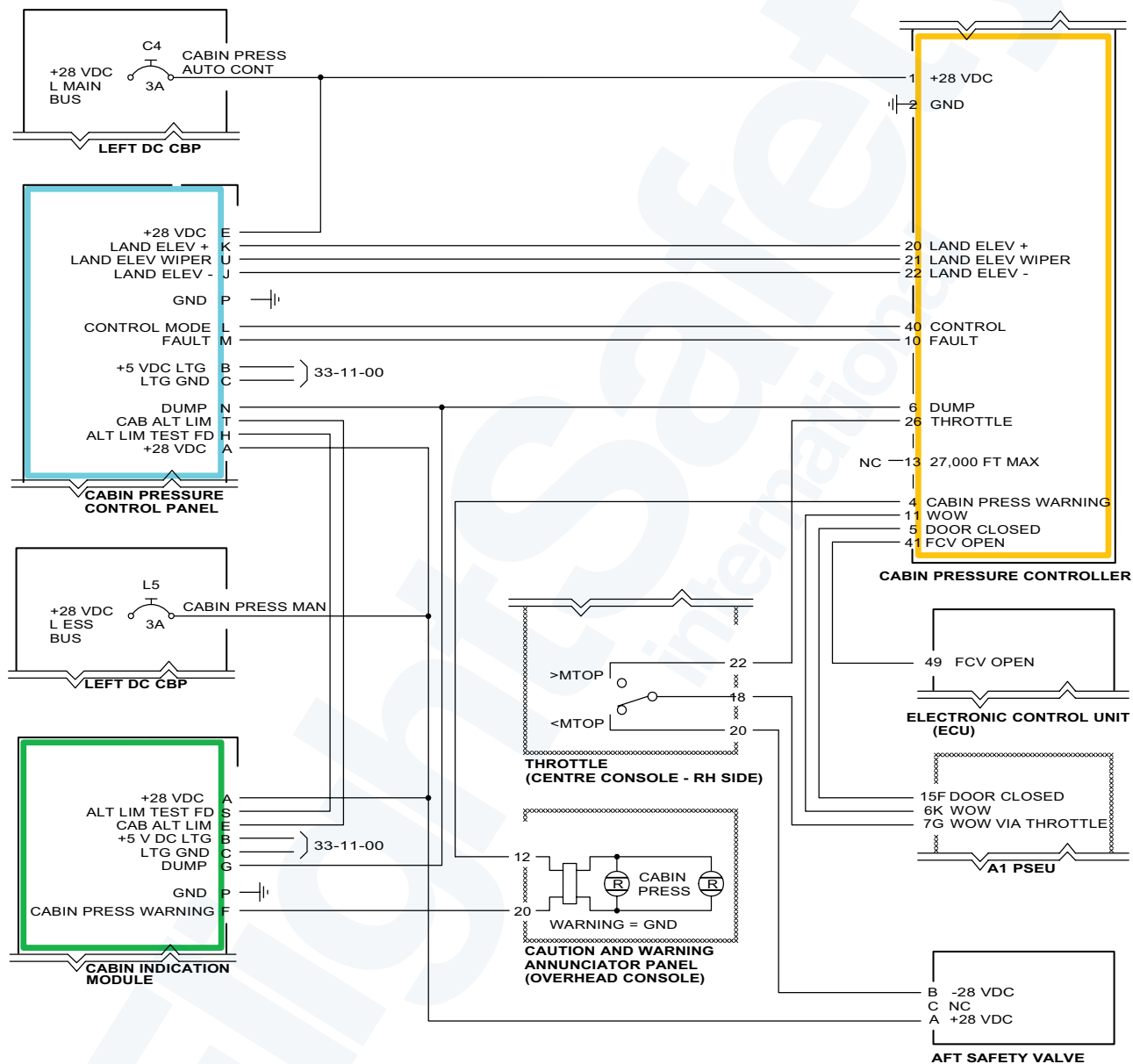
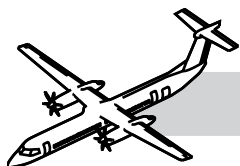
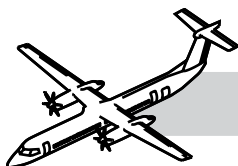


Figure 21-110. CPCS - Electrical Schematic



Pressure Dump Mode

Refer to Figure 21-110. CPCS - Electrical Schematic.

The fast depressurization function is possible from the automatic and the manual modes:

The AUTO/MAN/DUMP switch set to DUMP position.

- The CPCP connector P2 pin N sends a dump discrete to the CPC connector P1 pin 6 and to the CIM connector 2131-P1 pin G.
 - During automatic mode, an automatic dump signal is also sent through the CPCP connector P2 pin F (manual open) channel directly to the aft outflow valve (automatic motor) connector P4 pin C.
- During manual mode, the crew must set and hold the MAN DIFF to the DECR position to send the (valve open) signal from the CPCP connector P2 pin F directly to the aft outflow valve connector P4 pin C.

Maintenance Practices

Pressurize the Aircraft for Maintenance

AMM Task 21-00-00-863-801 provides the steps to pressurize the aircraft and optionally verify the fuselage leak rate is within limits. The Tools required for this procedure are:

- Stop watch
- Calibrated pressure gauge 0-25 psi
- Rubber plugs to prevent air leaking through the drain masts

NOTE

1. Special precaution must be made to avoid cabin differential > 5.5 psid as the safety valves can open at 5.8 psid and may cause injury to personnel and damage to the aircraft.
2. Always use a calibrated absolute pressure gauge to determine the cabin internal pressure has reached 5.5 psi above what was recorded in the cabin prior to the start of the pressurization run.
3. Do not open doors or hatches during pressurization run or until the internal pressure is dropped to 0 psid.

The pressurization procedure is summarized as follows:

- Ensure all door are closed and locked
- Start and run both engines with #2 PLA > 60°
- Use MANUAL pressurization mode to close the outflow valve, making sure the RATE indication does not exceed -300 FPM
- Monitor the hand held pressure gauge and the CIM to watch for when cabin differential pressure has reached 5.5 psid
- Observe the time taken for DIFF to drop from 5.0 to 3.0 is > 194 seconds
- If the leakage rate is lower than the AMM limit, inspect for leaks and repair as necessary.

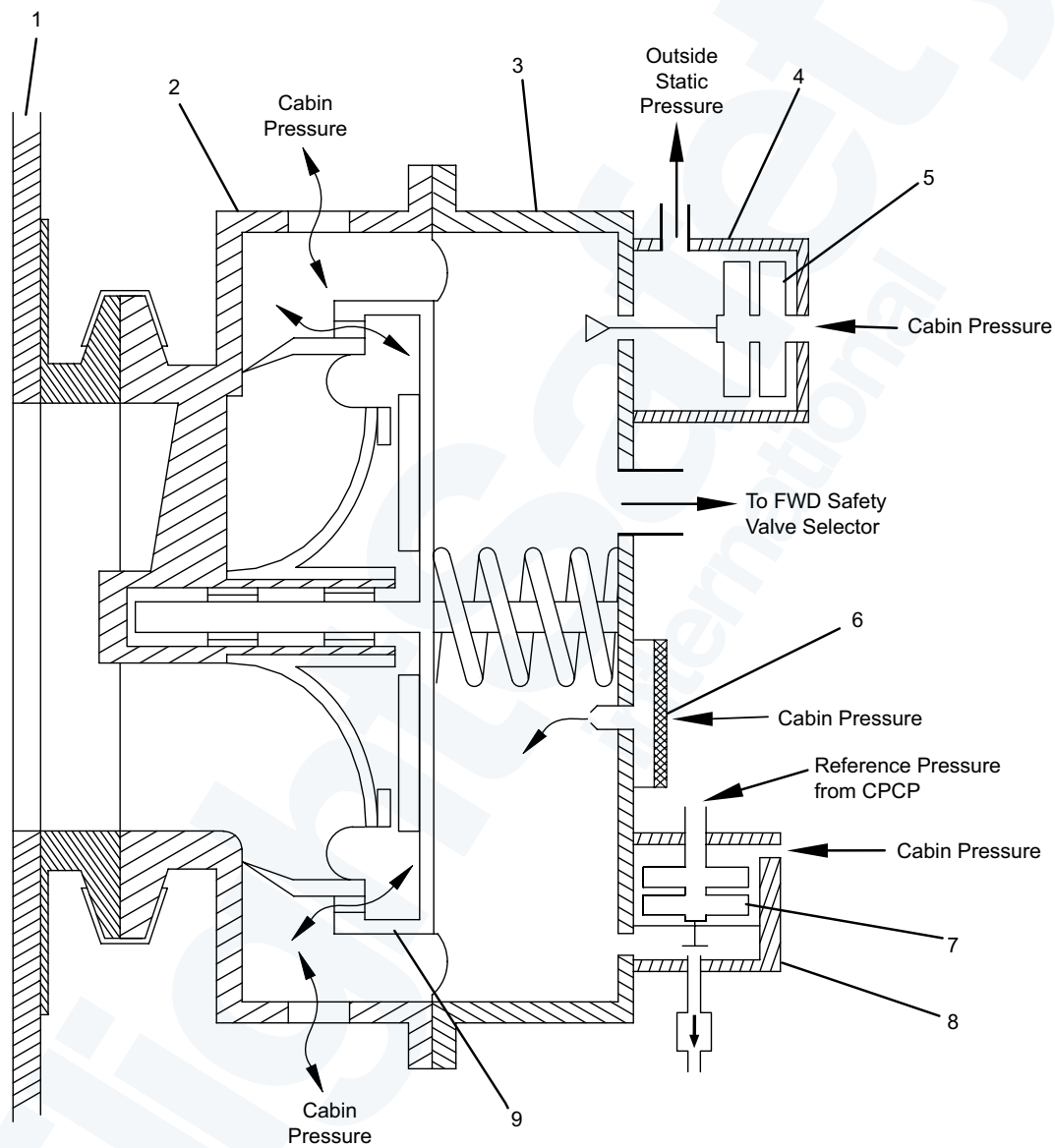
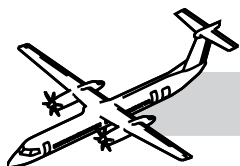
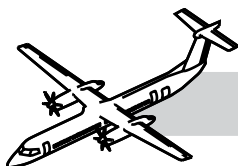


Figure 21-111. Forward Safety Valve - Detail



Emergency Mode / Smoke Removal

NOTES

Refer to:

- Figure 21-111. Forward Safety Valve - Detail.
- Figure 21-112. Forward Safety Valve Operation.

There are two emergency selectors that send vacuum reference pressure to the Forward Outflow valve which causes the valve to open and expel the cabin air/smoke.

- FWD OUTFLOW selector (on the CPCP) can be used to modulate the forward outflow valve.
- FORWARD SAFETY VALVE selector (on the right side panel) opens and closes the forward outflow valve.

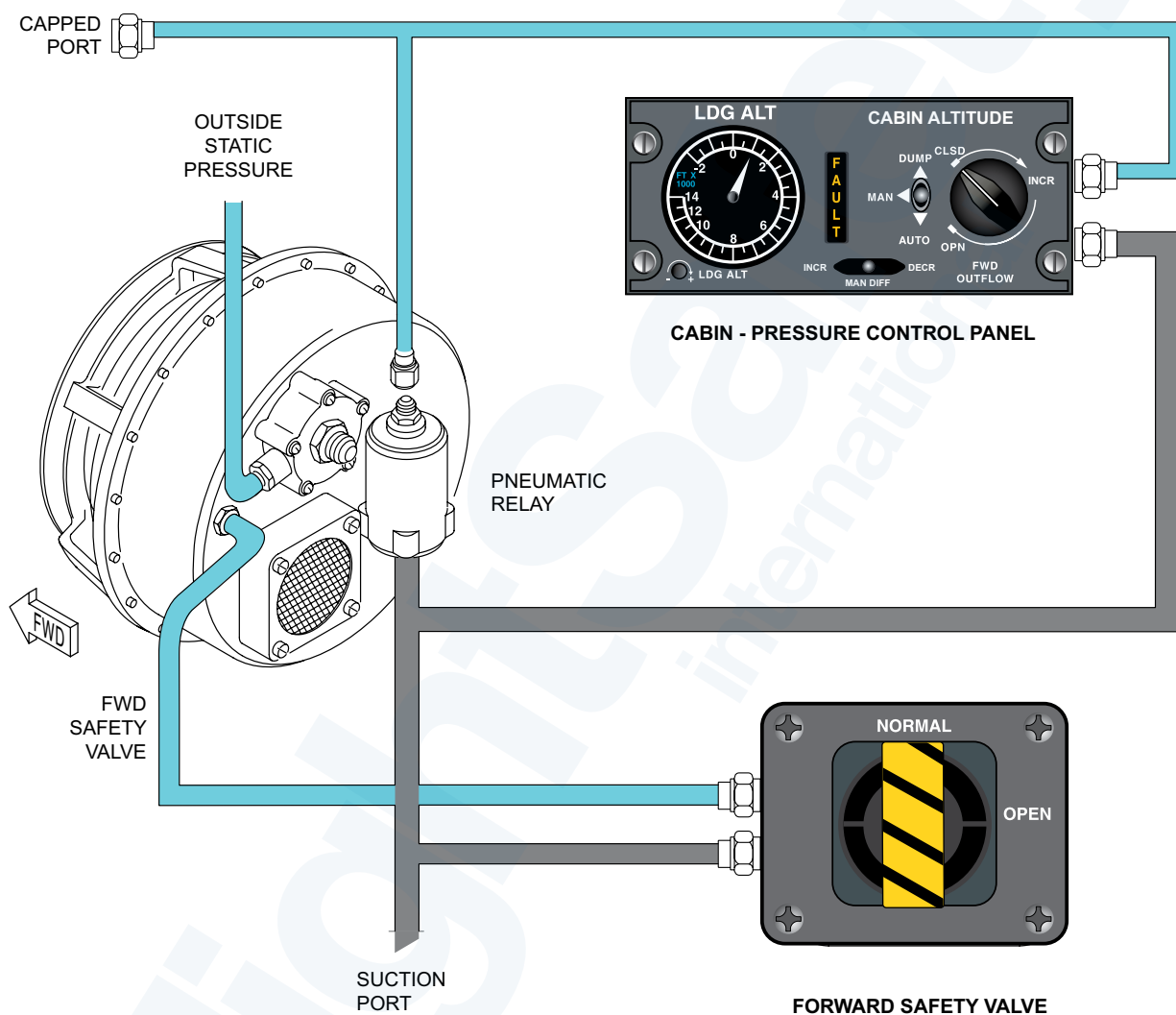
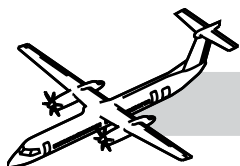
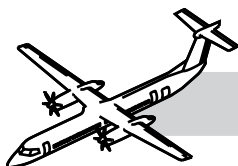


Figure 21-112. Forward Safety Valve Operation



FWD OUTFLOW OPERATION

NOTES

Refer to Figure 21-112. Forward Safety Valve Operation.

- CPCP AUTO-MAN-DUMP switch set to MAN.
- This disables the automatic mode of the aft outflow valve. The aft outflow valve in manual mode does not move unless selected with the MAN DIFF switch.
 - CPCP FWD OUTFLOW selector set to the OPEN position.
 - The selector valve connects the vacuum reference pressure into the manometric capsule in the pneumatic relay which collapses and retracts the poppet from the suction port vacuum.
 - The low pressure in the cover of the forward safety valve causes the main poppet to retract.
 - When the main poppet retracts, the valve exhausts the cabin air and smoke.

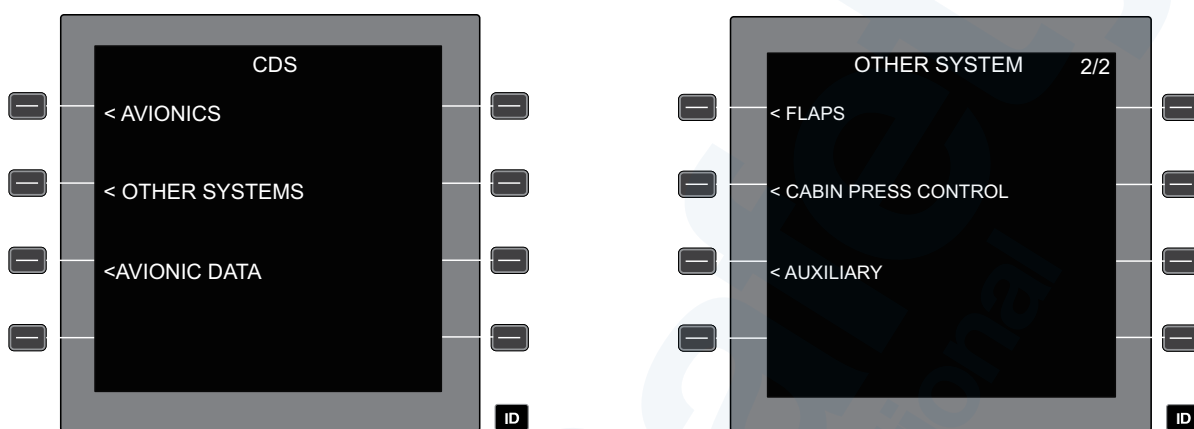
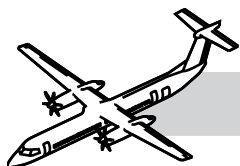


Figure 21-113. CDS Other System Screen

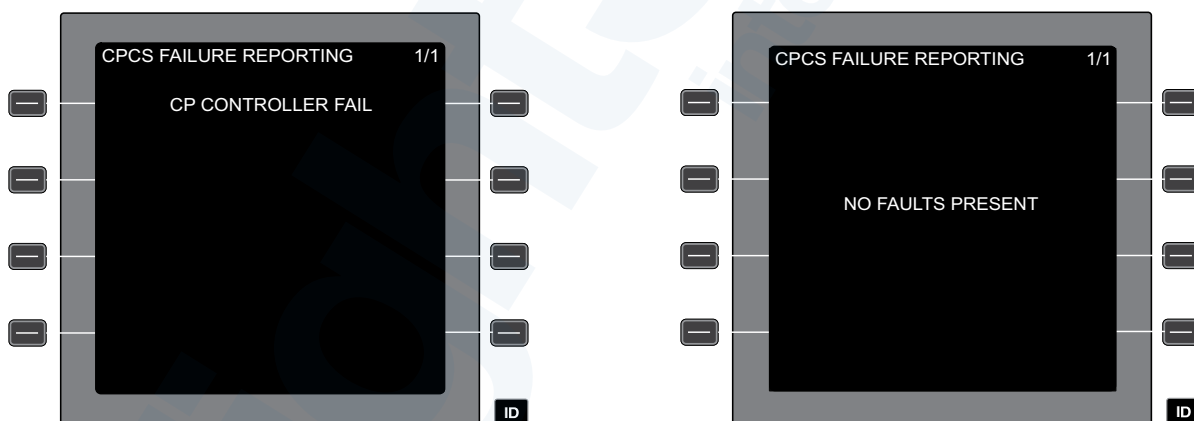
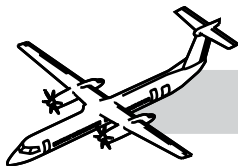


Figure 21-114. CDS CPCS Failure Reporting Screen



MAINTENANCE CONSIDERATION

System Limitations

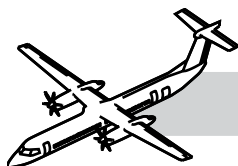
As per section 2-6-1 of the Q400 Airplane Flight Manual (DOT), the following limitations apply:

- Normal cabin pressure differential limit.....5.46 ± 0.1 psi
- Maximum cabin pressure differential limit.....5.95 psi
- Maximum cabin pressure differential limit during taxiing, take-off and landing0.5 psi
- Maximum cabin altitude (pressurized light) 8000 ft

Maintenance Troubleshooting

There is a separate CDS page dedicated for the pressurization system - Cabin Pressure Control page. You can access it by drilling down the Other Systems menu item as in Figure 21-113 and Figure 21-114 depict cabin pressurization related fault and a no fault scenarios.

To clear existing ECS or CPC system faults, maintenance personnel have to use the NVM Rotary Switch to select the appropriate system before pushing the NVM Reset Pushbutton Switch.



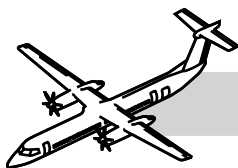
21-00-00 SPECIAL TOOL & TEST EQUIPMENT

- Panduit Corporation, Tool No. GS-2B
- GSB2100006, Vacuum Pump
- GSB3411011(or equivalent) Barfield Pitot-Static Test Set
- GSB5180001/5180005 or equivalent Electrical Bonding Test Set
- GSB2111004-7 (or equivalent) Thermometer
- GSB2400001 (or equivalent) Multimeter, hand held
- GSB2111004 Kit - Fluidized Bath

21-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 21-22-00-900-801 Discard the Cabin Air Distribution Recirculation Filter (MRB# 212200-201)
- AMM 21-22-22-000-801 Removal of the Recirculation Filter
- AMM 21-22-22-400-801 Installation of the Recirculation Filter
- AMM 21-22-74-710-801 Operational Check of the Aft Baggage Compartment Vent Shut-Off Valve
- AMM 21-25-00-900-801 Discard the Front Fuselage LCD Cooling Filters (MRB#212200-203)
- AMM 21-25-00-710-801 Operational Check of the Avionics Cooling Fans (MRB# 212500-201)
- AMM 21-31-41-960-802 Replacement of the Safety Valve Filters
- FIM 21-30-00-810-801 CABIN PRESS (Warning) – Fault Isolation
- FIM 21-30-00-810-802 FAULT, Cabin Pressurization Control Panel (Caution) – Fault Isolation
- FIM 21-30-00-810-803 AFT OUTFLOW VALVE FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-804 CPC/OF VALVE DATA FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-805 CPC/ADU ARINC FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-806 CPC/CIM ARINC FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-807 CIM/ADU ARINC FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-808 CP CONTROL PANEL FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-809 CP CONTROLLER FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-813 INDICATION MODULE FAIL (Status) – Fault Isolation
- FIM 21-30-00-810-814 CABIN PRESSURE LOSS OR FLUCTUATIONS–Fault Isolation
- AMM 21-31-00-720-801 Functional Test of the Pressurization Control
- AMM 21-31-16-720-801 Functional Test of the Aft Outflow Valve



- AMM 21-52-00-710-801 Operational Check of the Pack Flow Control Shut-off Valve (MRB #215200-201)
- FIM 21-60-00-810-801 CABIN PACK HOT (Caution) – Fault Isolation
- FIM 21-60-00-810-802 FLT COMPT PACK HOT (Caution) – Fault Isolation
- FIM 21-60-00-810-803 FLT COMPT DUCT HOT (Caution) – Fault Isolation
- FIM 21-60-00-810-804 CABIN DUCT HOT (Caution) – Fault Isolation
- FIM 21-60-00-810-806 CAB BYP VLV (Status) – Fault Isolation
- FIM 21-60-00-810-809 PACK FCV (Status) – Fault Isolation
- FIM 21-60-00-810-810 PACK ILT MISCMP, channel 1 (Status) – Fault Isolation
- FIM 21-60-00-810-811 PACK ILT PR, channel 1 (Status) – Fault Isolation
- FIM 21-60-00-810-812 PACK ILT TMP, channel 1 (Status) – Fault Isolation
- AMM 21-61-00-710-801 Operational Check of the Temperature Control, Electronic Control Unit (MRB# 216000-201)
- AMM 21-61-00-710-803 Operational Test of the ECS Temperature Control